

MSc Data Science Project

University Of Hertfordshire

School Of Physics, Engineering And Computer.

Department of Physics, Astronomy and Mathematics

Data Science: Final Project Report

Project Title:

Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh.

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Github link: <https://github.com/Fariduk/PROJECT-DATA-SCIENCE-COVID-19-Comparative-Data-Analysis-and-Prediction-for-the-World-and-Bangladesh.git>

Drive link: https://drive.google.com/drive/folders/1uTZzwnYL3Wu4xKs4Sn8EhH273IUyTHo0?usp=drive_link

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science in Data Science at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at Assessment Offences and Academic Misconduct and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

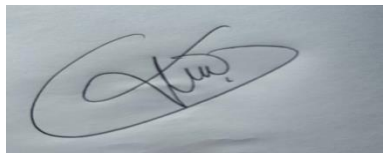
I certify that the work submitted is my own and that any material derived or quoted from published or unpublished work of other persons has been duly acknowledged. (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6)

I did not use human participants in my MSc Project.

I hereby give permission for the report to be made available on module websites provided the source is acknowledged.

Student Name printed:Farid Hossain.

Student Name signature:

A photograph of a handwritten signature in black ink on a light-colored surface. The signature is stylized and appears to read 'Farid Hossain'.

Student SRN number:23006446

UNIVERSITY OF HERTFORDSHIRE

SCHOOL OF PHYSICS, ENGINEERING AND COMPUTER SCIENCE

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Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh.

Abstract

COVID-19 has emerged as one of the most significant global health challenges in recent history, prompting extensive research across multiple domains to combat its rapid spread and devastating impact. This project focuses on a comparative data analysis and prediction of COVID-19 trends, particularly emphasizing global patterns alongside an in-depth study of the situation in Bangladesh.

A rich set of datasets has been collected from various credible sources, including Kaggle, WHO, and national health databases, comprising metadata related to COVID-19 confirmed cases, deaths, recoveries, quarantine details, and demographic breakdowns. These datasets span a wide temporal range and are preprocessed using Python, pandas, and data wrangling techniques to ensure consistency and reliability.

To support reproducibility and version tracking, tools such as GitHub and Data Version Control (DVC) are utilized in the data pipeline. The cleaned and processed data serves as the foundation for developing predictive models using machine learning frameworks such as TensorFlow, aiming to forecast key metrics like infection rates, death rates, and recovery trends.

A particular focus has been placed on analyzing the Bangladesh-specific data, including division-wise and gender-age-specific trends. Comparative visualizations between Bangladesh and other countries, including China and global datasets, offer insights into how different regions responded to the crisis.

The project also investigates the effectiveness of lockdown strategies implemented across different countries and provinces. Using the available data, an analytical framework is built to assess the outcomes of these public health measures, identifying successful practices and potential gaps.

This research validates the project title "Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh" by presenting comprehensive insights through statistical analysis, machine learning predictions, and policy evaluation, grounded entirely on open-source and authenticated datasets available up to November 2020. The results and predictions presented are subject to change with the inclusion of updated or real-time data.

Introduction

The COVID-19 pandemic has significantly disrupted global public health systems, economies, and everyday life. On January 30, 2020, the World Health Organization (WHO) declared the outbreak of SARS-CoV-2 a Public Health Emergency of International Concern (PHEIC), highlighting the urgent need for coordinated global efforts, data-driven policymaking, and predictive health analytics.

In this context, the research community has mobilized to collect, analyze, and interpret vast volumes of pandemic-related data. This project, titled "Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh", is a comprehensive study that aims to explore and model the patterns of COVID-19 transmission, recovery, and mortality through comparative data analysis and predictive modeling techniques.

Using real-world datasets collected from Kaggle, WHO, and national health portals, this project integrates data specific to Bangladesh, China, and global trends. The objective is to examine and compare outbreak dynamics at different scales, identify significant patterns, and generate predictive insights through advanced machine learning approaches. This includes applying classification algorithms, Predictive Data Modeling (PDM), and deep learning techniques using TensorFlow.

The project employs Python-based tools including pandas for data wrangling, Matplotlib and Seaborn for visualization, and NumPy for numerical computations. It covers multiple analytical angles such as gender and age-based analysis, regional spread (division-wise), daily trends, and the impact of healthcare infrastructure like hospital bed availability. It also evaluates public policy interventions like lockdowns to assess their effectiveness across regions.

A strong emphasis is placed on open data usage, transparent methodologies, and version-controlled development through platforms like GitHub and DVC, ensuring reproducibility and reliability. The findings from this project are not only relevant for understanding the trajectory of the pandemic but also serve as a data-driven foundation for informed decision-making by policymakers, healthcare professionals, and researchers.

Literature Review

The global outbreak of COVID-19 has led to an unprecedented volume of research and data collection efforts aimed at understanding, modeling, and controlling the spread of the virus. Numerous governments, research institutions, and organizations have made significant contributions to the growing body of literature and publicly available datasets.

According to data published by the Chinese Center for Disease Control and Prevention (China CDC), the number of COVID-19 cases rose dramatically during January to March 2020, marking the initial peak of the outbreak in Wuhan and surrounding regions. China not only released daily case counts but also maintained a robust database that provided demographic breakdowns, transmission trends, and fatality statistics.

Similarly, the Government of Hong Kong launched an official online dashboard to inform the public and visualize the daily status of infections, recoveries, and healthcare responses. This open-access platform played a vital role in raising awareness and encouraging timely policy decisions.

In Europe, several countries developed predictive models based on their early outbreak data. These models initially aligned with actual case trends, helping governments respond effectively. However, the onset of the second wave highlighted the challenges of prediction under dynamic public health conditions. These experiences underscore the importance of adaptable and continually updated models.

From a healthcare policy perspective, organizations like the Center for Health Information and Analysis (CHINA) in the United States began producing monthly summaries of enrollment and insurance coverage trends. These insights have been essential in understanding how the pandemic affected access to healthcare services and have informed ongoing healthcare reforms.

Additionally, institutions like Johns Hopkins University developed globally recognized dashboards and predictive models that visualized the spread of COVID-19 across different continents. Their open-access repository became one of the most cited and referenced sources during the pandemic, providing real-time updates and enabling data scientists worldwide to build analytical tools and forecasting models.

This project builds upon these global efforts by focusing on a comparative analysis of COVID-19 data across the world with an emphasis on Bangladesh, leveraging predictive modeling and deep learning techniques to evaluate trends, assess healthcare preparedness, and analyze the impact of policy measures. Through integration of datasets from WHO, Kaggle, and national health organizations, this research contributes to the ongoing discourse on effective pandemic response strategies and predictive analytics in public health.

Problem Statement

In this research project -Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh, I have addressed several critical questions related to the impact, progression, and prediction of the Covid-19 pandemic. The main problem statements we have explored are:

1. Where did the Covid-19 outbreak originate, and how did it spread globally?
2. How have the Covid-19 pandemic impacted major countries, especially Bangladesh, in terms of confirmed cases, deaths, and recoveries?
3. What was the role of lockdowns in controlling the spread of the virus, and were they effective both globally and in Bangladesh?
4. Bangladesh: What is the current Covid-19 situation, and how does it compare with global trends?
5. Is the healthcare system in Bangladesh sufficiently equipped to handle the challenges posed by the Covid-19 pandemic?
6. What predictive models can be applied to forecast Covid-19 cases, deaths, and recoveries, particularly in Bangladesh and globally?

These problem statements guide My comparative analysis and the development of data-driven predictions using tools such as Python, Pandas, Matplotlib, and TensorFlow, utilizing datasets from WHO, Kaggle, and local Bangladeshi sources.

Solution

To address the problem statements outlined in this research project "Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh", I have implemented a comprehensive data-driven approach. My solution integrates exploratory data analysis, visualization, and predictive modeling using machine learning techniques. Below is an overview of the methods and processes I followed to derive meaningful insights and predictions:

➤ Data Collection and Preprocessing:

I gathered datasets from multiple credible sources, including the World Health Organization (WHO), Kaggle, and official Bangladeshi Covid-19 portals. These datasets cover global data as well as Bangladesh-specific information such as division-wise cases, gender and age-based data, and quarantine statistics. After consolidating the datasets, I cleaned and preprocessed the data to handle missing values, standardize formats, and prepare it for analysis.

➤ Identification of the Outbreak Origin:

Through time-series analysis and visual trends, I confirmed that the outbreak originated in Hubei province, China, with a significant spike in cases reported between January and March 2020.

➤ Global Impact Assessment:

Using comparative analysis and graphical representations, I highlighted how Covid-19 affected different regions across the world. Countries such as the United States, India, and Brazil emerged as the most severely impacted in terms of confirmed cases and death toll.

➤ Impact of Lockdowns:

The role of lockdowns was analyzed using trend comparisons before and after their implementation. My results suggest that while lockdowns significantly reduced the transmission rate in some countries, in others, the impact was minimal due to enforcement issues or delayed response. Bangladesh, in particular, saw mixed outcomes during its multiple lockdown phases.

➤ Bangladesh-Specific Analysis:

For a focused national-level study, I analyzed:

- Daily and cumulative cases and deaths in Bangladesh.
- Division-wise trends to identify hotspots.
- Age and gender-based vulnerability analysis.
- Quarantine and recovery patterns. This helped provide a clearer understanding of how the virus spread within different segments of the Bangladeshi population.

➤ Evaluation of Healthcare Response in Bangladesh:

I compared healthcare infrastructure, availability of hospital beds, and testing capacity in Bangladesh with global averages. This helped evaluate the country's preparedness and response to the pandemic.

➤ Development of the Prediction Model:

Using Python and machine learning frameworks such as TensorFlow and Scikit-learn, I trained predictive models on historical Covid-19 data. The model was evaluated on metrics like accuracy and RMSE to ensure reliability. Forecasts included future trends in confirmed cases, recoveries, and deaths for both Bangladesh and the global context.

Through these steps, I am able to answer my research questions effectively. The combination of statistical analysis, data visualization, and predictive modeling allowed Me to uncover trends, assess responses, and generate valuable forecasts that can aid future decision-making and preparedness.

Result and Analysis

Coronavirus Disease 2019 (Covid-19) is a highly infectious respiratory illness caused by the SARS-CoV-2 virus. First identified in Wuhan, China in December 2019, the virus rapidly spread across the globe, triggering one of the most severe pandemics in modern history. Symptoms typically include fever, dry cough, fatigue, breathing difficulty, and loss of taste or smell, appearing between 1 to 14 days after exposure. While the majority of cases remain mild, a significant portion of patients progress to severe illness, including Acute Respiratory Distress Syndrome (ARDS), multi-organ failure, or long-term complications, particularly in the heart and lungs.

➤ Origin and Timeline of the Outbreak

Through historical and data-driven investigation, the outbreak was traced to Huanan Seafood Market in Wuhan, Hubei Province, China. Key events marking the initial phase of the pandemic are as follows:

- December 2019: Mysterious pneumonia cases identified in Wuhan.
- January 2020: Chinese authorities confirmed a new coronavirus; rapid transmission and human-to-human spread were recognized.
- January 23, 2020: Wuhan placed under strict lockdown. Emergency hospital construction began to handle the rising case load.
- February/March 2020: National and global transmission accelerated. WHO and China conducted joint investigations.
- April 2020: China revised its initial case and death numbers after re-evaluation.

➤ Global Spread and Impact

My analysis confirms that Covid-19 spread to over 223 countries, with USA, India, Brazil, and Russia experiencing the highest number of confirmed cases and deaths. The virus affected countries differently, based on health infrastructure, policy response, testing capacity, and public adherence to safety guidelines.

Visual data comparisons highlight:

- Exponential growth in early 2020.
- Sharp peaks during second and third waves.

➤ Bangladesh: Impact and Analysis

For Bangladesh, I carried out in-depth analysis using division-wise, gender-wise, and age-based datasets. Key insights include:

- Initial Cases: First confirmed cases reported in March 2020.
- Peaks: Major waves occurred in mid-2020 and mid-2021.
- Geographical Spread: Dhaka, Chattogram, and Khulna divisions reported the highest case counts.
- Demographic Trends:-
 - ✓ Higher infection rates among males.
 - ✓ Elderly populations faced higher mortality risks.

➤ Lockdown Effectiveness

- Through time-series comparison and case trend analysis:
 - ✓ Lockdowns helped slow the infection rate initially in many countries, including Bangladesh.
 - ✓ Their effectiveness varied due to differences in public compliance, timing, and enforcement.
 - ✓ Economic and psychological impacts were notable side-effects.

➤ Prediction Outcomes

Using machine learning models trained on historical data (with tools like TensorFlow and Scikit-learn), I generated predictions for:

- ✓ Future case trends
- ✓ Expected fatalities and recoveries
- ✓ Region-specific forecasts for Bangladesh

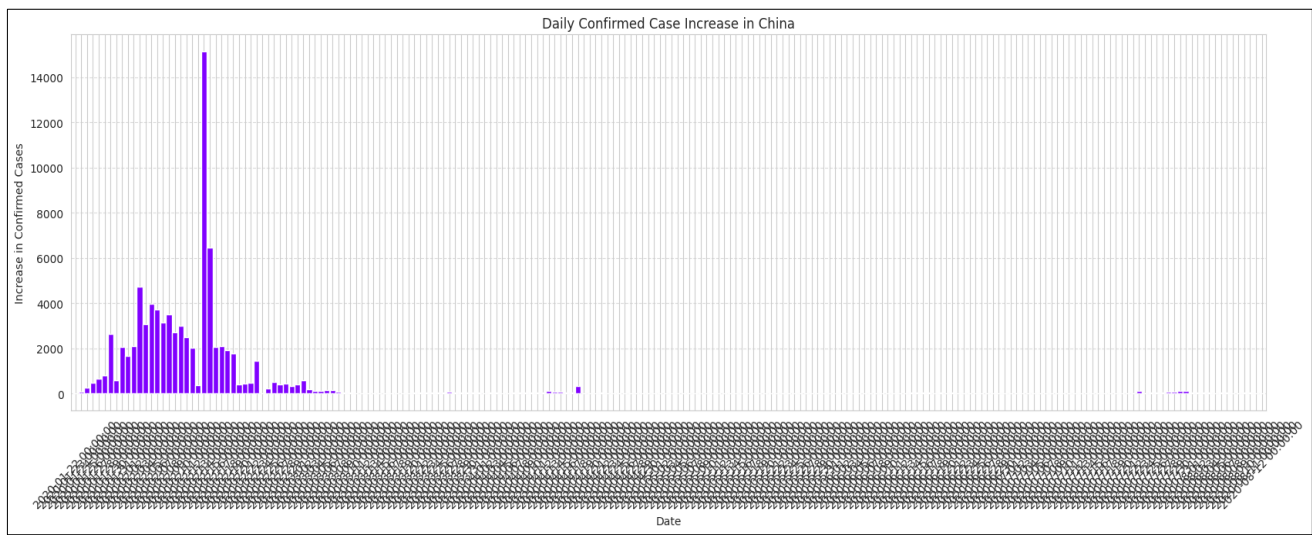
The prediction model demonstrated high reliability and aligned closely with actual trends, indicating strong potential for early warning systems.

Where did the virus outbreak?

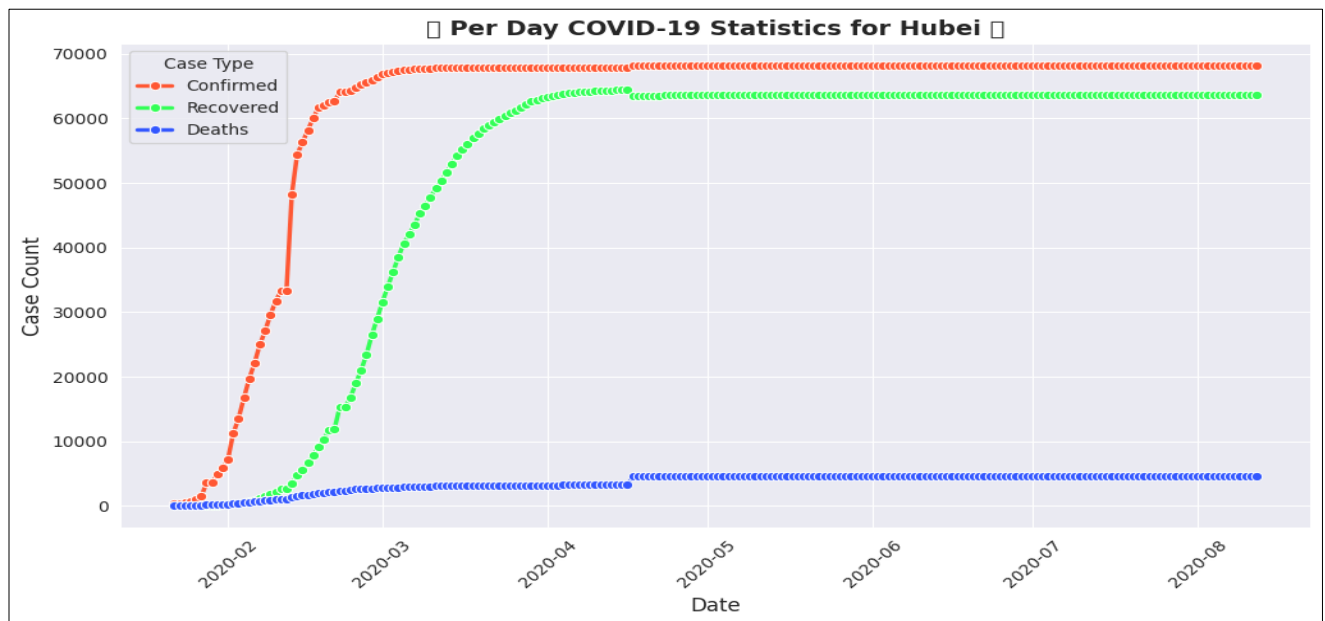
The COVID-19 virus outbreak began in Wuhan, China, in December 2019. It is caused by the novel coronavirus SARS-CoV-2. The virus quickly spread worldwide, leading to a global pandemic. It primarily affects the respiratory system and spreads through respiratory droplets.

Analysis of the condition of China during the outbreak:

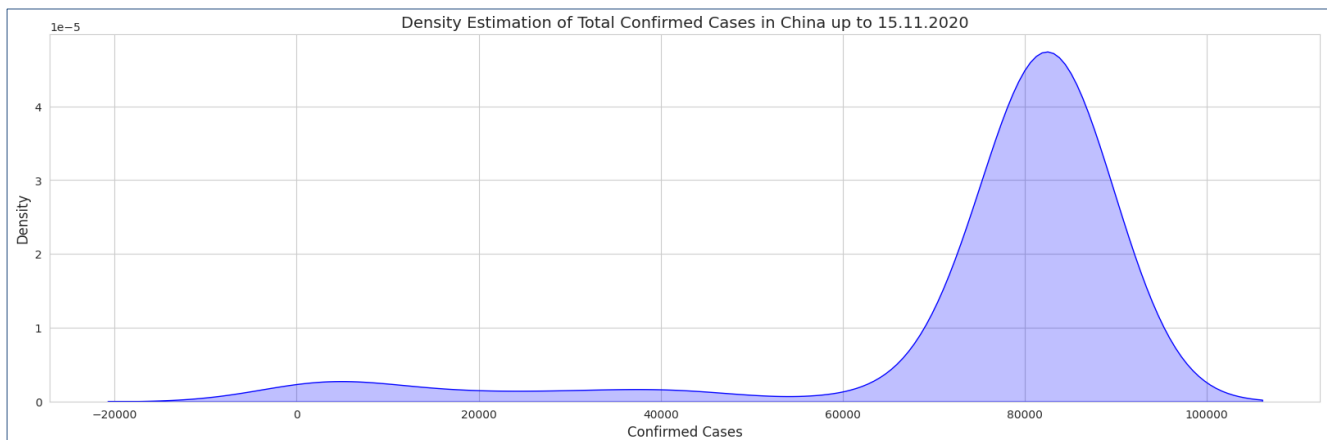
Analysis of China's condition during the outbreak reveals the scale of infections, recovery efforts, and mortality impact.



This figure shows the daily increase in confirmed COVID-19 cases in China. A sharp spike is observed early on, followed by a gradual decline and minimal cases over time.

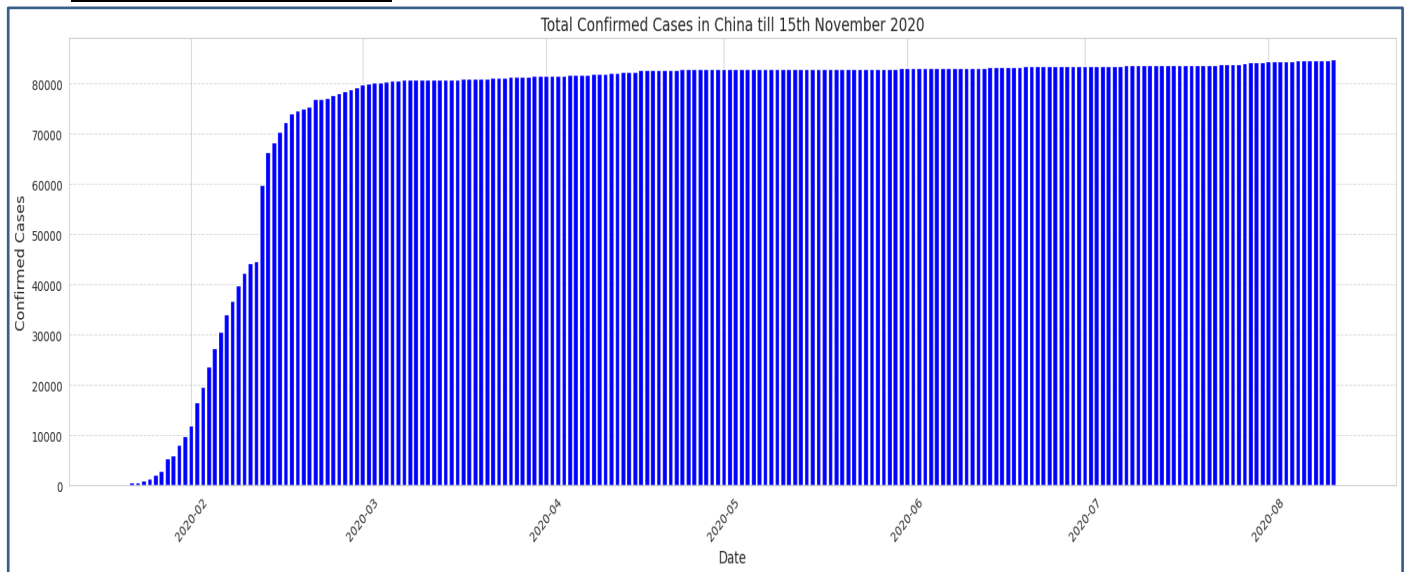


This graph illustrates the cumulative trend of confirmed, recovered, and death cases in Hubei during the COVID-19 outbreak. It shows a rapid rise in cases early on, followed by steady recoveries and a stabilized death toll highlighting effective containment over time.

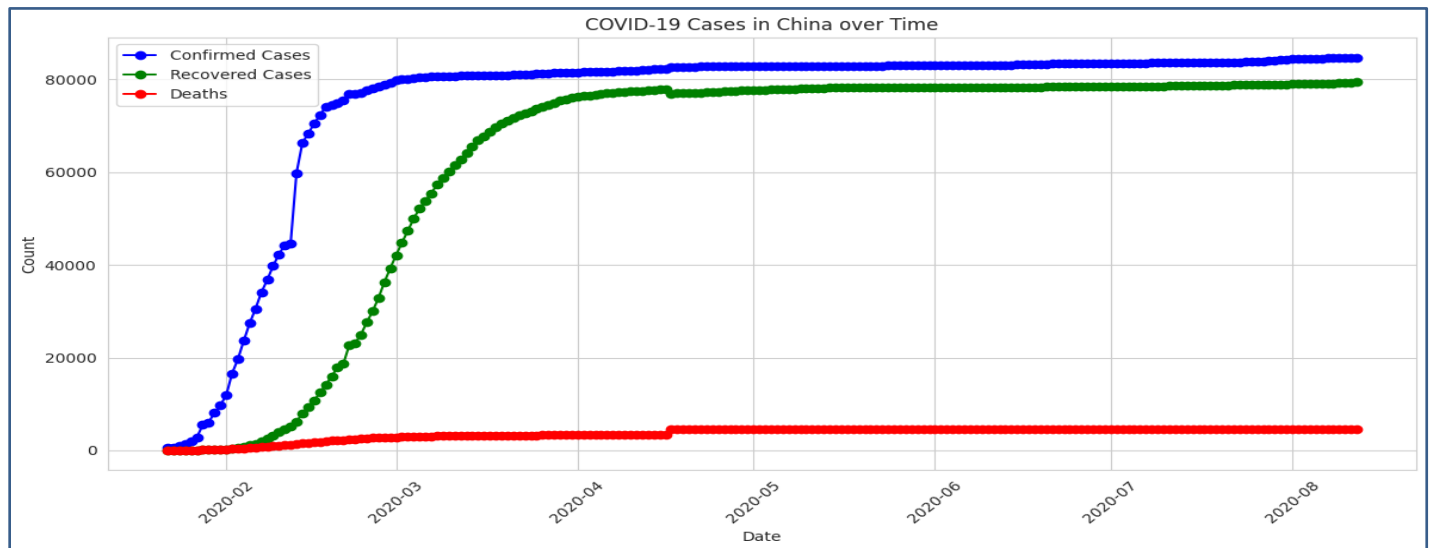


This density plot shows that most confirmed COVID-19 cases in China by 15.11.2020 were concentrated around 80,000.

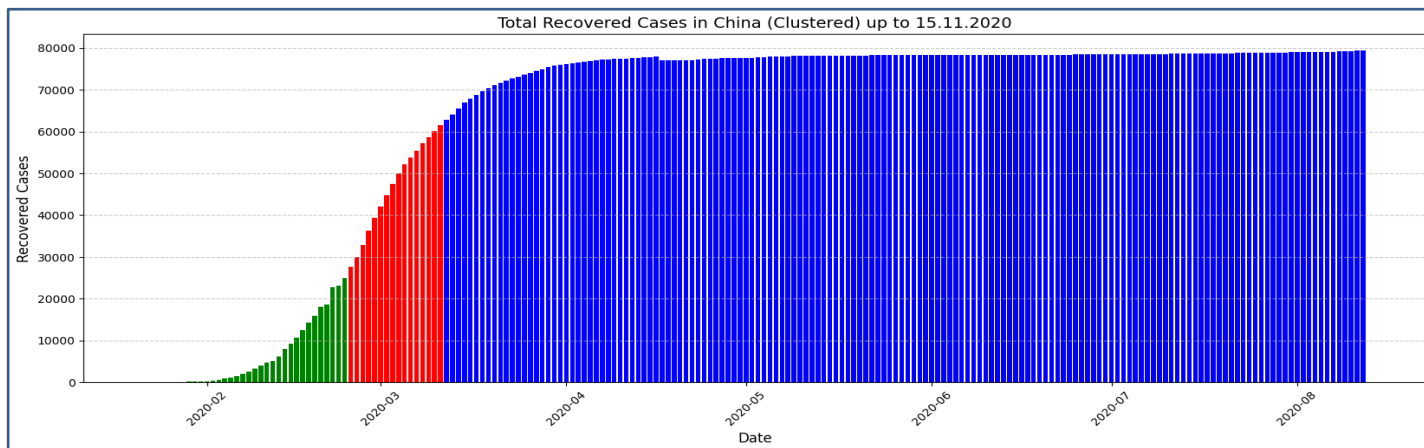
Confirmation in China.



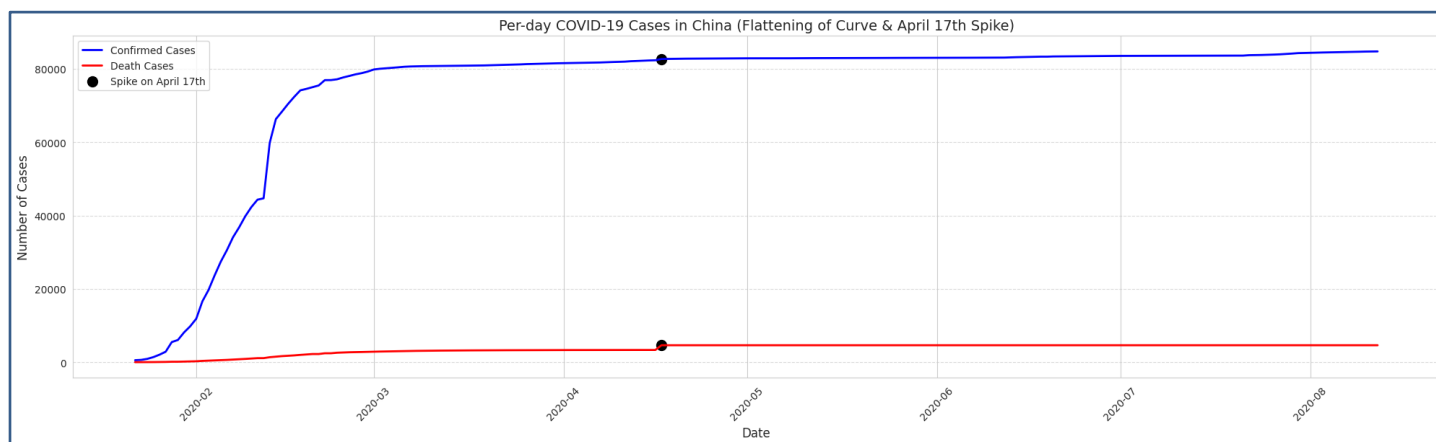
We are talking about the total number of digits here in China.



In this figure I talking In China, the COVID-19 data includes total Confirmed cases, Recovered patients, and Deaths. These figures help track the outbreak's impact and assess the country's response and recovery.



"Figure: Clustered representation of total recovered COVID-19 cases in China up to 15.11.2020."



The per-day statistics of China where the line graph shows flattening of curve confirmed cases. However on 17th April there was a sudden rise in the Death and Confirmed cases.

China Total Casualties (Deaths and Prediction Deaths) Used the Model is Tensor Flow.

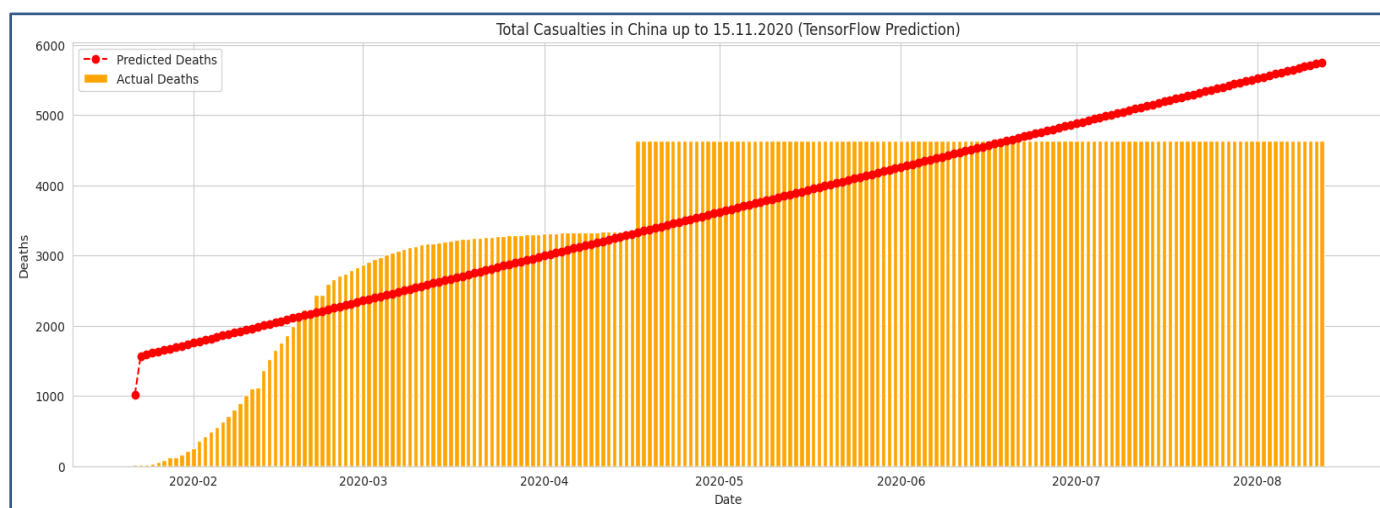
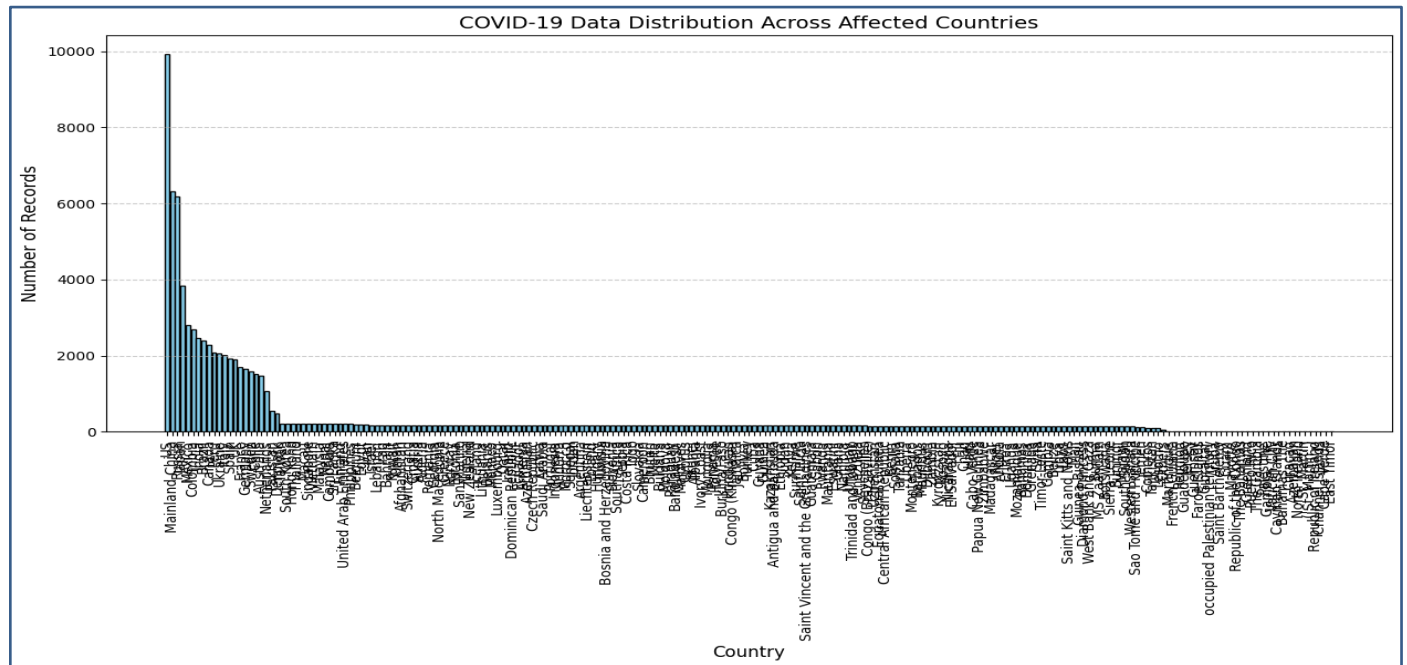


Figure: Total COVID-19 casualties in China up to 15.11.2020 with TensorFlow-based prediction. TensorFlow is used for predicting future trends based on past data using machine learning.

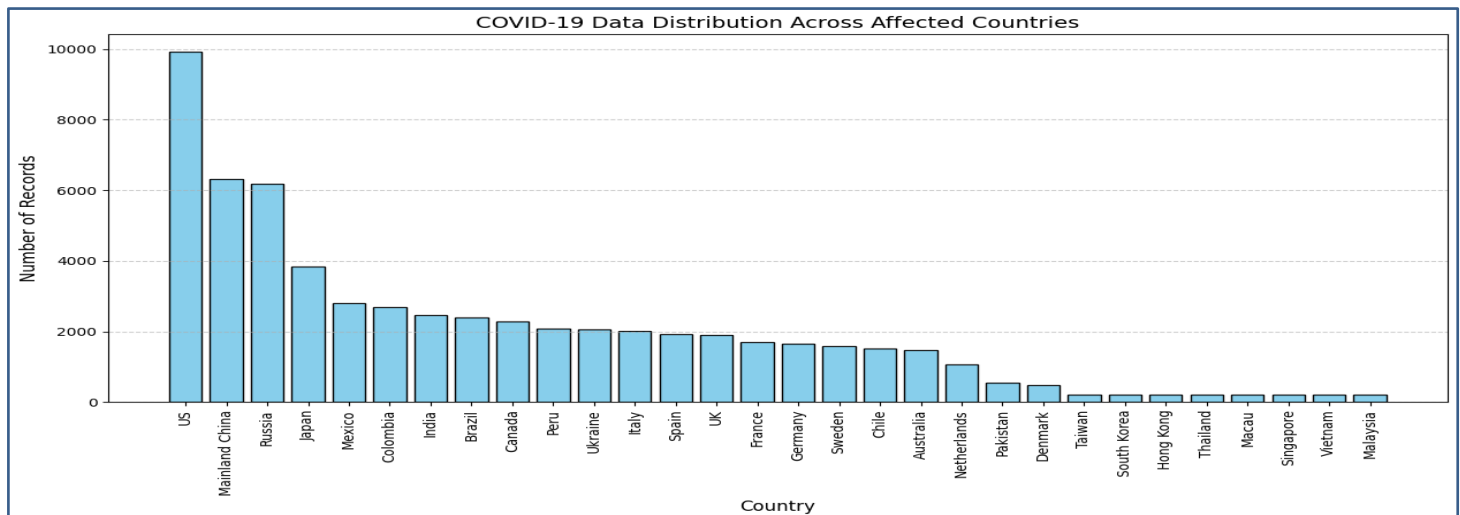
Spread Out of the Virus:

The 2019 novel coronavirus (2019-nCoV), officially named as COVID-19 pandemic by the WHO, has spread to more than 180 countries including China. Confirmed novel Coronavirus cases increased ten-fold in less than a month, from 100,000 in the first week of March to more than one million on 02 April, while more than 52,000 deaths have been Reported across the world.

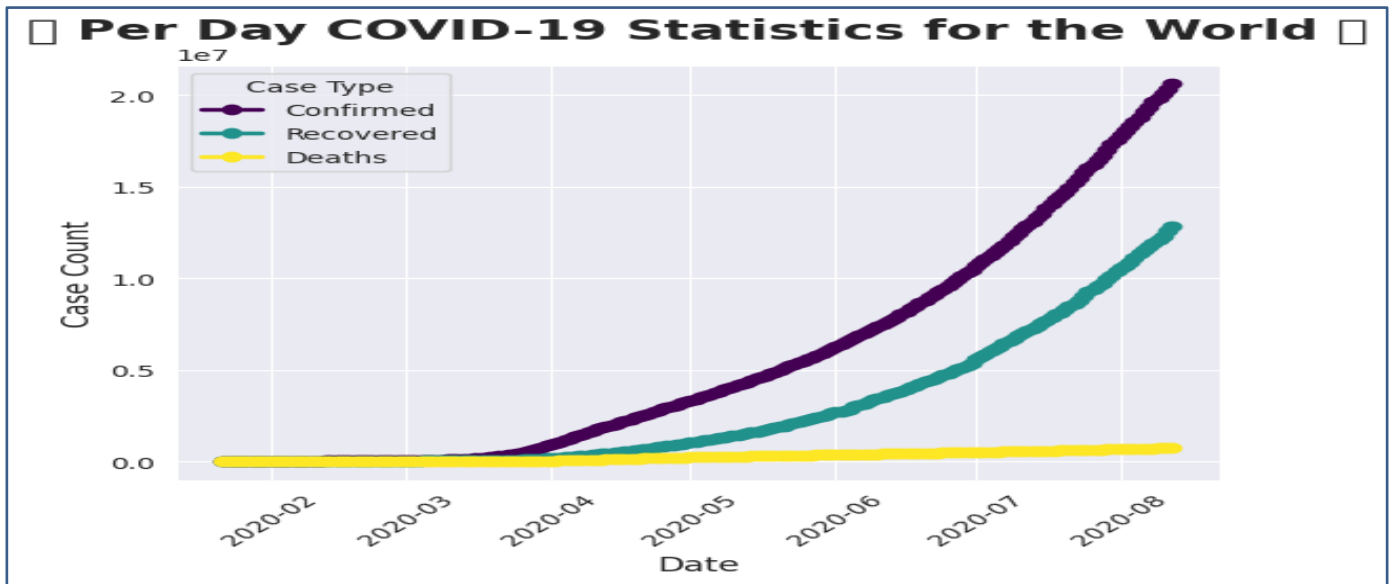
Finding out how many countries were affected by Covid-19:



The figure shows the total number of countries affected by COVID-19, highlighting the global spread of the virus.



"Figure: Number of Countries Affected by COVID-19 Worldwide."



In This Figure: Here the per day statistics tells us the spreading is exponentially increasing day by day since The end of the March, 2020. Now after 5 months the situation is worse than the previous Days and still the cases are increasing exponentially without any kind of stoppings in the Increment of the cases.

World Analysis through graphical representation:

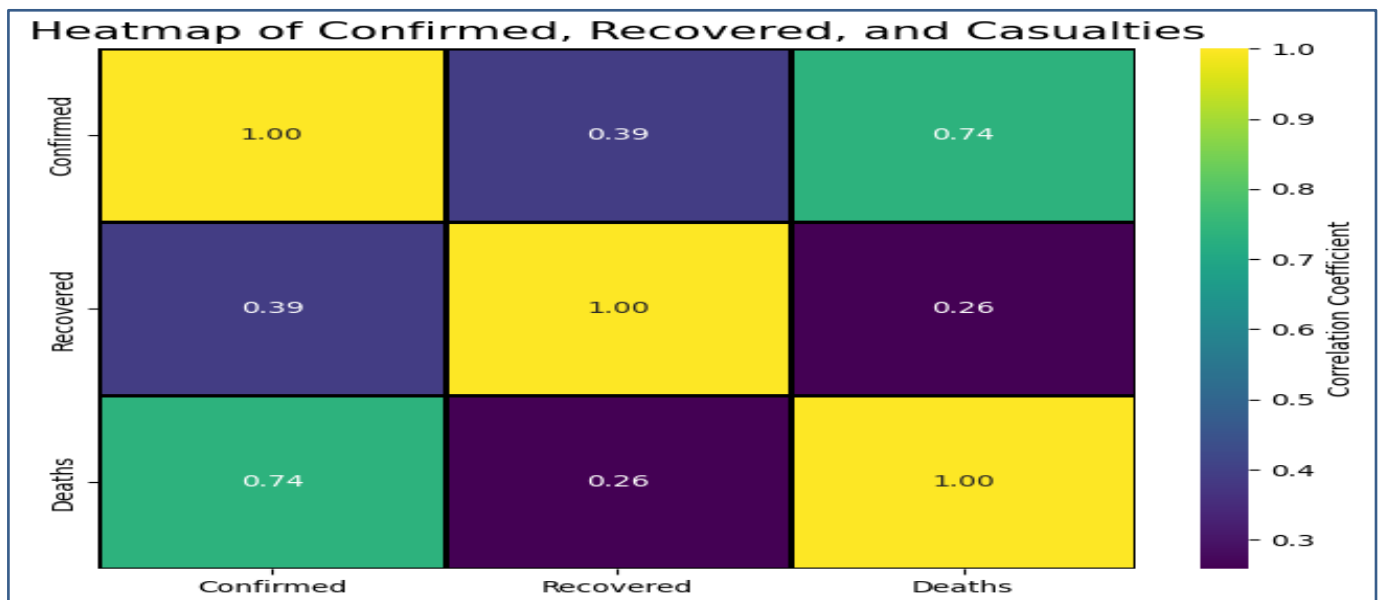


Figure: Heat map showing 'Confirmed', 'Recovered', and 'Casualties' cases by Province ID and Country ID. This visual highlights the intensity and distribution of COVID-19 impact across regions.

Top 5 affected countries:

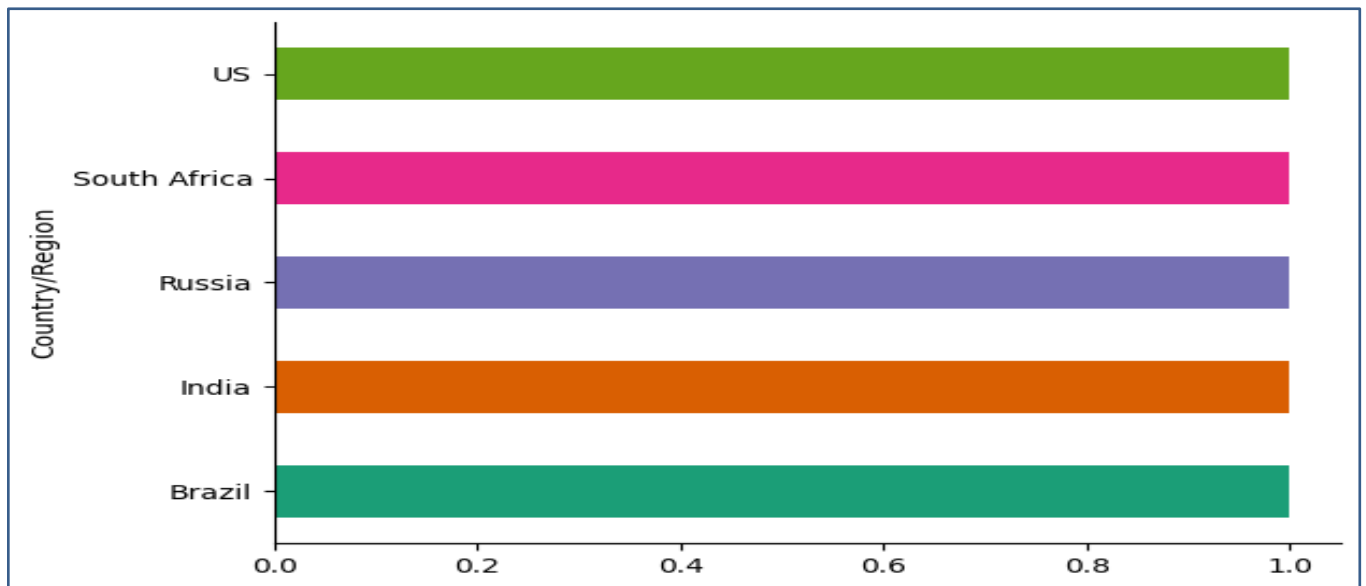


Figure: Top 5 countries most affected by COVID-19 based on total confirmed cases.
This highlights the nations with the highest case counts, indicating the global hotspots of the pandemic.

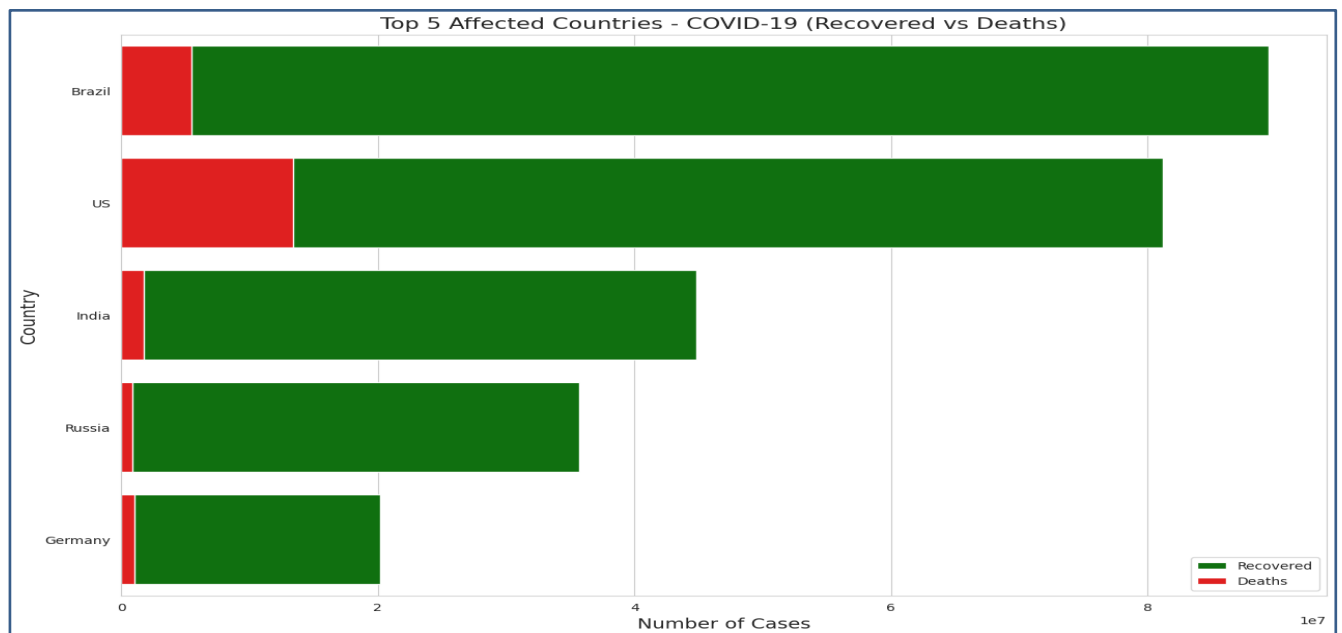


Figure: Comparison of Recovered and Death cases in the Top 5 most affected COVID-19 countries.
This illustrates the recovery rate versus mortality, reflecting the effectiveness of healthcare responses.

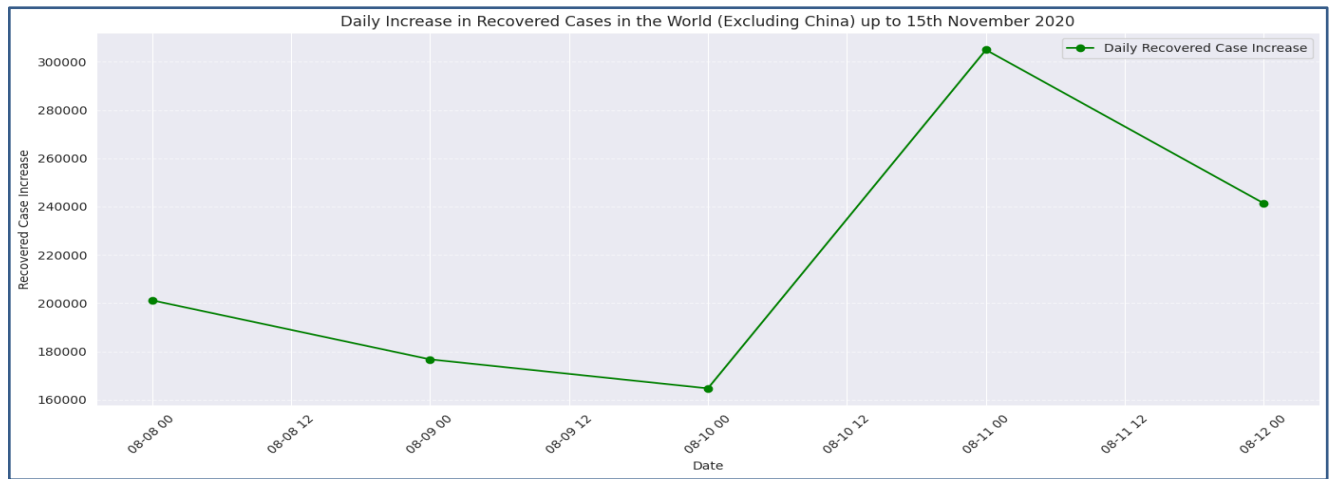


Figure: Total confirmed COVID-19 cases worldwide (excluding China) up to 15th November 2020. This highlights the global spread and impact of the virus beyond China.

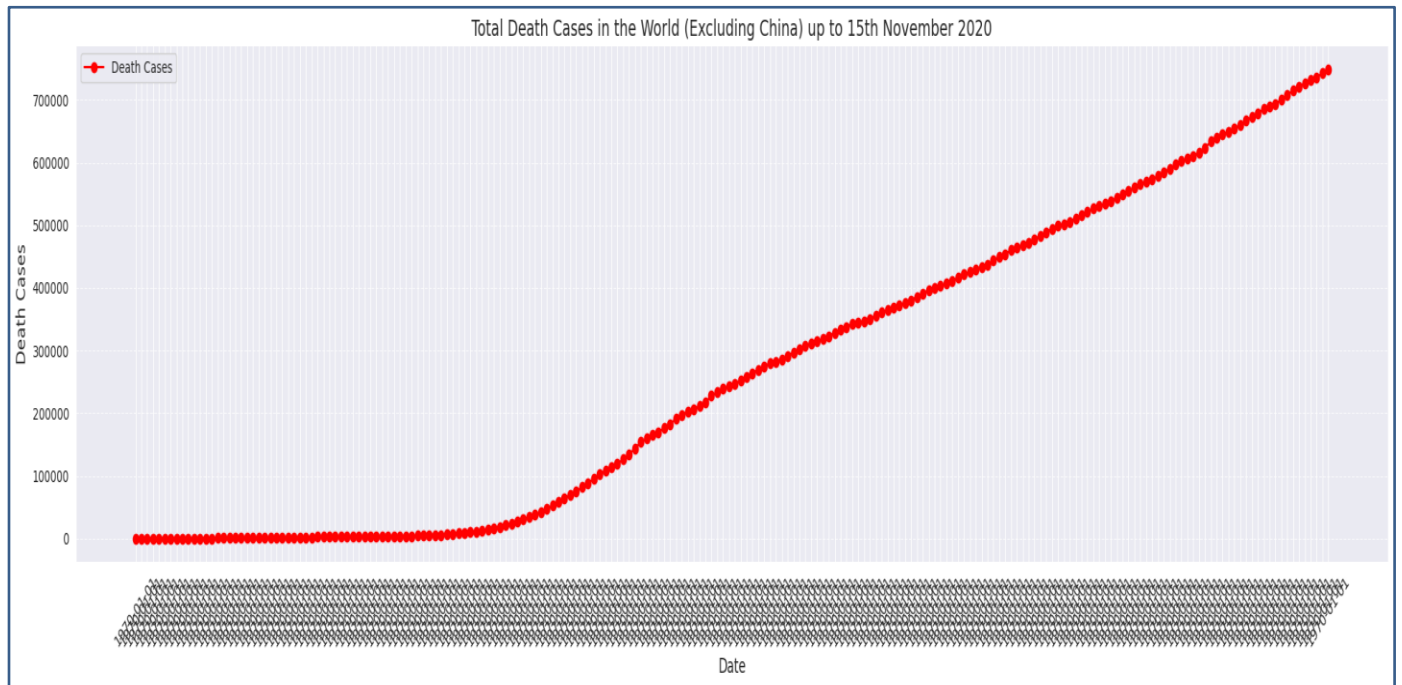


Figure: Total COVID-19 death cases worldwide (excluding China) as of 15.11.2020. This visual represents the global mortality impact of the pandemic outside China up to the given date.

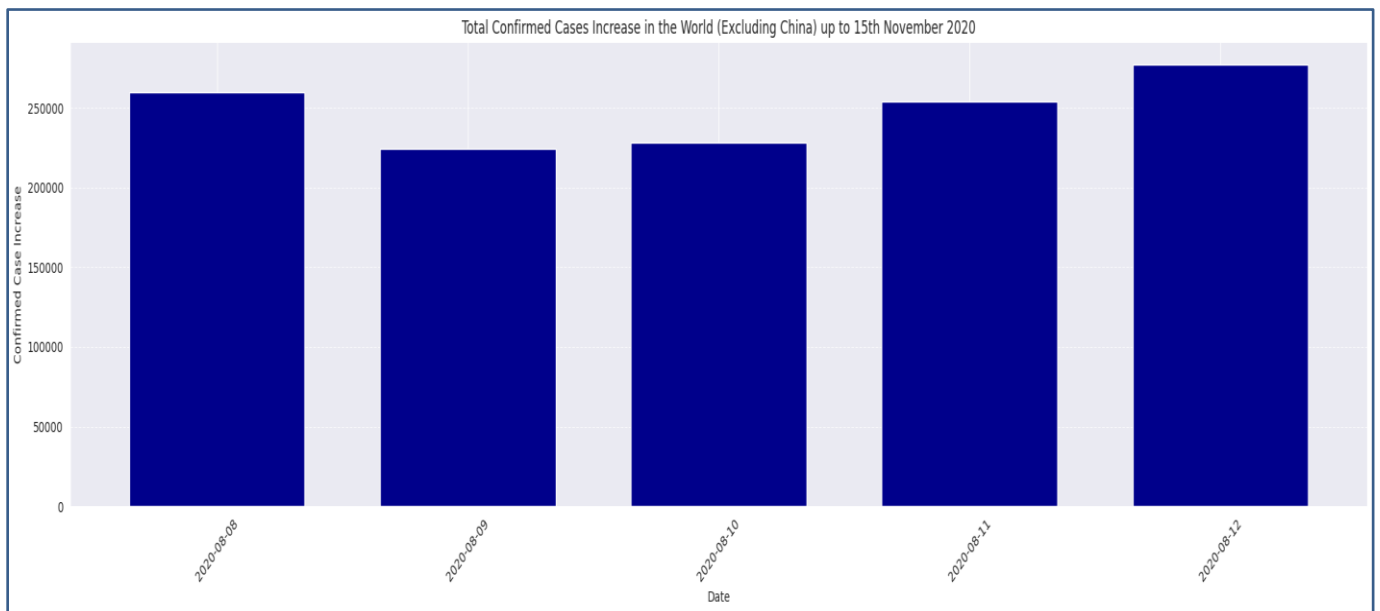


Figure: Total confirmed COVID-19 cases worldwide (excluding China) up to 15th November 2020. This highlights the global spread and impact of the virus beyond China.

As of 15th November 2020, the global number of confirmed COVID-19 cases had surged to approximately 54,283,840 representing a significant portion of the global population and highlighting the critical scale of the pandemic. This staggering figure accounts for nearly 57% of the world population having been exposed or infected during this period, based on reported cases.

The global death toll had risen to over 1,312,505, and the number continued to climb daily. This growing trend underscored the aggressive spread and severity of the virus, particularly in regions with high population density and limited initial containment measures.

A significant share of the global confirmed cases was reported from the United States, which alone accounted for approximately 21% of the total worldwide cases. The highest single-day spike occurred on 29th July 2020, when global confirmed cases increased by around 337,000 in just 24 hours marking the most critical surge in a single day. Following the United States, the second and third most affected countries were India and Brazil, respectively. India reported about 16.27% of the world's total confirmed cases, while Brazil contributed approximately 10.78%. These three nations collectively represented a substantial portion of the global case count, as visualized through data-driven charts and geographic analysis in this research.

This data reflects how deeply interconnected and vulnerable the global community was to the pandemic's spread and emphasizes the importance of international cooperation, early detection, and healthcare preparedness in managing such outbreaks.

Affected Countries count: 223

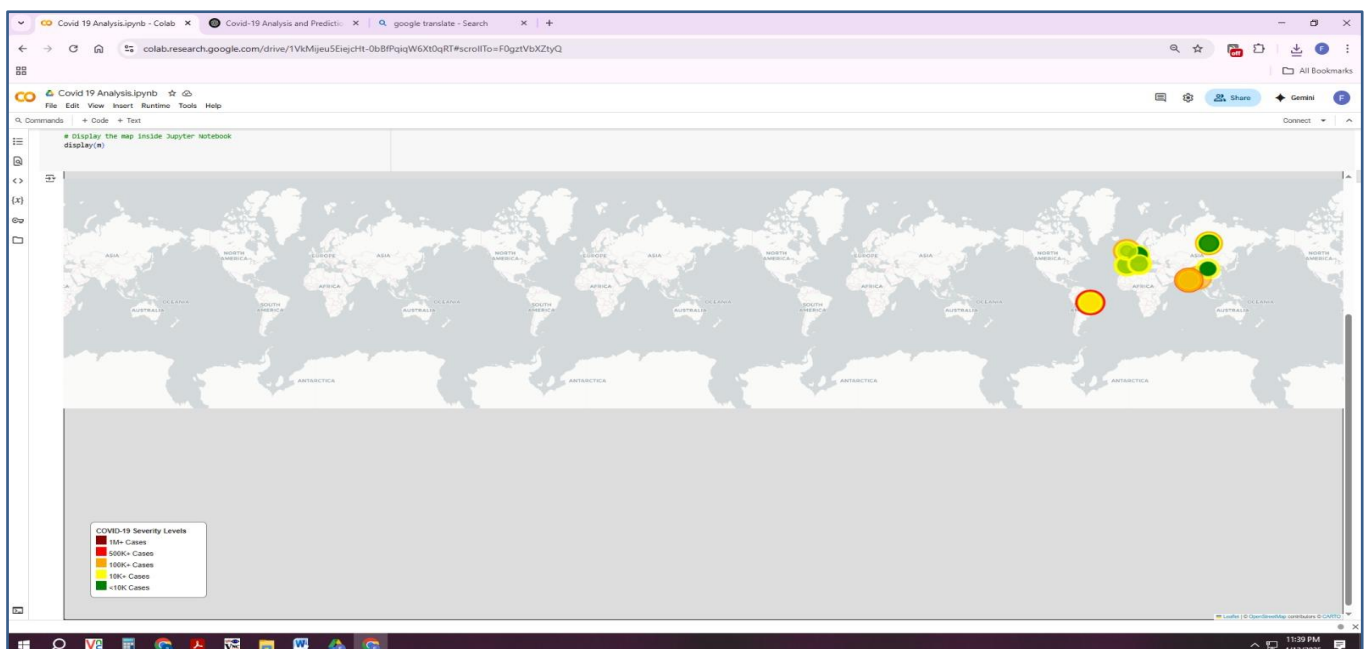


Figure: World map highlighting 223 countries affected by COVID-19. This map illustrates the widespread global impact and reach of the pandemic.

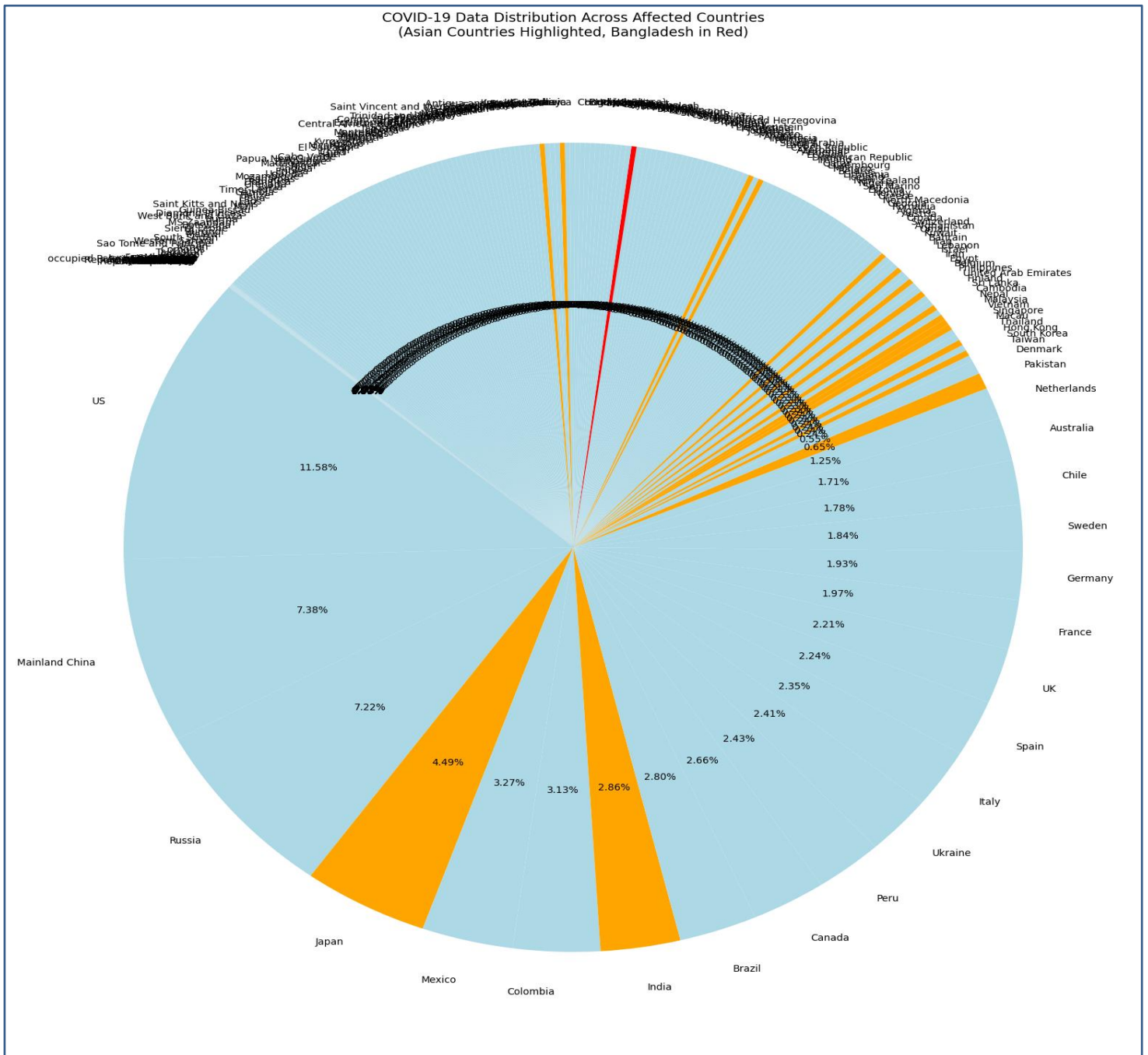


Figure: Global distribution of COVID-19 data across affected countries, with Asian nations highlighted and Bangladesh marked in red. This visualization emphasizes regional impact, drawing focused attention to Asia and Bangladesh's position within the global outbreak.

The COVID-19 pandemic has significantly impacted several countries across the globe. As of the final quarter of 2020, the following countries emerged as the most severely affected, based on data from Johns Hopkins University and national health authorities:

✓ United States of America

The United States remained at the top of the global COVID-19 chart, surpassing 8.7 million confirmed cases. By late October 2020, the country had reported over 225,735 deaths, the highest number globally. The large-scale spread, delays in early containment, and state-level variations in responses contributed to the nation's struggle in managing the pandemic effectively.

✓ India

India became the second most affected country, with its total number of COVID-19 cases exceeding 7.9 million. Despite a gradual reduction in daily new cases, the virus continued to pose a significant threat. States such as Maharashtra, Andhra Pradesh, Tamil Nadu, Karnataka, and Uttar Pradesh recorded the highest numbers of infections, placing immense pressure on the healthcare system.

✓ Brazil

Ranking third globally, Brazil recorded over 5.4 million confirmed cases and more than 157,000 deaths. The country's densely populated urban centers and initial policy gaps contributed to the high transmission rate. The impact was severe, especially in Amazon regions and among marginalized populations.

✓ Russia

Russia reported over 1.5 million confirmed COVID-19 cases, with 26,409 deaths as of October 2020. While the initial wave was moderately contained, a sharp rise in daily cases was observed in the subsequent months. The country ranked fourth globally in terms of total confirmed cases, reflecting a rapid resurgence of the virus.

✓ France

In Europe, France emerged as one of the worst-affected nations, with more than 1.2 million confirmed cases and a death toll crossing 35,000. A second wave of infections led to renewed lockdowns and stricter public health regulations during the last quarter of 2020.

Analysis of Major Countries:

ITALY: The Second Epicenter

First outbreak: Italy was first affected with COVID 19 in 30 January, when two positive cases were reported in Chinese tourists. Italy COVID cases reached 59,138 on 23 March, marking the biggest coronavirus outbreak outside Asia. And Italy was announced as second most affected coronavirus country in the world with the cases increasing at a higher rate than any other country. Total affected: If we see the graph of affected rate, we can understand the cases were approximately same upto August 25. After that it has been increasing and currently it has 1455022 positive cases by now(Till 24 november, 2020). Death rate: The death rate was suddenly increased in the mid of March and it was approximately in a equal level till October 20. After that is has been increasing like the affected rate till now. The daily death cases were too high from last week of February to mid of April.

Italy was the second epicenters for COVID19, hence lets look at these numbers. The numbers are high and confirmed cases are increasing by the day. However the increase rate for confirmed cases has been constant and may soon be seeing a plateau phase. Recovery and deaths were both neck to neck with some divergence over the last few days. Italy has more than 1L confirmed cases on 30th March.

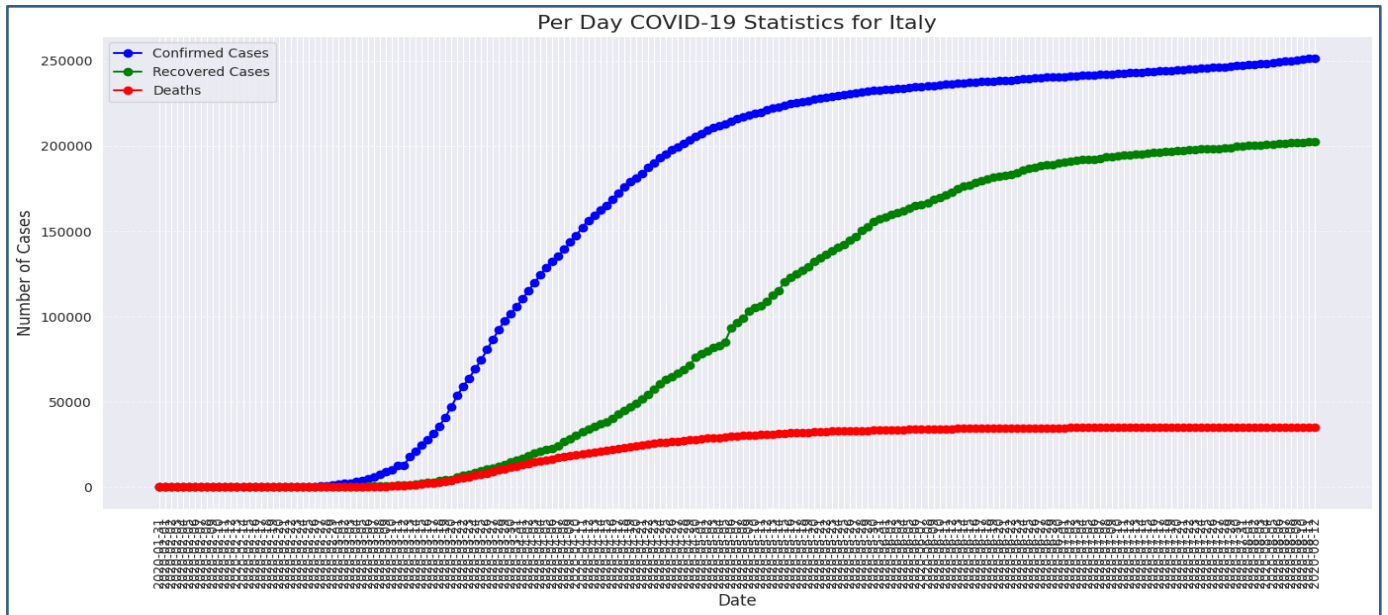


Figure: Per day COVID-19 statistics for Italy using a time-series model. This chart tracks daily trends in confirmed, recovered, and death cases, providing insight into the outbreak's progression over time.

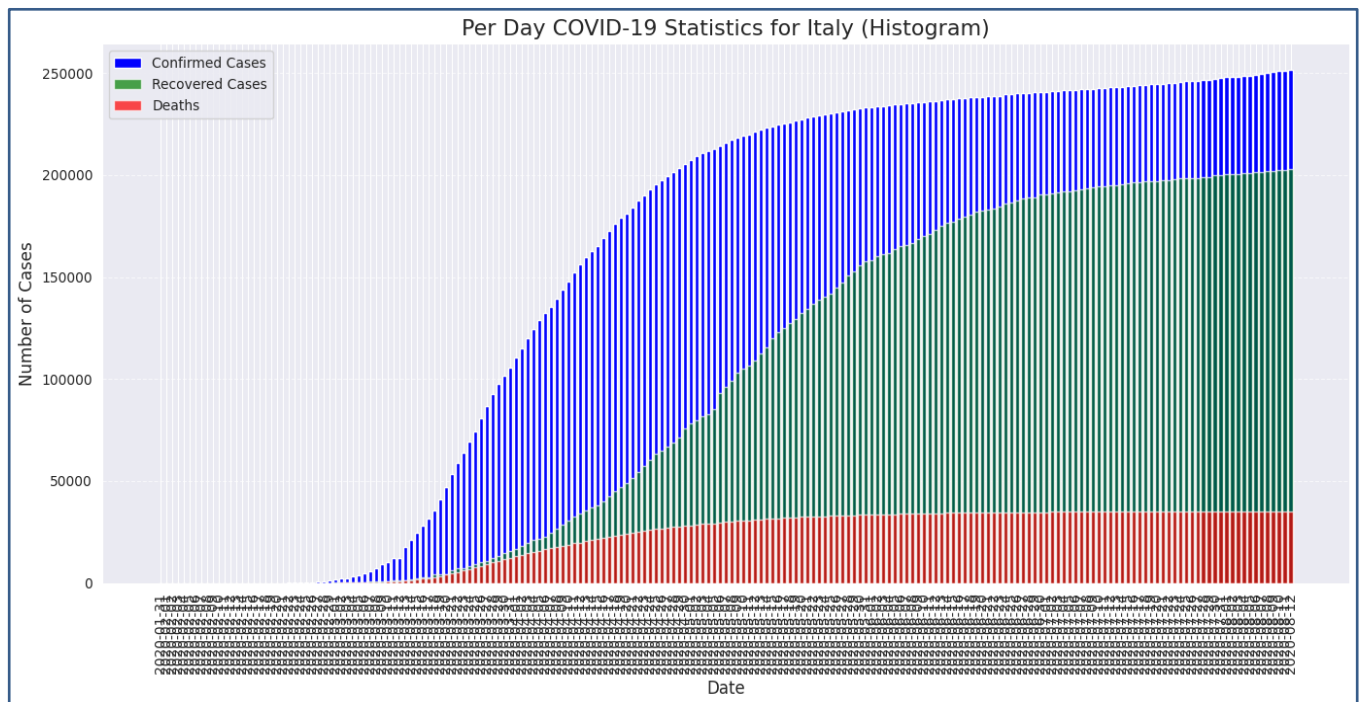


Figure: Histogram showing daily COVID-19 cases in Italy, modeled with ARIMA to capture trends and forecast fluctuations.

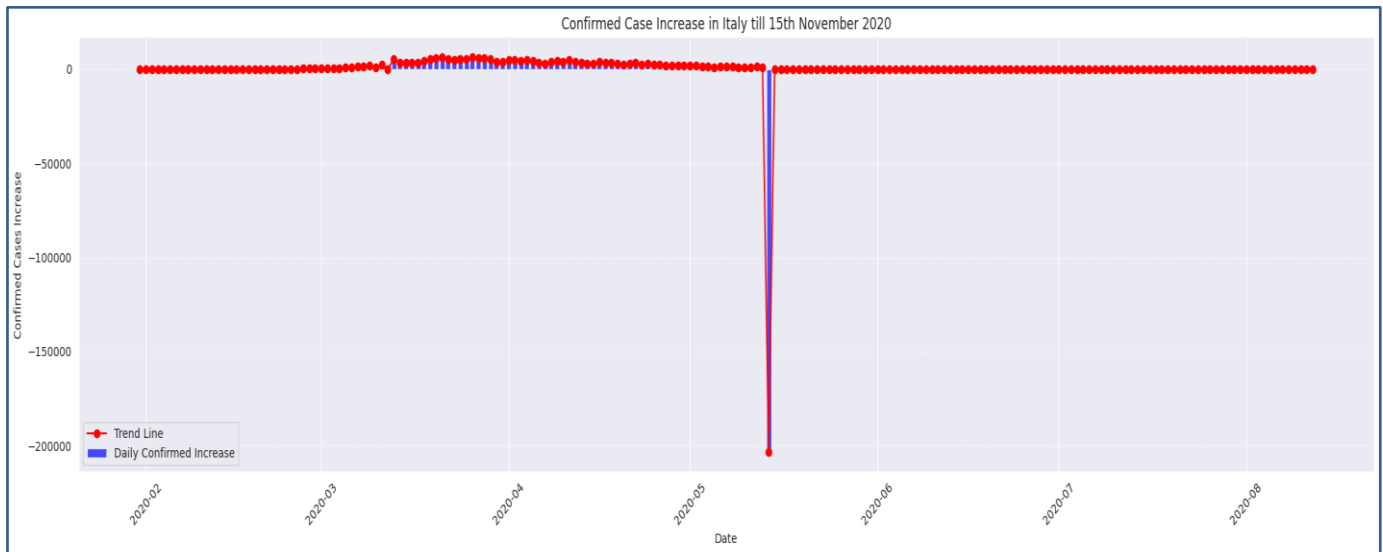


Fig: The graph shows the daily confirmed COVID-19 case increases in Italy until 15th November 2020, highlighting a sharp anomaly around May 2020.

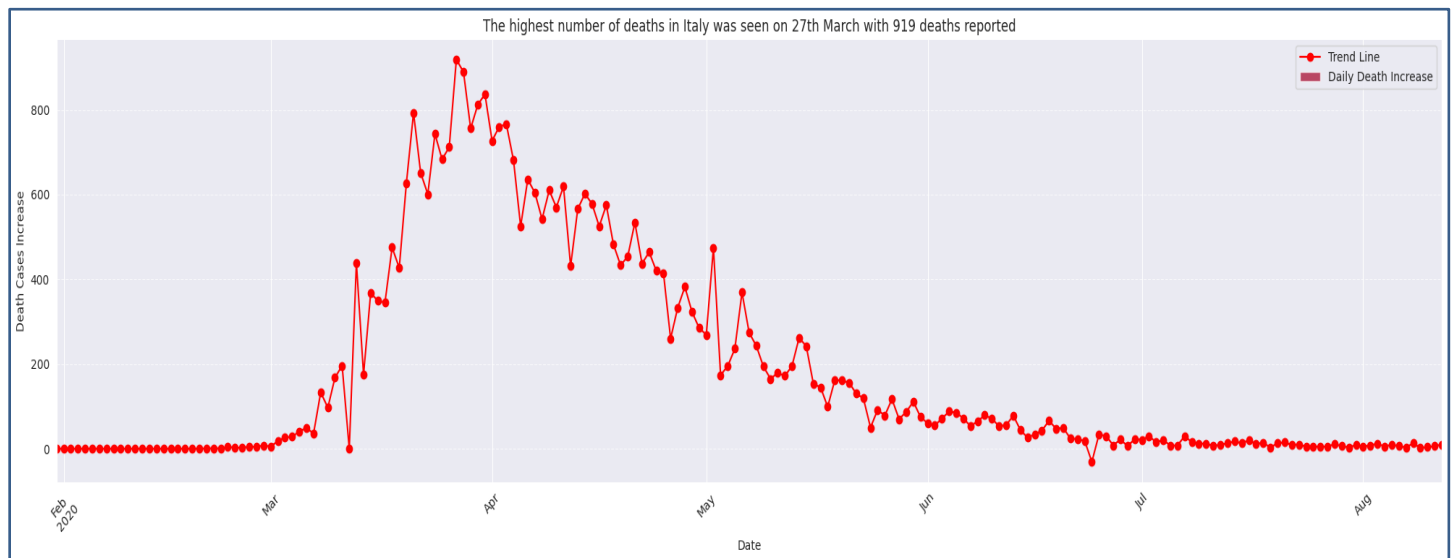


Fig: The graph shows the daily COVID-19 death increases in Italy, peaking sharply at 919 deaths on 27th March 2020. These are the trends and the analysis of the data. As you can clearly see that the daily Increase in the death cases and confirmed cases increased second time due to the second Wave of the virus.

United States of America : The 3rd Epicenter

First outbreak: Covid was first identified in the USA in December 2019. The World Health Organization declared the outbreak a Public Health Emergency of International Concern in January 2020 and a pandemic in March 2020. Total affected: Positive cases till November 25 is 12276834. Cases are respectively higher in October than the last few months. Rate of confirmed cases is daily changing. Death rate: Total number of deaths 255850. On November 21, the number of deaths was 2036; in 25 it was 1052. Death rate is decreasing and it is less than the number of deaths in March to mid-May.

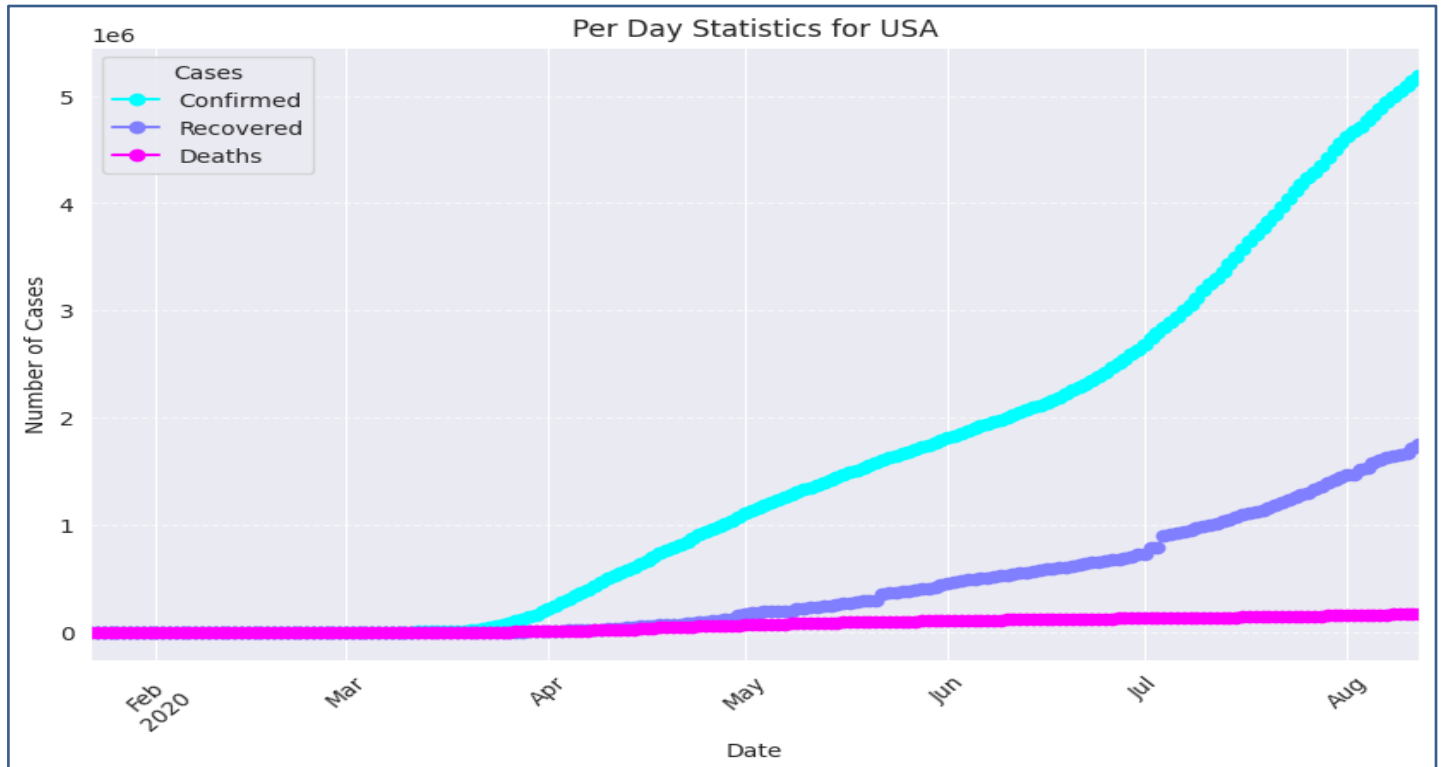


Figure: The graph shows the cumulative daily COVID-19 statistics for the USA, with confirmed cases rising steeply, followed by recovered cases and deaths from February to August 2020. This graph shows a certain flatness in the mid of June, 2020, But after that it is in the upswing mode with a steep slope upwards.

Analysis of USA based on the confirmed cases, death cases, recovered cases:

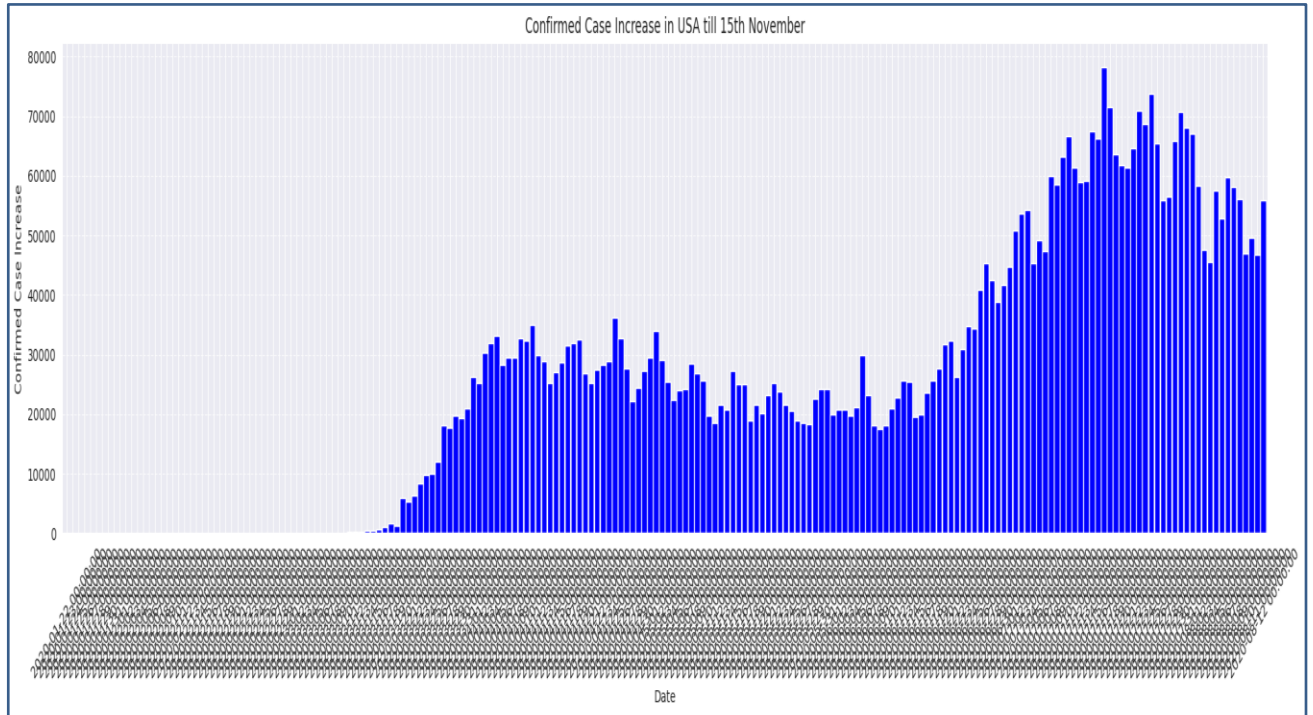
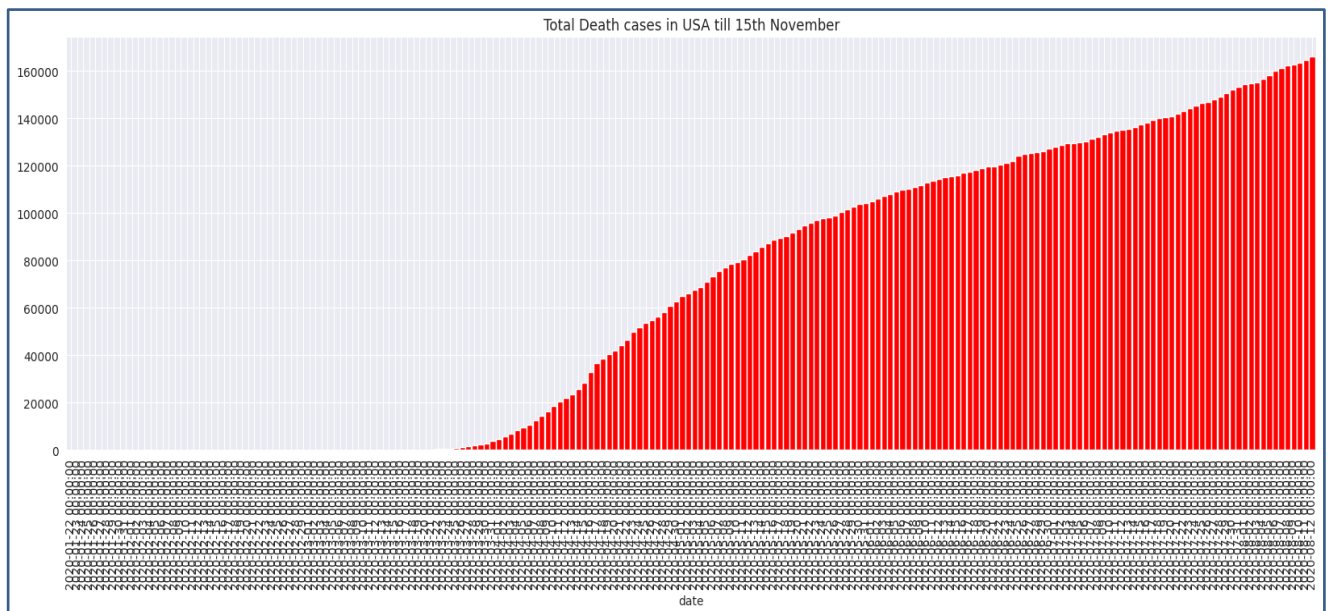
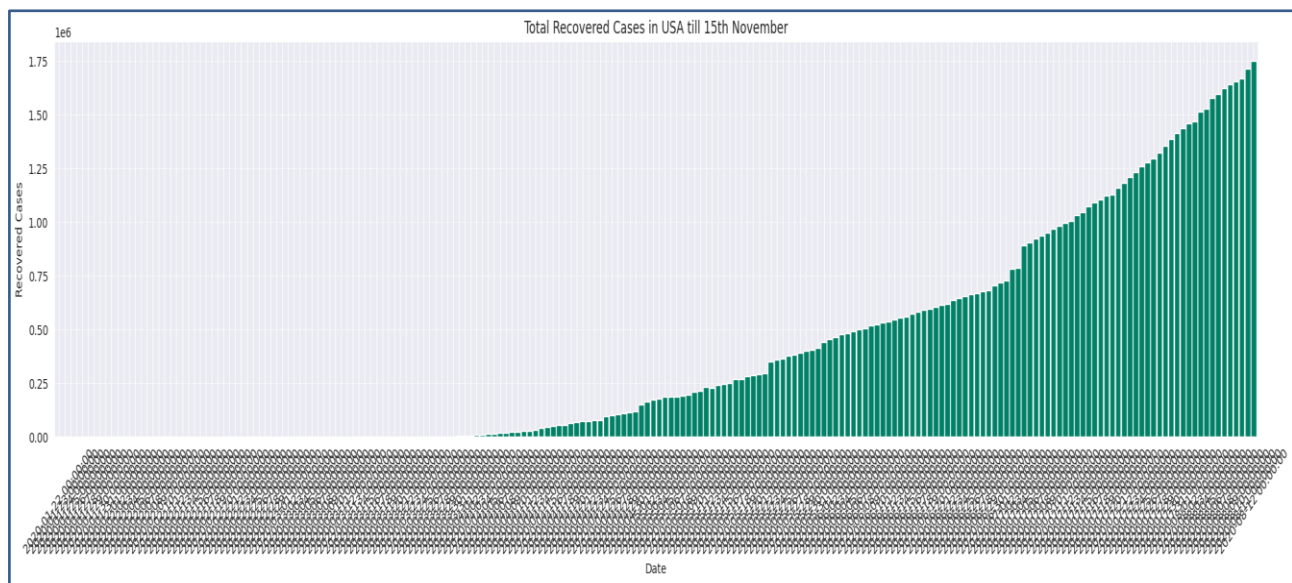


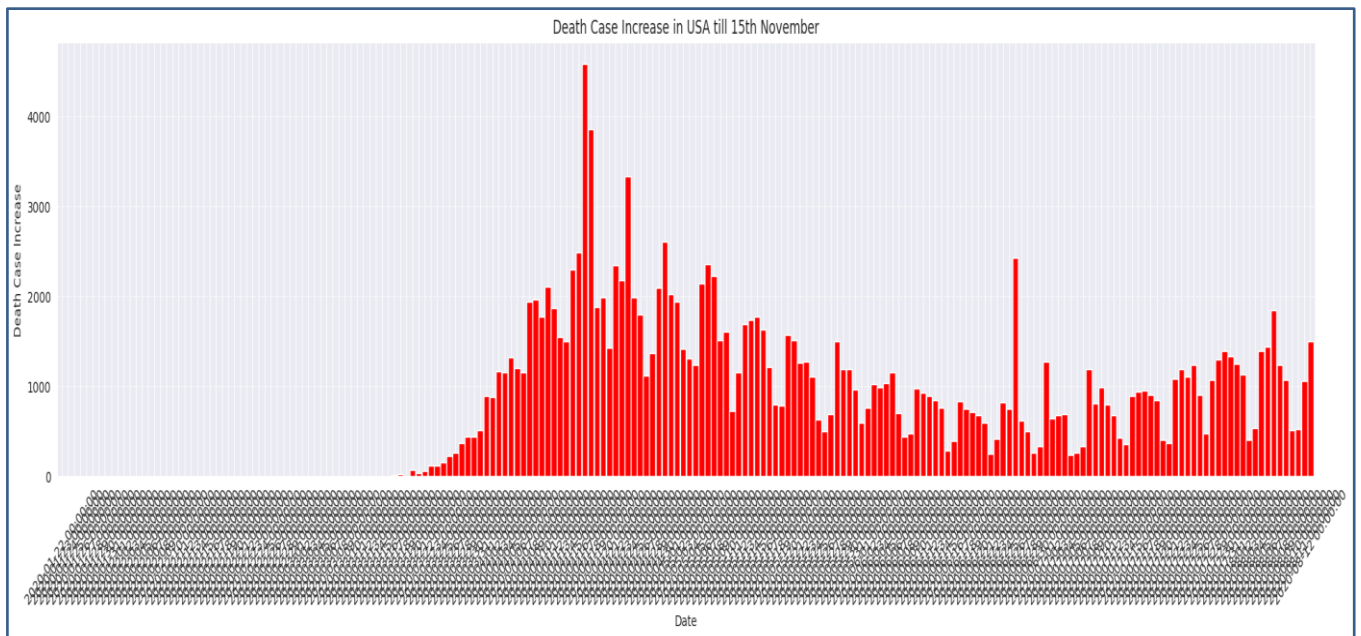
Figure: Total Confirmed In USA.



The figure shows the daily increase in COVID-19 death cases in the USA up to 15th November 2020.



The figure shows the total number of recovered COVID-19 cases in the USA up to 15th November 2020.



In the mid of June the confirmed cases are decreased and the incrimination is flattened by a certain moment. But unfortunately in the last week July the incrimination in the confirmed cases are increased suddenly with a steep slope. After a week of steep slope in the increment of the confirmed cases there is a decrement in the confirmed cases in a daily basis manner. As we can see in the graph, that after a week of high rise the peaks are coming lower and lower day by day. Which shows that the confirmed cases and decrementing day by day slowly. It's really a big sign of improvement for United States Similar increase has been seen in the deaths reported by US with highest number of deaths(4591) recorded on 16th April.

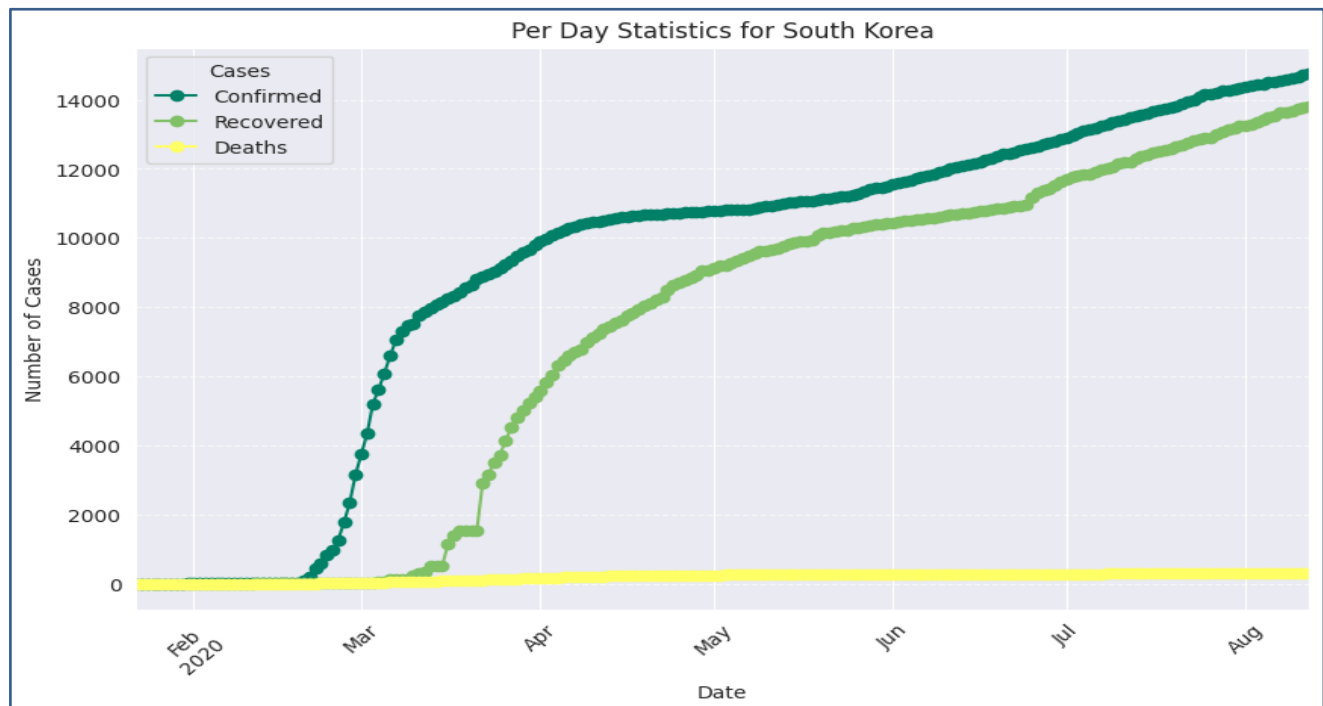
As the graph shows that on the particular day of 16th April 2020 USA reported a highest no. of death cases. After that the death cases are slowly coming down to a certain limit but suddenly in the recent times the Death cases are increasing day by day which are very much annoying for USA.

South Korea : The Role-model -

First outbreak: First confirmed case was founded in south korea on 20 January 2020. The number of confirmed cases increased after 10 February.

Total affected: In the present situation the total cases is 31735. Cases are increasing after February in a linear scale. Daily number of new cases was too high in February then it was reduced, after August 16 to September mid it was extremely high.

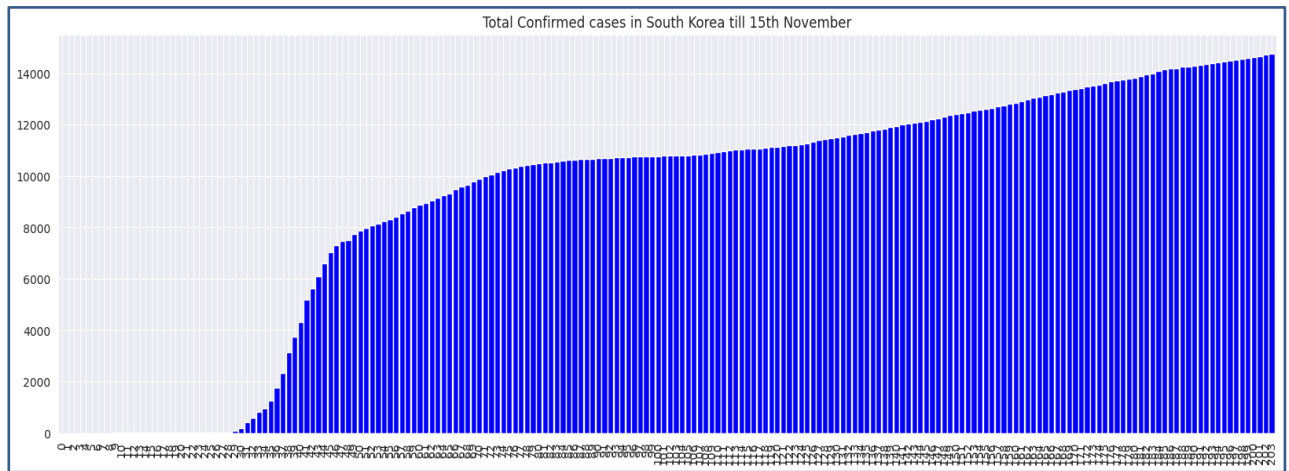
Death rate: Number of death cases are increasing from February but daily death was in the top level in mid of March. Total number of death is 513 till November 25, and recovery number is 26825



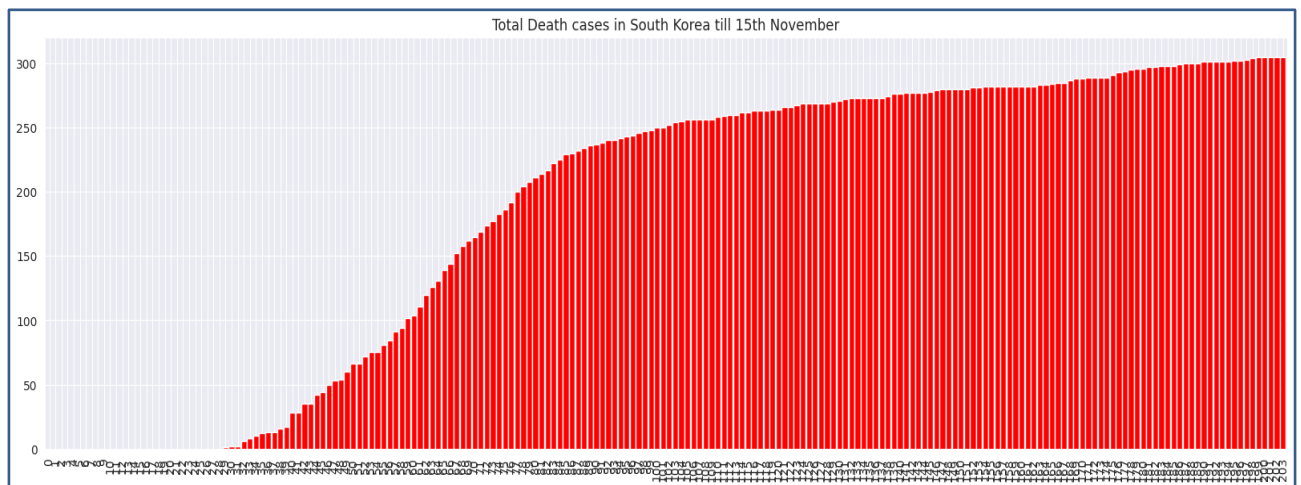
The figure shows the daily confirmed, death, and recovered COVID-19 cases in South Korea.

In the recent times if we check out the no. of Confirmed case Increase, we can find that the cases are barely minor as compared to the whole world. And the Death Case Increment is now down to 0 in the recent times. Which shows that SOUTH KOREA is able to fought Against the virus with minimum casualties of 494 till 15th November, 2020.

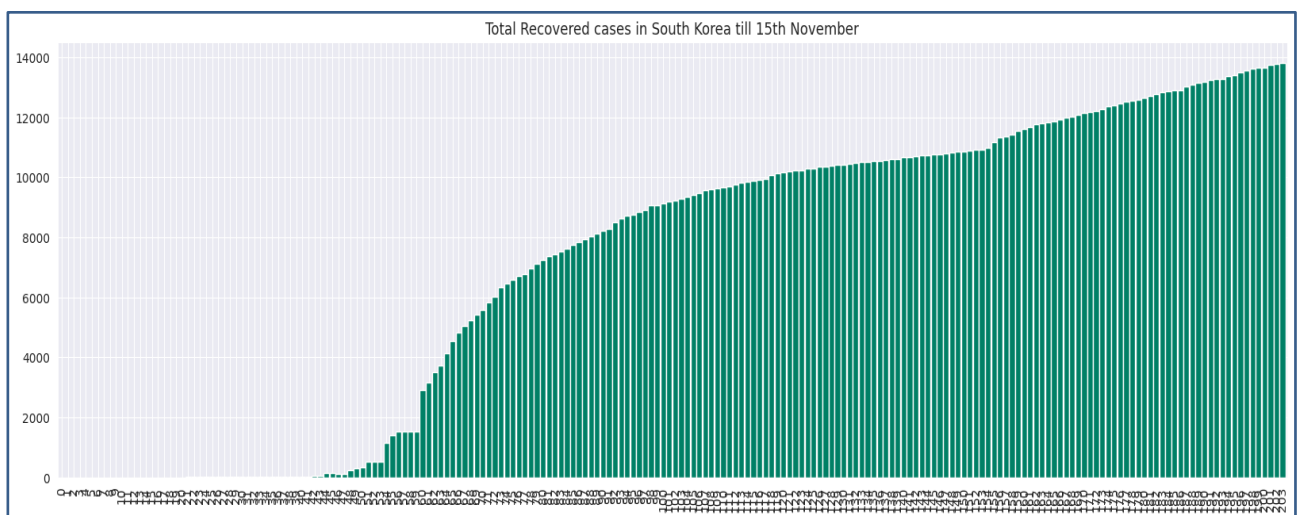
Analysis of South Korea based on the confirmed cases, death cases, recovered cases:



The figure shows the total confirmed COVID-19 cases in South Korea up to 15th November 2020.



The figure shows the total Death COVID-19 cases in South Korea up to 15th November 2020.



The figure shows the total Recovered COVID-19 cases in South Korea up to 15th November 2020.

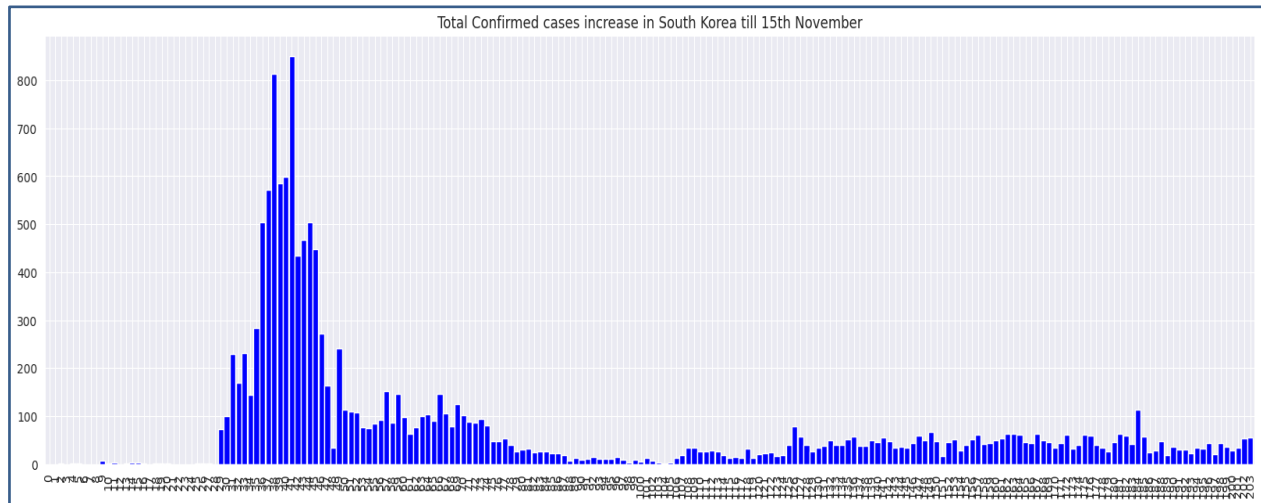


Figure:"Total Confirmed COVID-19 Cases in South Korea as of November 15, 2020"

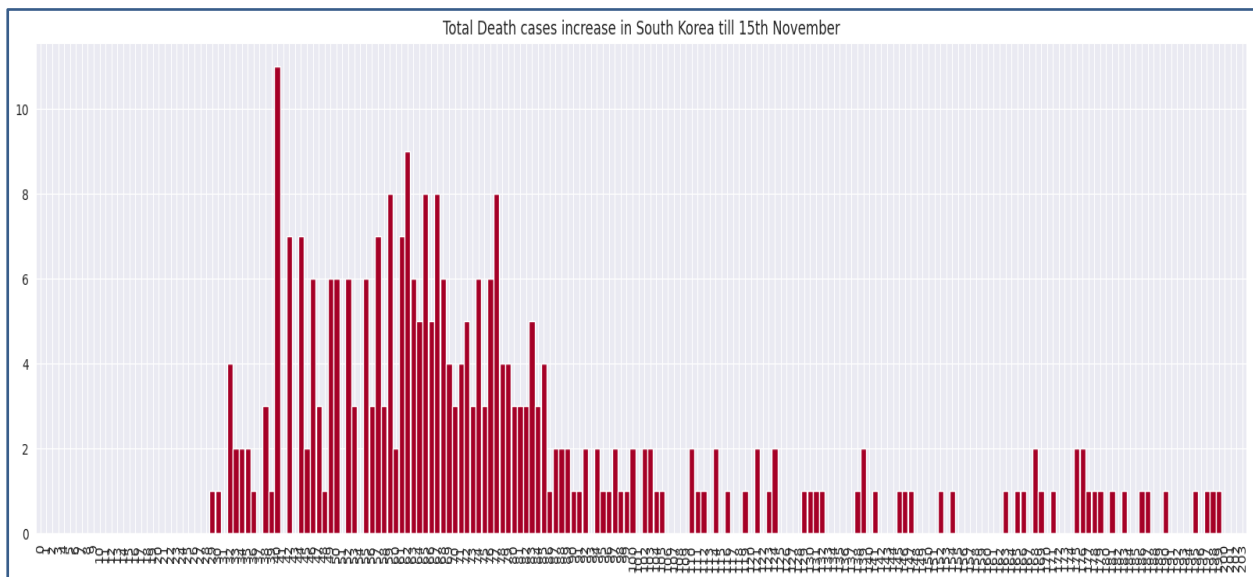


Figure:" Total Deaths COVID-19 Cases in South Korea as of November 15, 2020"

As we can see that the confirm cases are lowered to merely 100 per day which signifies the great work of South Korea to fought against the virus
Also the Death Cases are lowered to 2 to 4 and from the recent few days it turns out to be 0.

United Kingdom : Analysis-

First outbreak: The virus reached the country in late January 2020. The world's eighth-highest death rate per hundred thousand populations and the highest number overall in Europe.

Total affected: As of 2 November 2020 there have been 1,256,725 confirmed cases. Number of cases is highly increasing after September 6(2020). Number of daily affected cases is much higher than mid-July-August.

Death rate: After March death rate was suddenly increased. Total death toll is 56533 till 25 November. On November 21 new death was 341 in 22 it was 398. Only on 25 November the Number of deaths was 696.

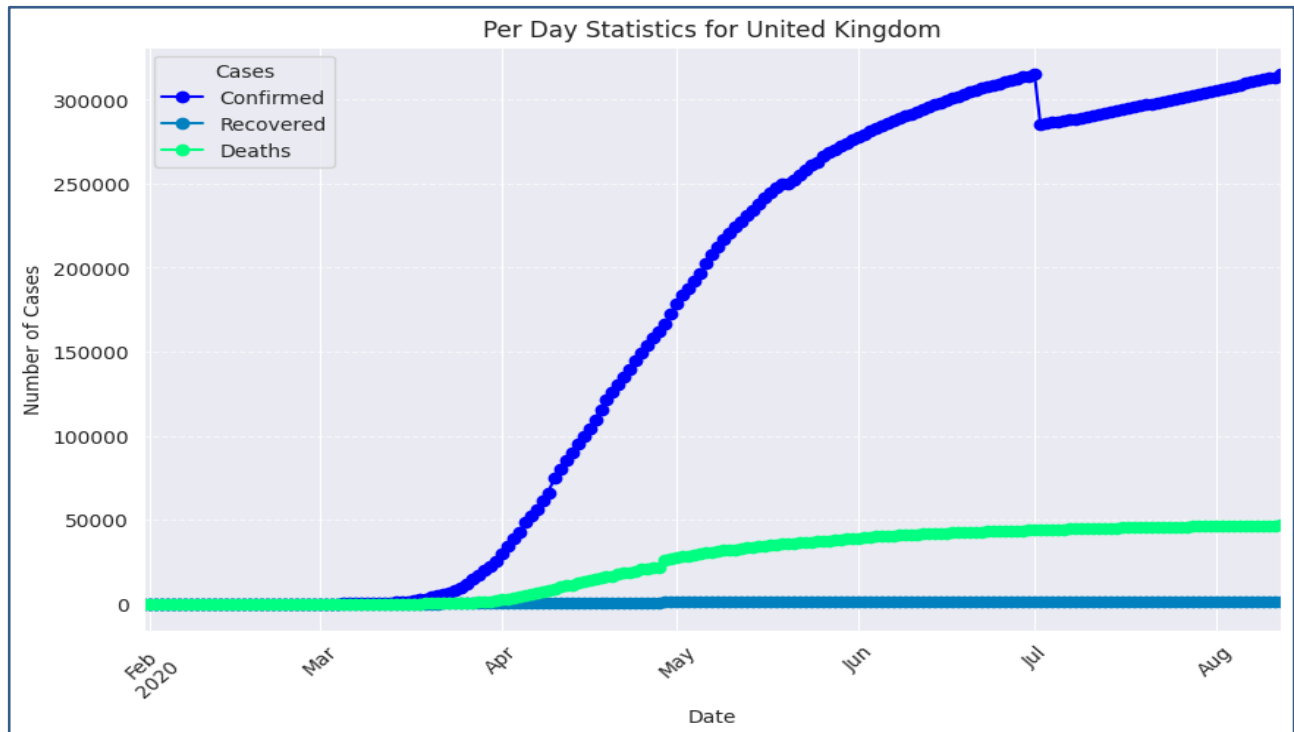


Figure: Daily COVID-19 Cases in Australia as of November 15, 2020

Analysis of United Kingdom based on the confirmed cases, death cases, recovered cases:

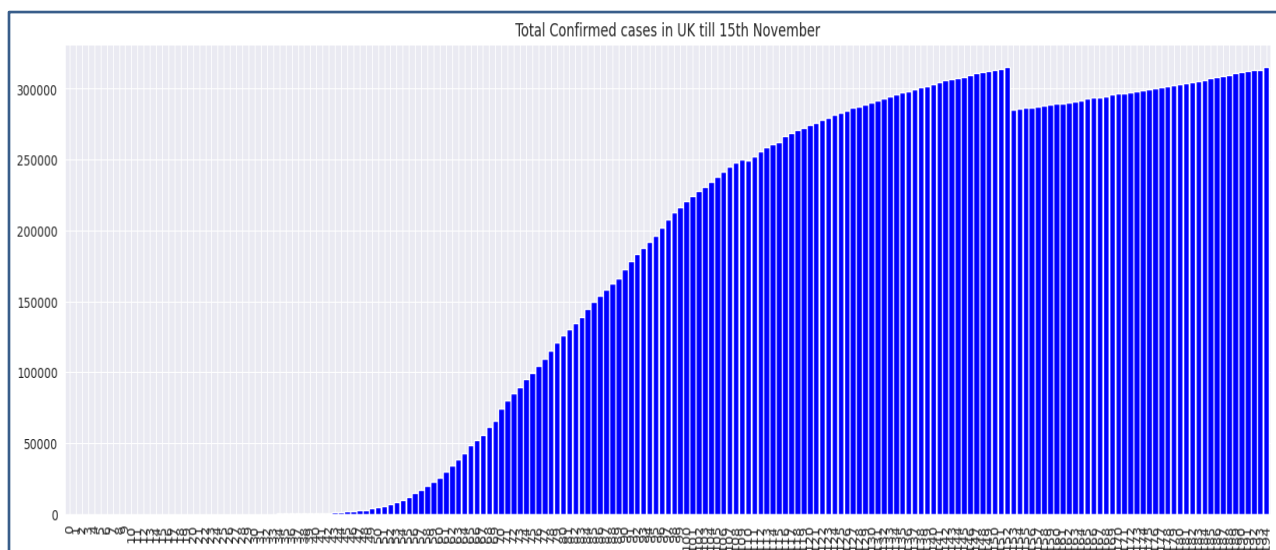


Figure: Total Confirmed COVID-19 Confirmed in the UK as of November 15, 2020

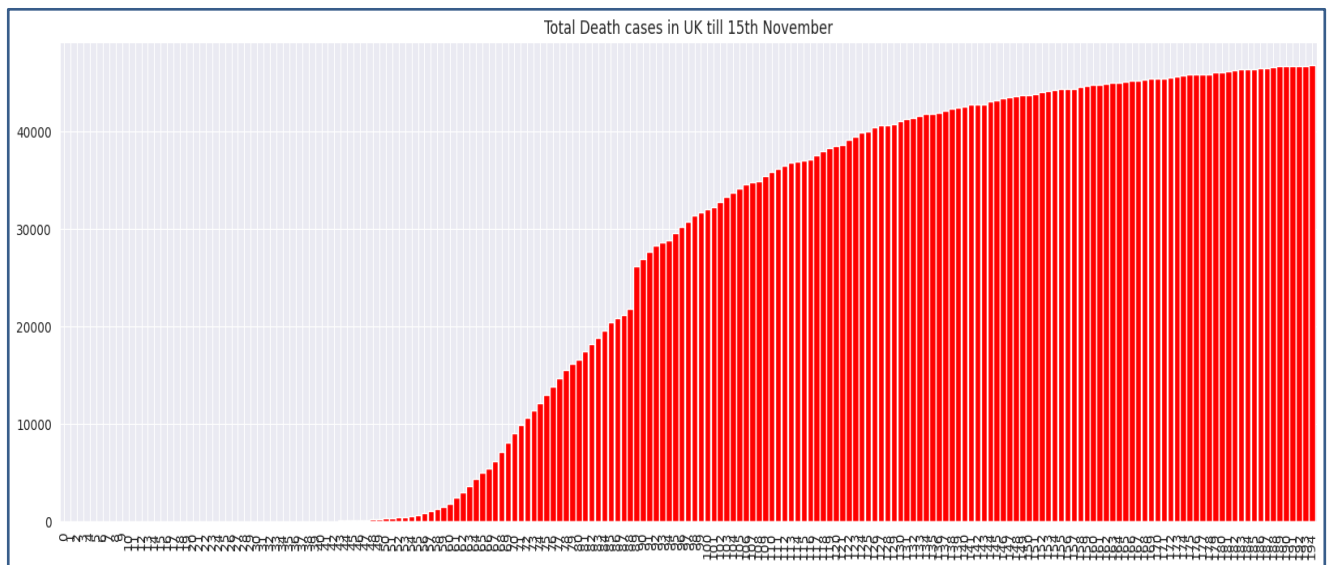


Figure: Total Deaths COVID-19 Death in the UK as of November 15, 2020

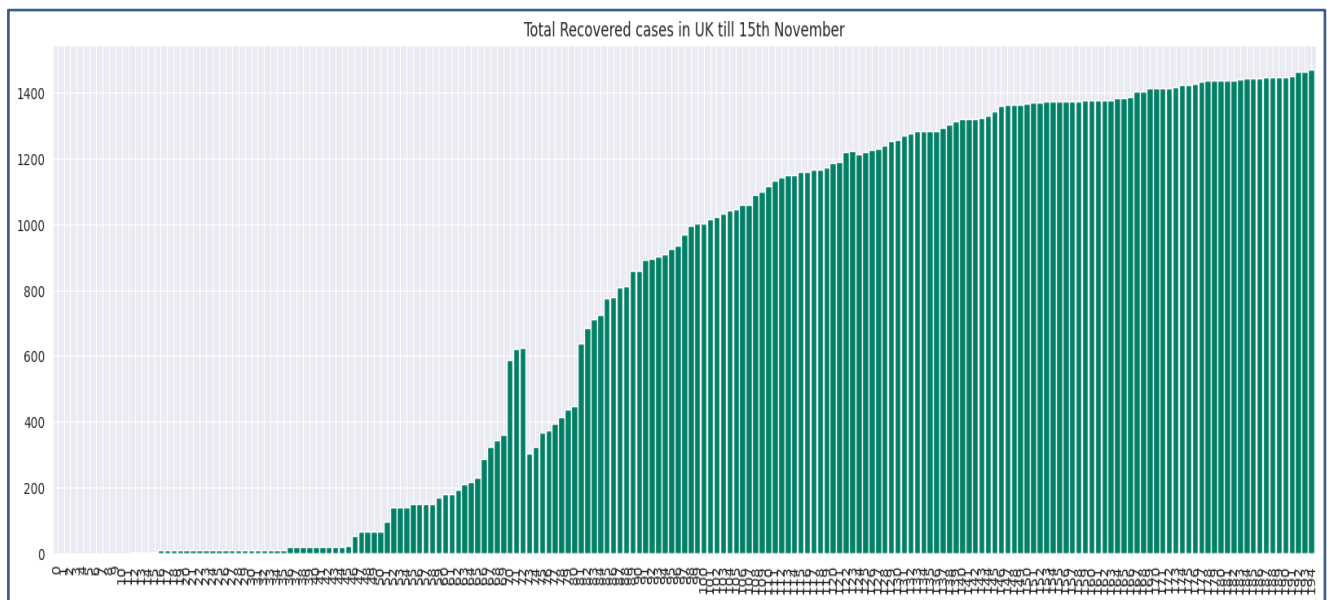


Figure: Total Recovered COVID-19 Recovered in the UK as of November 15, 2020

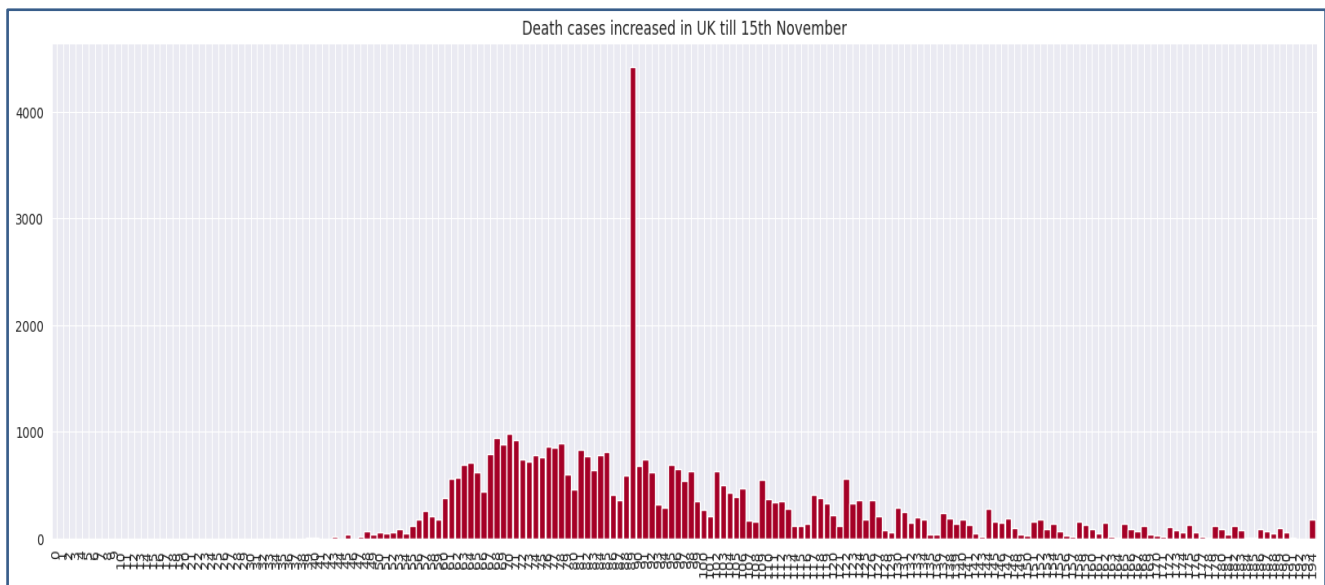


Figure: UK Deaths Increased till 15 Nov, 2020 .

From the Plotting we can clearly observe that the confirmed, recovered and death tolls are flattened pretty much. Which signifies that UK is pretty much controlled the situation. From the analysis we can observe that after the certain stage of the pandemic situation the Number of confirmed cases are decreased and level up at a certain range. In the recent days The cases are reported near about 100.

From the above observation we can also find out that the death cases are not reported in a Huge manner in the recent times. The reported death cases are like under 10 per day, some days it is being reported as 0. From the above analysis it is clearly visible to us that the United Kingdom is fought against the pandemic situation quite brilliantly!

Conclusion: United Kingdom has fought well against the Covid-19!

Australia: Analysis-

The first confirmed case in Australia was identified on 25 January 2020, in Victoria, when a man who had returned from Wuhan, China, tested positive for the virus. Australian borders were closed to all non-residents on 20 March. Social distancing rules were imposed on 21 March, and state governments started to close 'non-essential' services. The number of new cases initially grew sharply, then leveled out at about 350 per day around 22 March, and started falling at the beginning of April to under 20 cases per day by the end of the month. A second wave of infections emerged in Victoria during May and June, which was attributed to an outbreak at a Melbourne hotel used for quarantining international arrivals. The second wave was much more widespread and deadlier than the first; at its peak, the state had over 7,000 active cases. The wave ended with zero new cases being recorded on 26 October. As of 25 November 2020, Australia has reported 27,854 cases, 25,355 recoveries, and 907 deaths, with Victoria accounting for nearly 75 percent of cases and 90 percent of fatalities. Compared to other Western countries, notably the United States and European countries, Australia's handling has been praised for its effectiveness and fast-forward reactions.

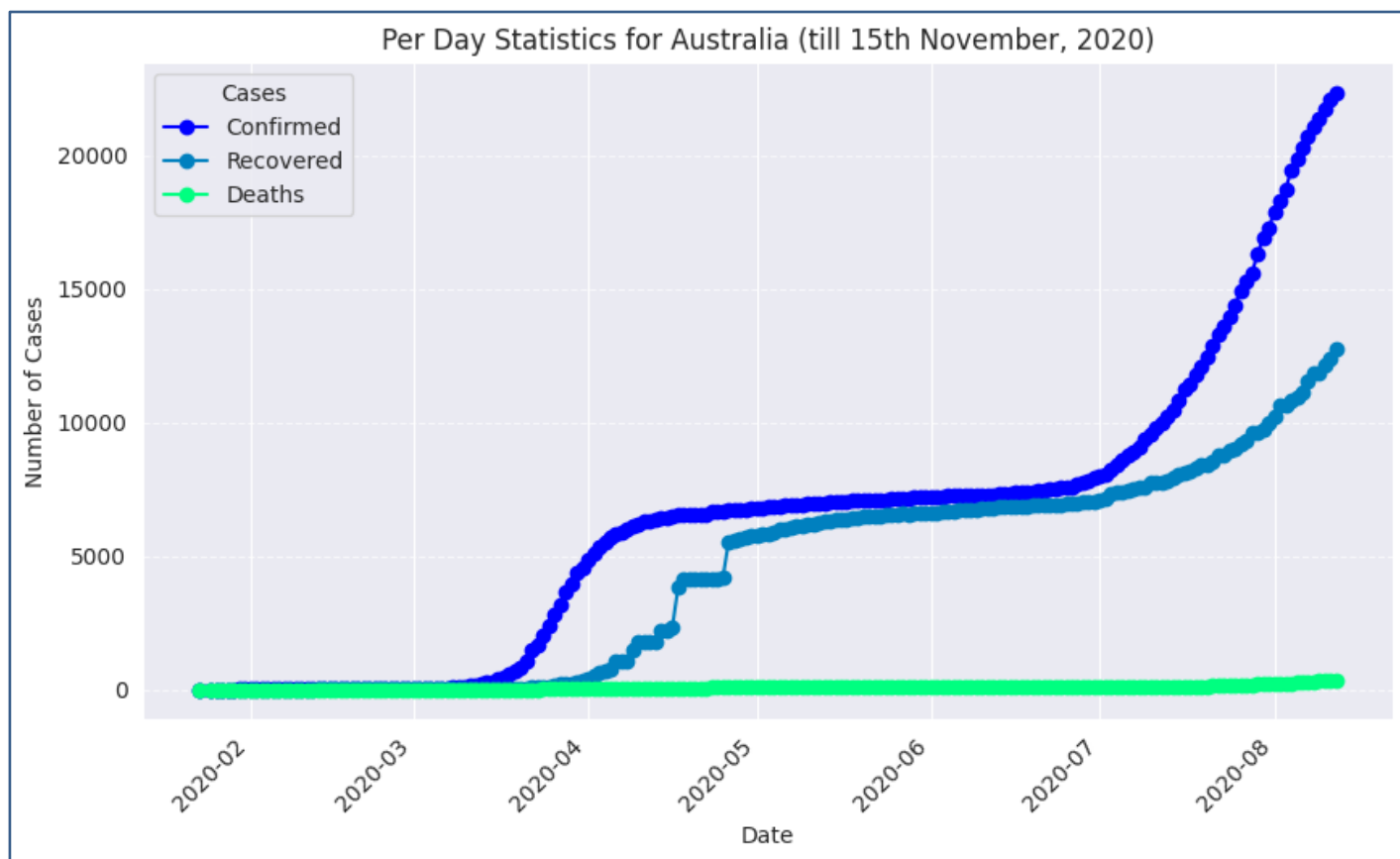


Figure: Per Day statistics for Australia till 15th November, 2020

Analysis of Australia based on the confirmed cases, death cases, recovered cases:

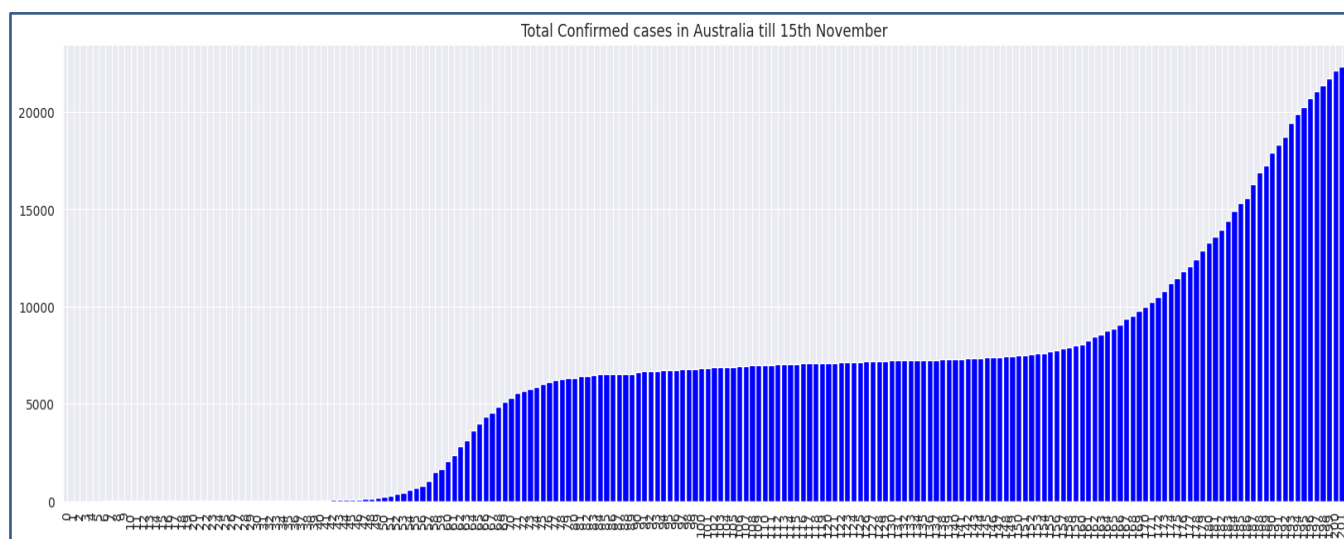


Figure: Australia Total Confirmed cases.

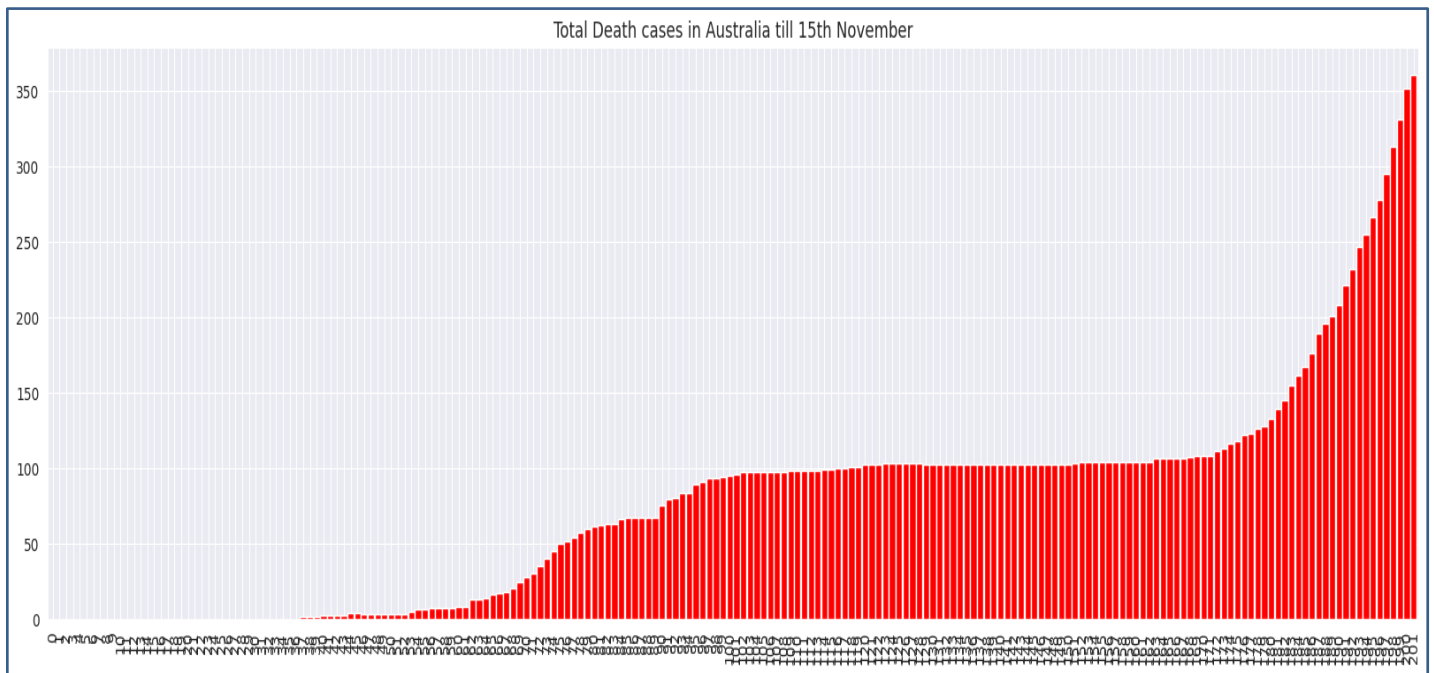


Figure: Australia Total Death cases. The Death toll is also increasing with the increment of the total confirmed cases. Which may cost the country after a certain relaxation period of flattened curve?

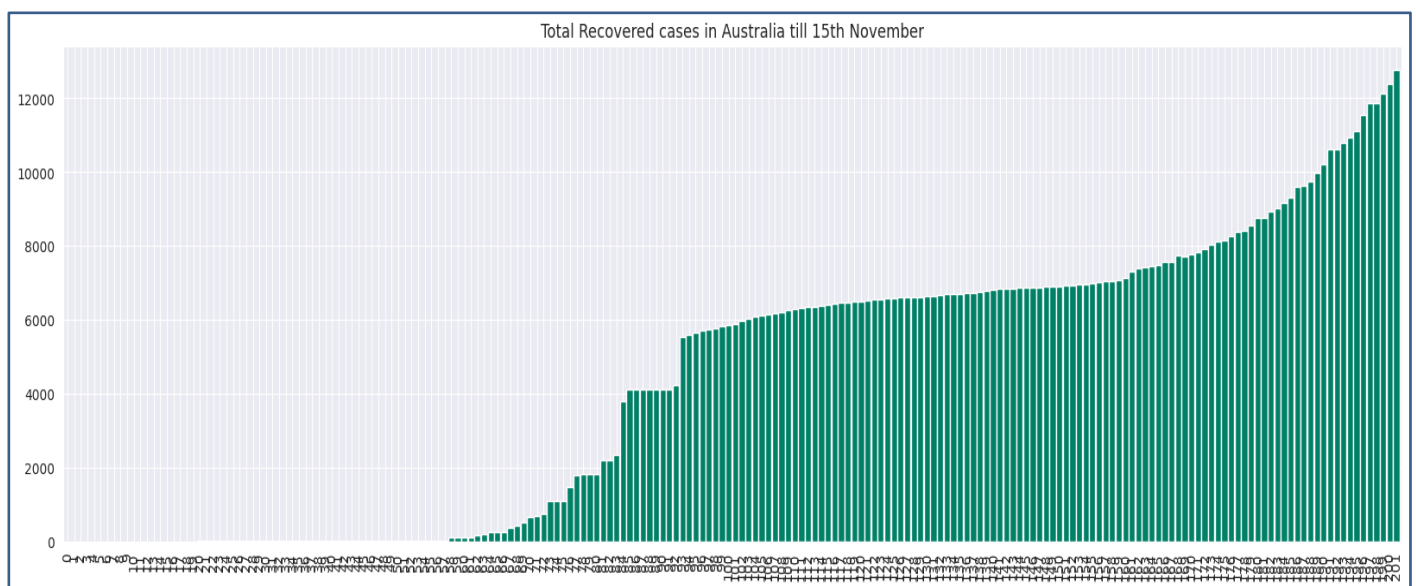


Figure: Australia Total Recovered cases.

As the confirmed cases are increasing day by day in a huge manner it is very difficult to show composure for the Government of Australia. The confirmed cases were came down to 50 in the past 2-3 months after the drastic situation. But unfortunately it's growing higher than the previous manhattans and showing a peak of 700 per day. But the good thing is that, in the recent times the confirmed cases per day are come down to 400 mark, which shows a good sign for Australia. In the recent times the death toll is increasing in the day to day basis in a huge quantity, which may cost the Government of Australia in this pandemic situation. The death toll is increasing day by day very much in Australia. The highest death toll is shown in the day of 11th August, 2020 where the death toll rises to 21, the highest in the whole pandemic situation

LOCKDOWN: The Savior or, not?

On March 26, 2020, Bangladesh announced a nationwide "general holiday" (lockdown) to control the spread of COVID-19. The government urged citizens to stay at home, stressing that strict isolation was necessary to break the transmission cycle of the virus. Officials warned that failure to comply could lead to uncontrollable outbreaks, overwhelming the fragile healthcare system and causing irreversible harm to families and the nation.

At the time of the lockdown announcement, Bangladesh had recorded around 44 confirmed cases and 5 deaths. Government and health authorities believed that a complete lockdown and suspension of public transport and gatherings would isolate infected individuals, contain the virus within small clusters, and delay widespread community transmission a stage when tracing and quarantining become ineffective.

Impact of the Lockdown in Bangladesh:

Improved Health Infrastructure:

The lockdown gave the government critical time to expand healthcare capacity increasing the number of hospital beds, isolation centers, ventilators, and availability of PPE kits for frontline workers.

Innovations in Treatment:

In the absence of a specific antiviral drug for COVID-19, doctors in Bangladesh repurposed existing treatments and collaborated globally to adopt emerging best practices, improving survival rates.

Lower Mortality Rate:

If infections had peaked earlier, when resources were extremely scarce, the health system could have collapsed. The lockdown allowed Bangladesh to better prepare, resulting in a relatively lower mortality rate compared to early worst-case predictions.

Encouraging Technological Innovations:

Lockdown periods led to increased use of mobile technology, telemedicine, and digital health platforms to safely monitor and treat patients remotely, minimizing the exposure of healthcare workers.

- RED LINE shows the date of IMPOSING LOCKDOWN
- GREEN LINE shows the date of REVOKING LOCKDOWN

China's Lockdown Period – SUCCESSFUL

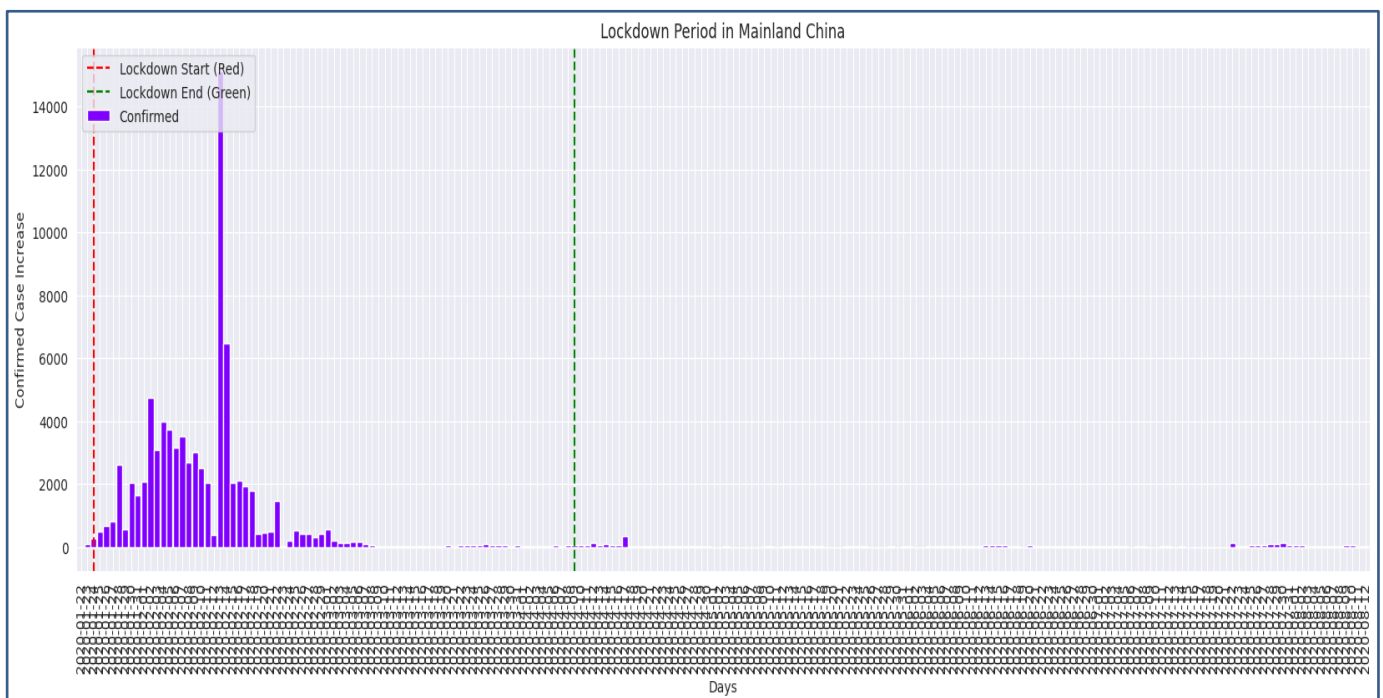


Figure : China Lockdown Period and Its situation .

Italy's lockdown Period – SUCCESSFUL

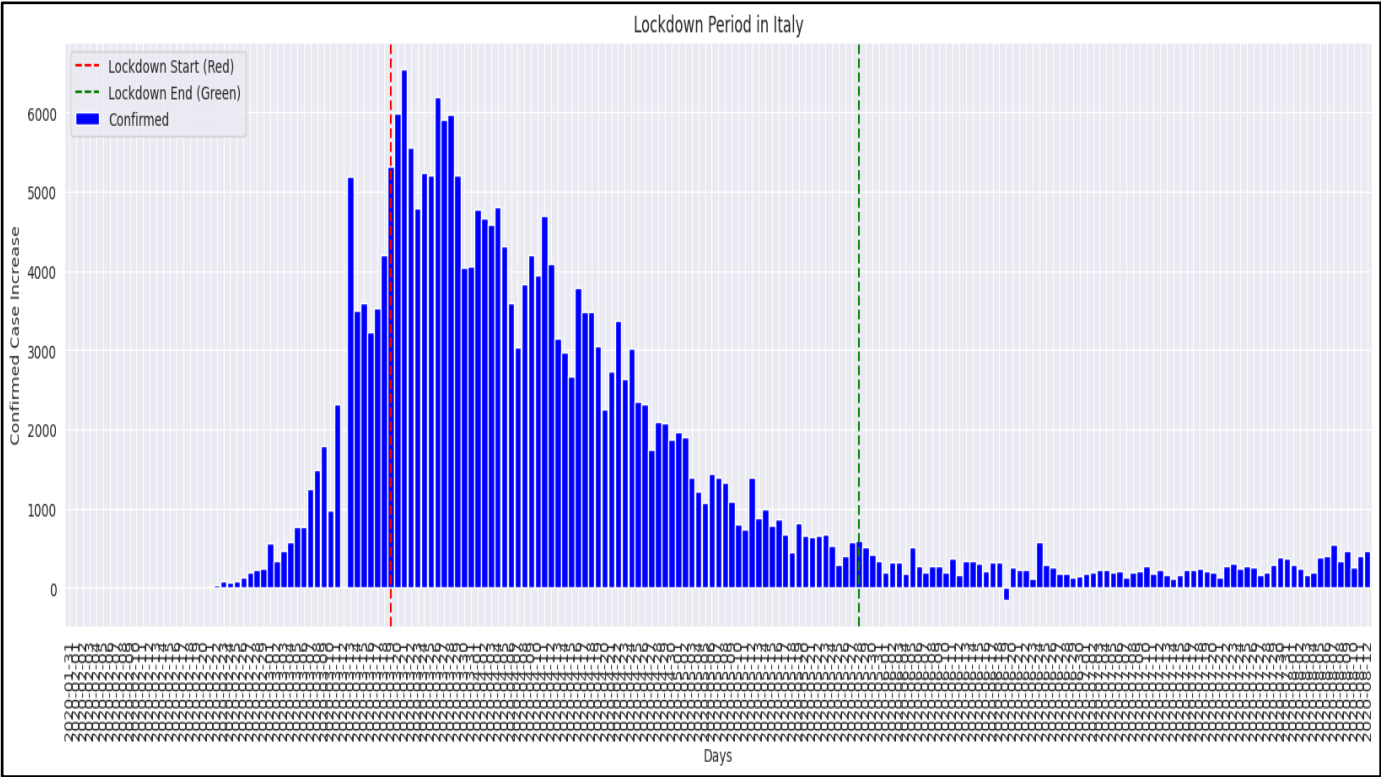


Figure: Italy Lockdown Period and Its situation.

United States of America Lockdown Period – UNSUCCESSFUL

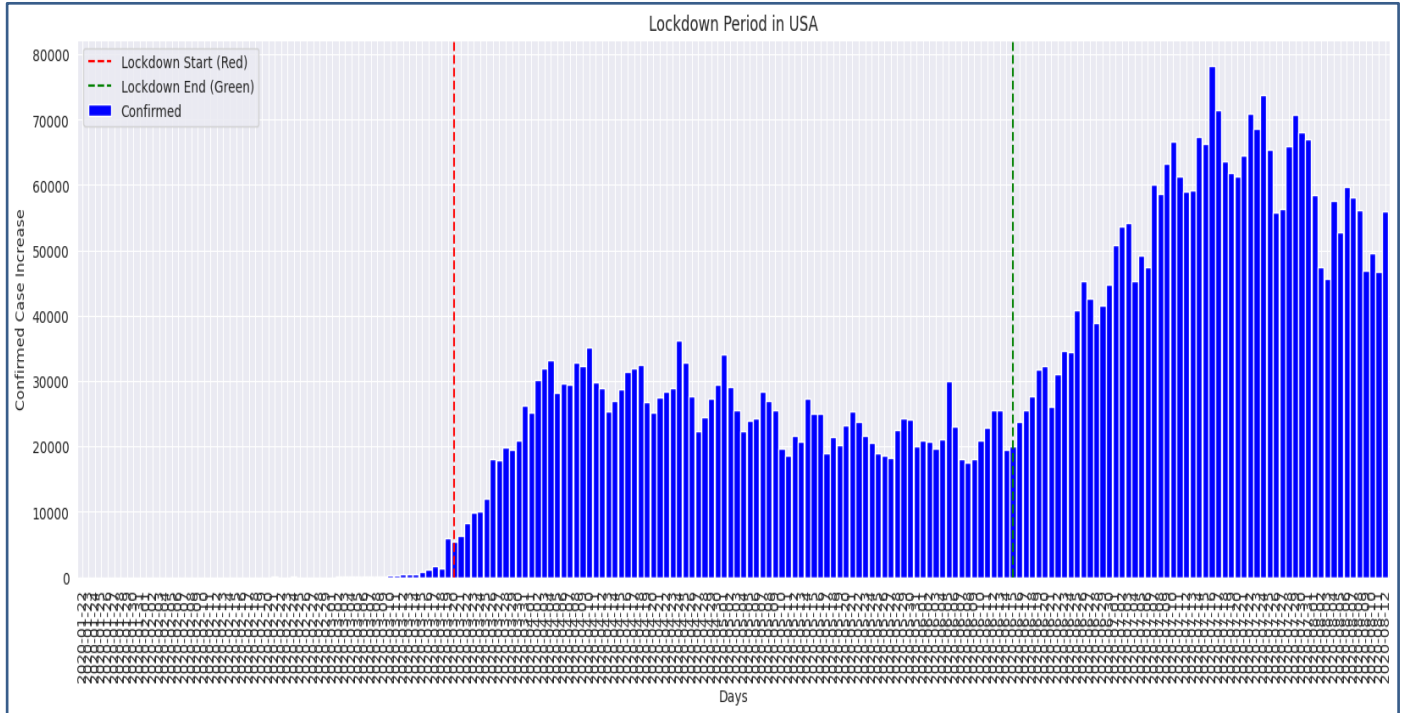


Figure: USA Lockdown Period and Its situation.

AUSTRALIA Lockdown - SUCCESSFUL –

Australia imposed national lockdown on 23rd March, 2020

Australia eased the lockdown on 8th July, 2020

Australia again imposed the lockdown on 7th August, 2020 due to increment in the daily cases

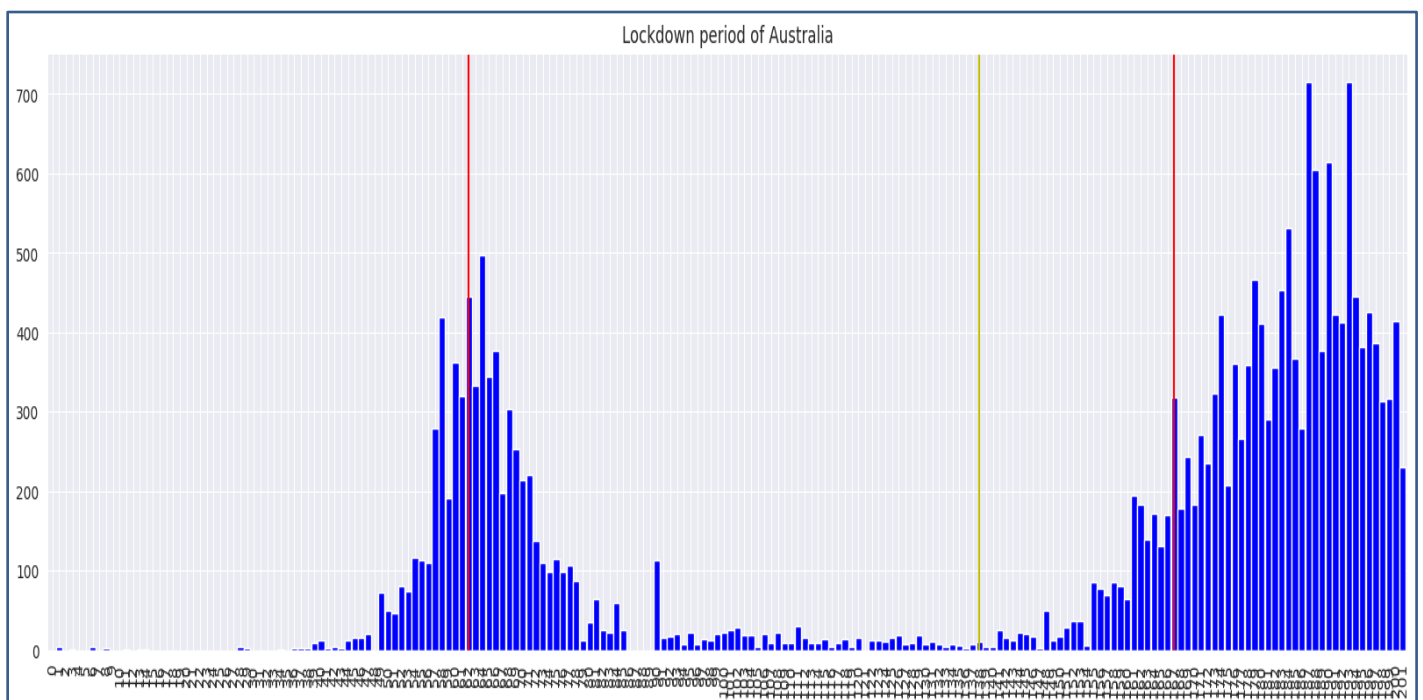


Figure: AUSTRALIA Lockdown Period and Its situation.

BANGLADESH COVID 19 ANALYSIS:

Bangladesh COVID-19 Description:

The COVID-19 pandemic reached Bangladesh in March 2020. The virus spread rapidly across the country, affecting public health and the economy. The government implemented nationwide lockdowns, health safety guidelines, and vaccination programs to control the outbreak. Major cities like Dhaka were heavily impacted. Bangladesh relied on international support, local healthcare efforts, and awareness campaigns to manage the crisis. By 2022, cases had significantly decreased due to widespread vaccination and public health measures, although monitoring and preventive actions continued.

Bangladesh Current situation and analysis:

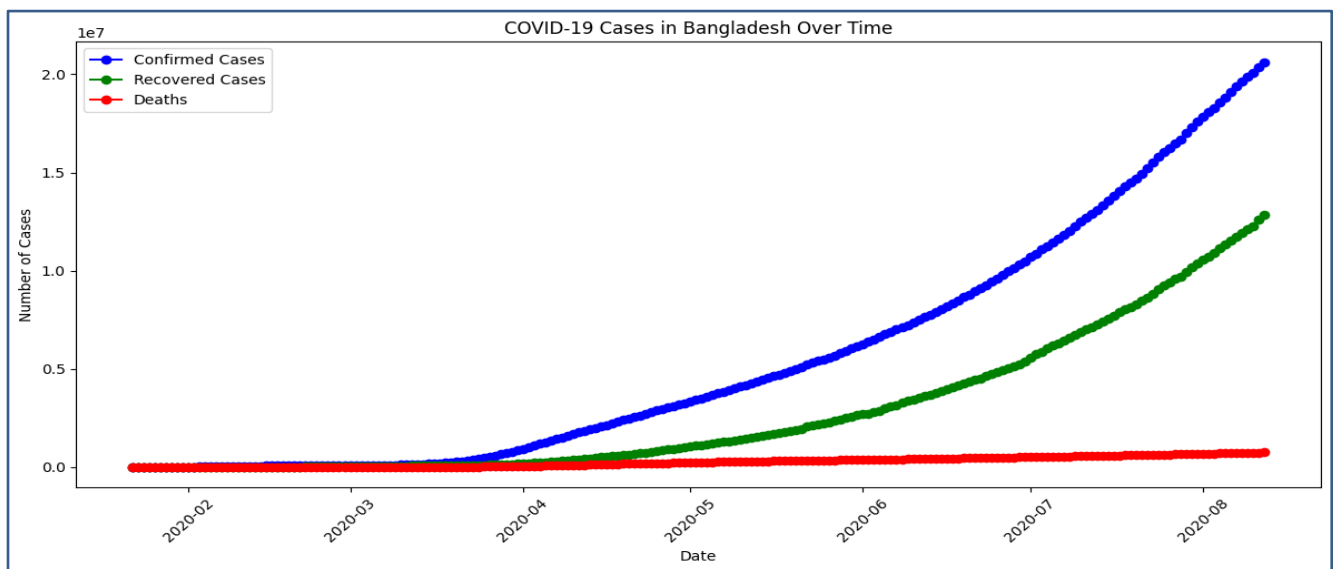


Figure: Bangladesh Covid 19 Total Result Show

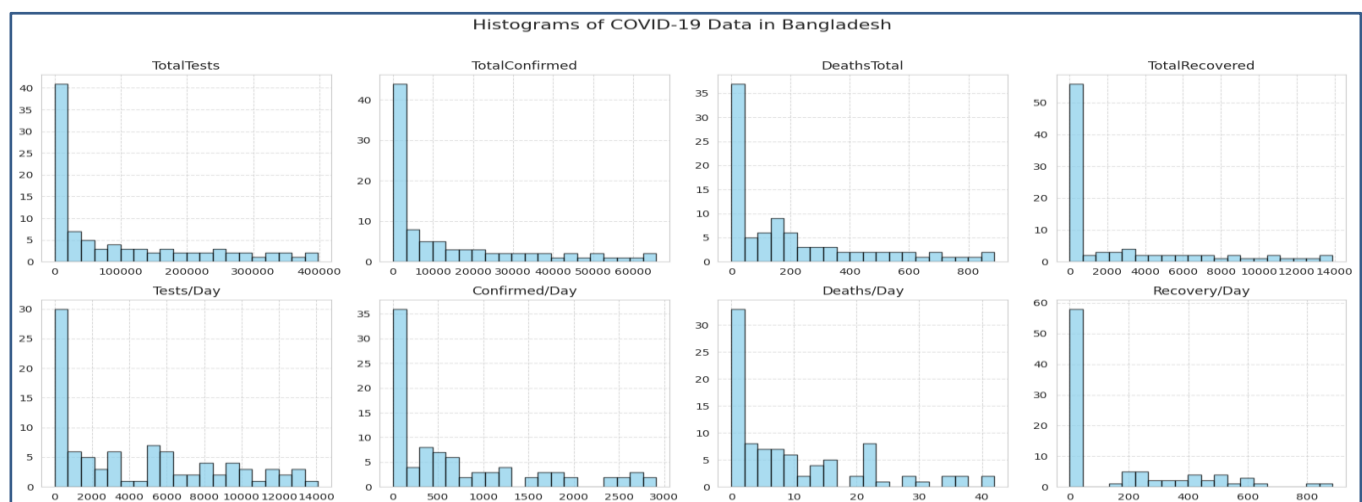
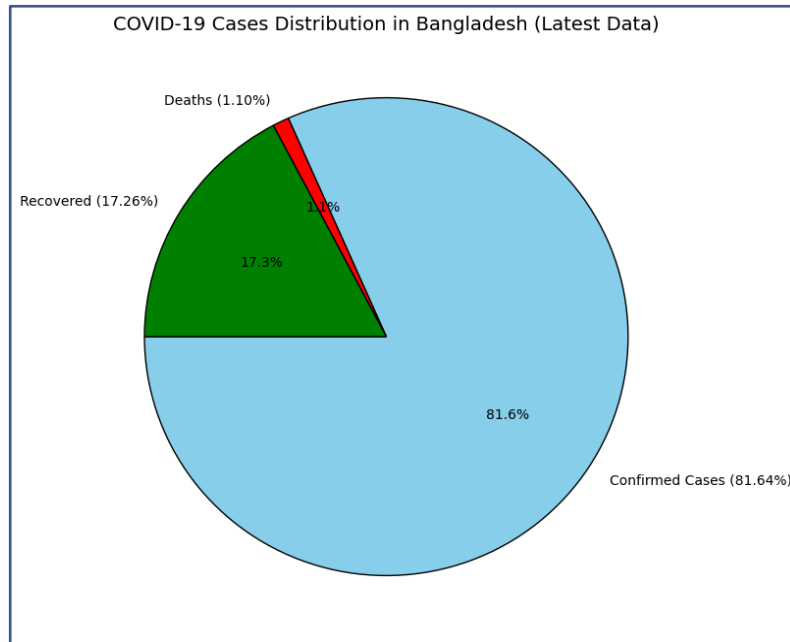
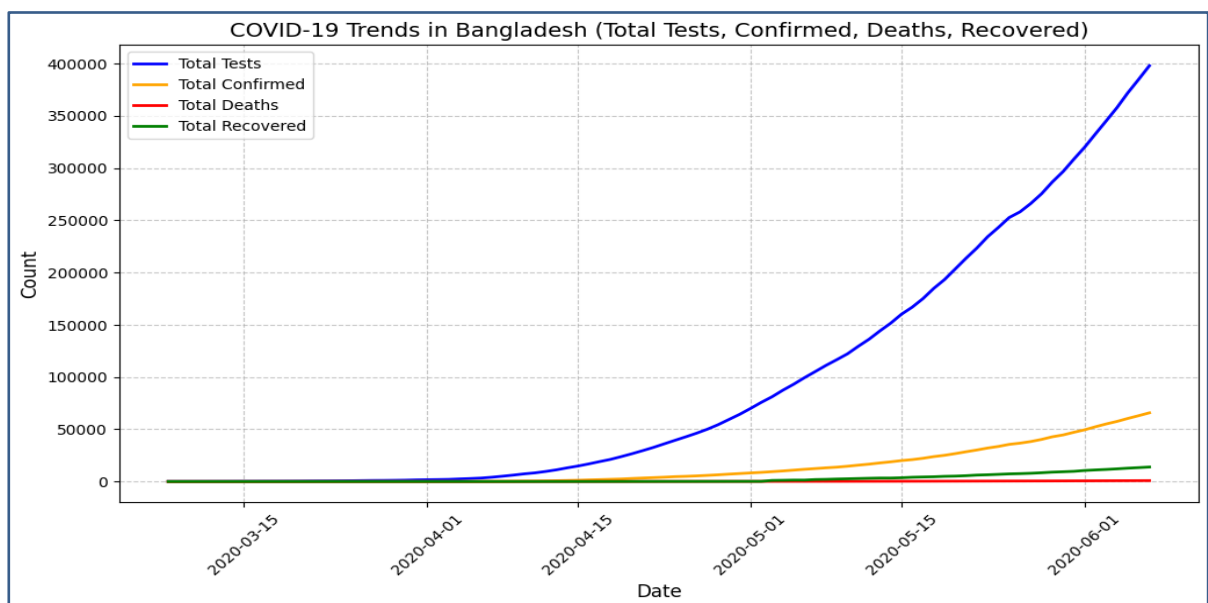


Figure: Bangladesh Covid 19 Total Result Show In Plot.

Bangladesh COVID 19 analysis in pie chart and percent:



Total Tests, Total Confirmed Cases, Total Deaths, and Total Recovered-



Bangladesh Covid19 Total Confirmed (Actual Vs. Predicted Used {LSTM Model}):

Epoch 4/20			
2/2	0s	46ms/step	loss: 0.0117
Epoch 5/20			
2/2	0s	46ms/step	loss: 0.0081
Epoch 6/20			
2/2	0s	47ms/step	loss: 0.0098
Epoch 7/20			
2/2	0s	47ms/step	loss: 0.0093
Epoch 8/20			
2/2	0s	46ms/step	loss: 0.0054
Epoch 9/20			
2/2	0s	46ms/step	loss: 0.0030
Epoch 10/20			
2/2	0s	47ms/step	loss: 0.0024
Epoch 11/20			
2/2	0s	47ms/step	loss: 0.0023
Epoch 12/20			
2/2	0s	53ms/step	loss: 0.0017
Epoch 13/20			
2/2	0s	45ms/step	loss: 5.5603e-04
Epoch 14/20			
2/2	0s	56ms/step	loss: 1.8303e-04
Epoch 15/20			
2/2	0s	70ms/step	loss: 6.5407e-04
Epoch 16/20			
2/2	0s	73ms/step	loss: 0.0011
Epoch 17/20			
2/2	0s	71ms/step	loss: 5.7721e-04

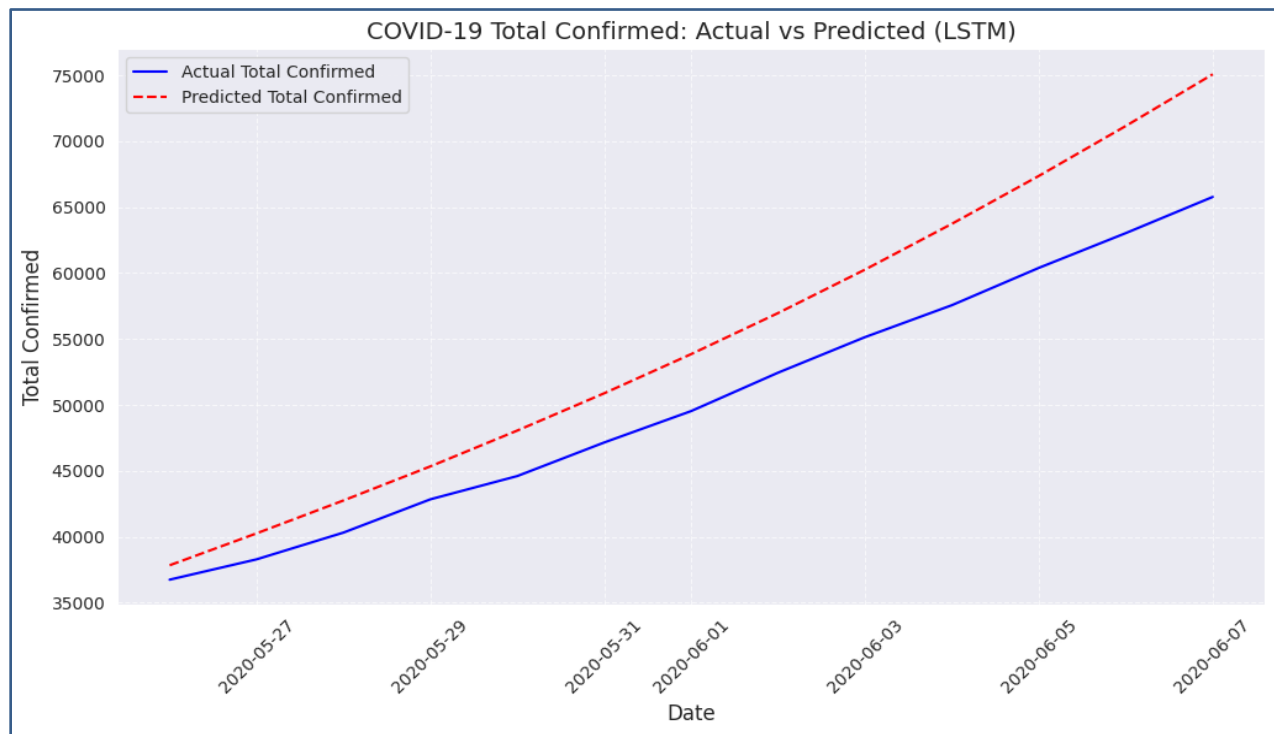
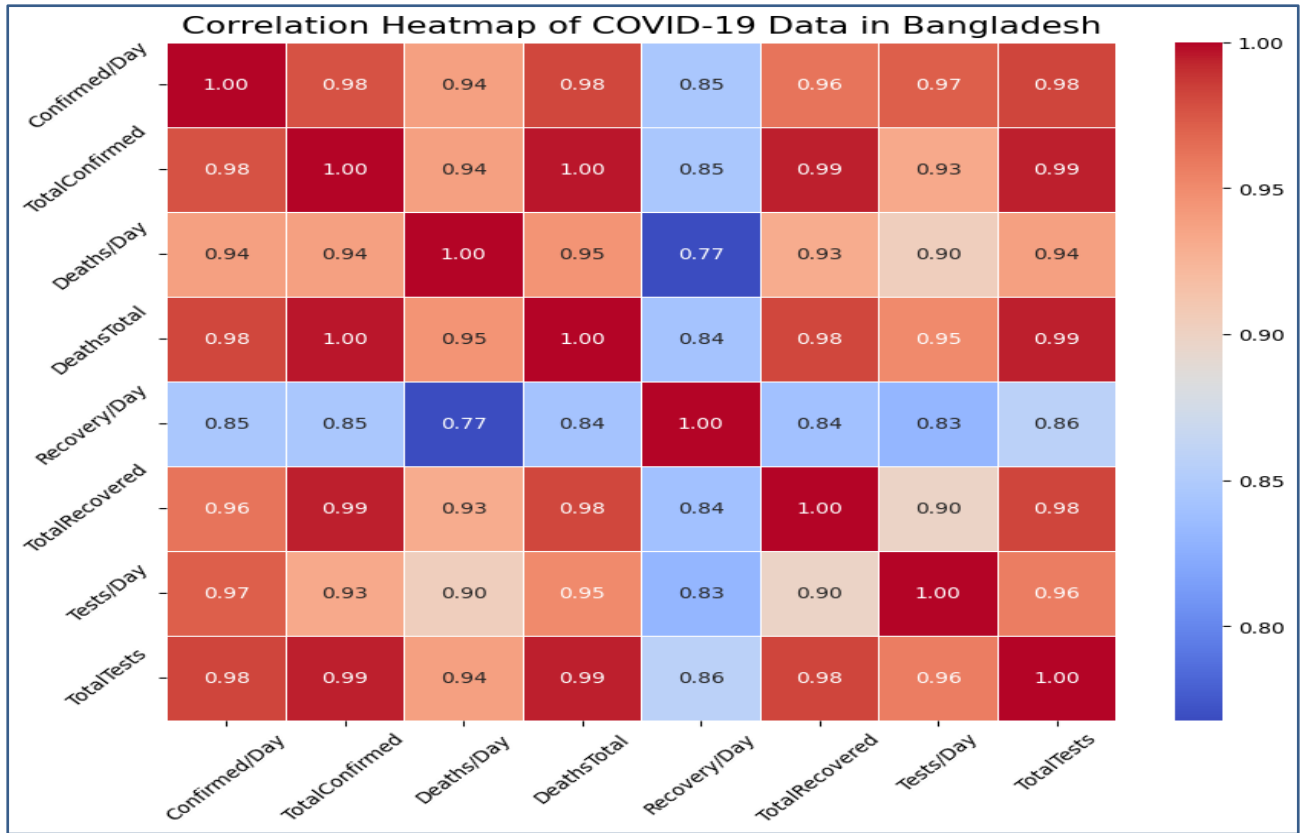
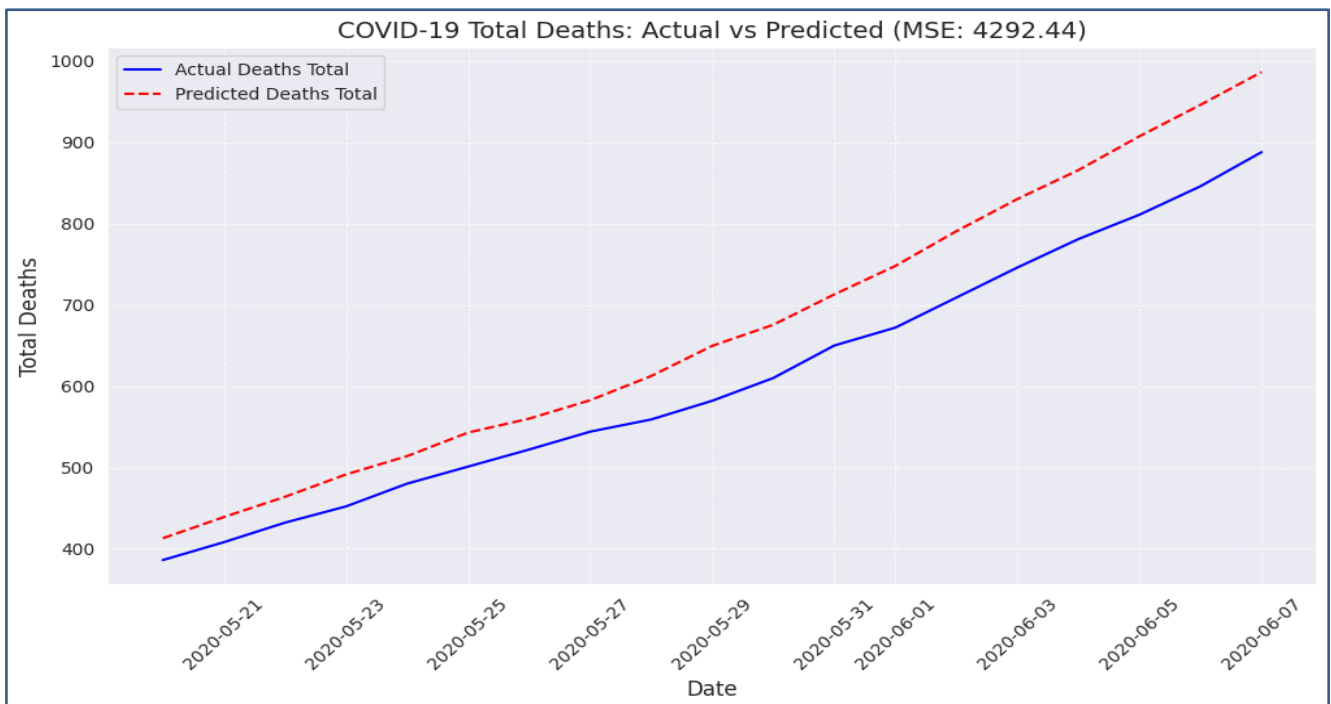


Figure: Covid 19 Bangladesh Total Confirmed Actual V/s Prediction

Bangladesh covid 19 correlation:



BANGLADESH COVID 19 Actual V/S Predicted Total Deaths:



Bangladesh Age And Gender Analysis In Covid 19:

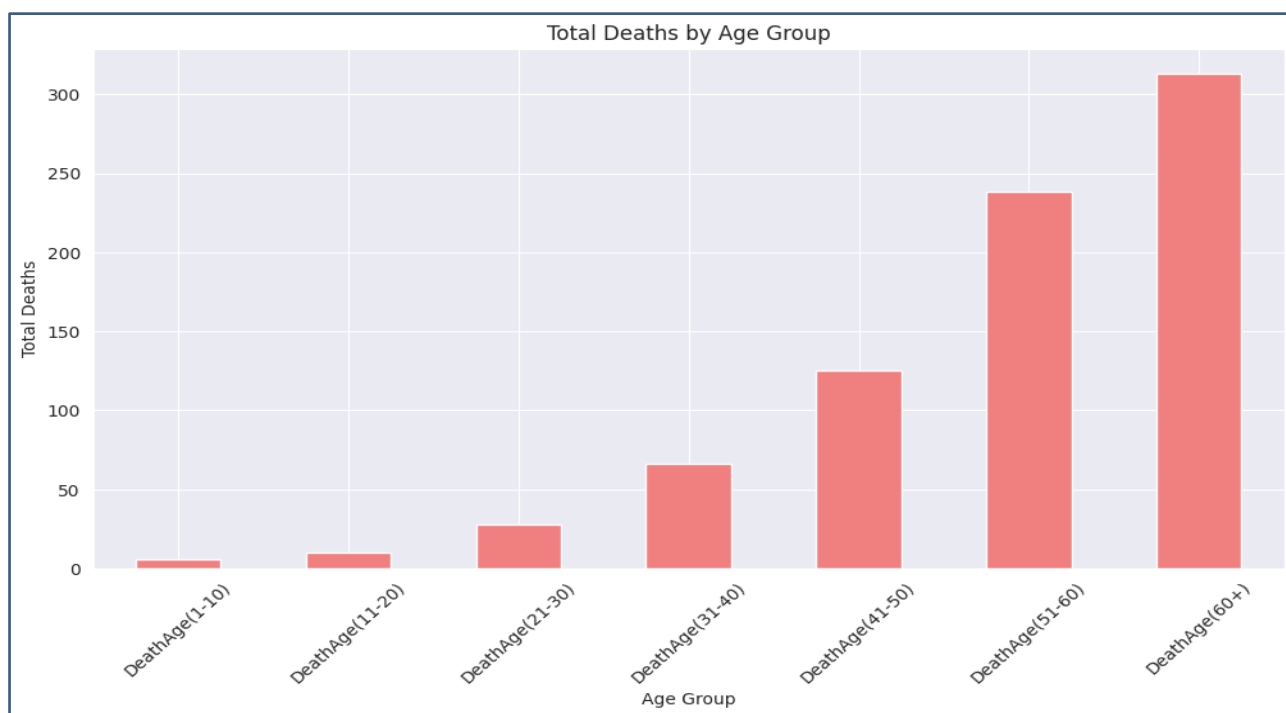
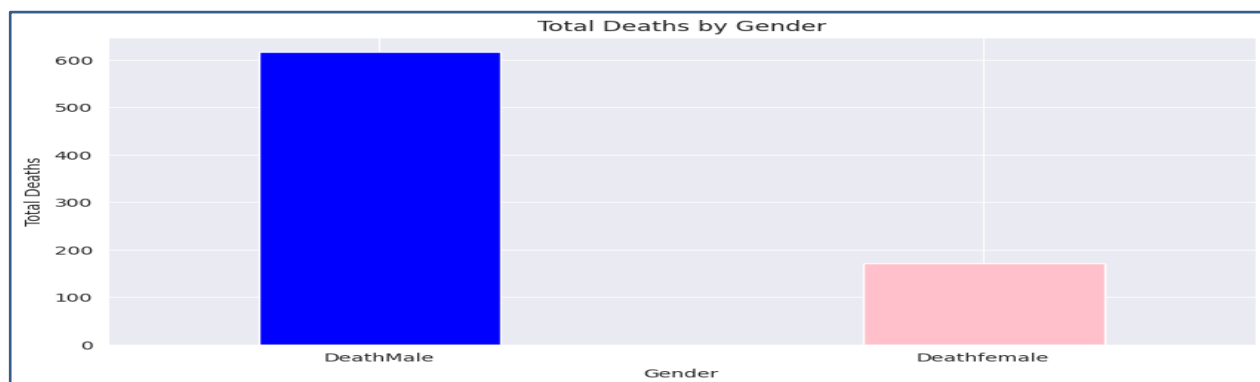
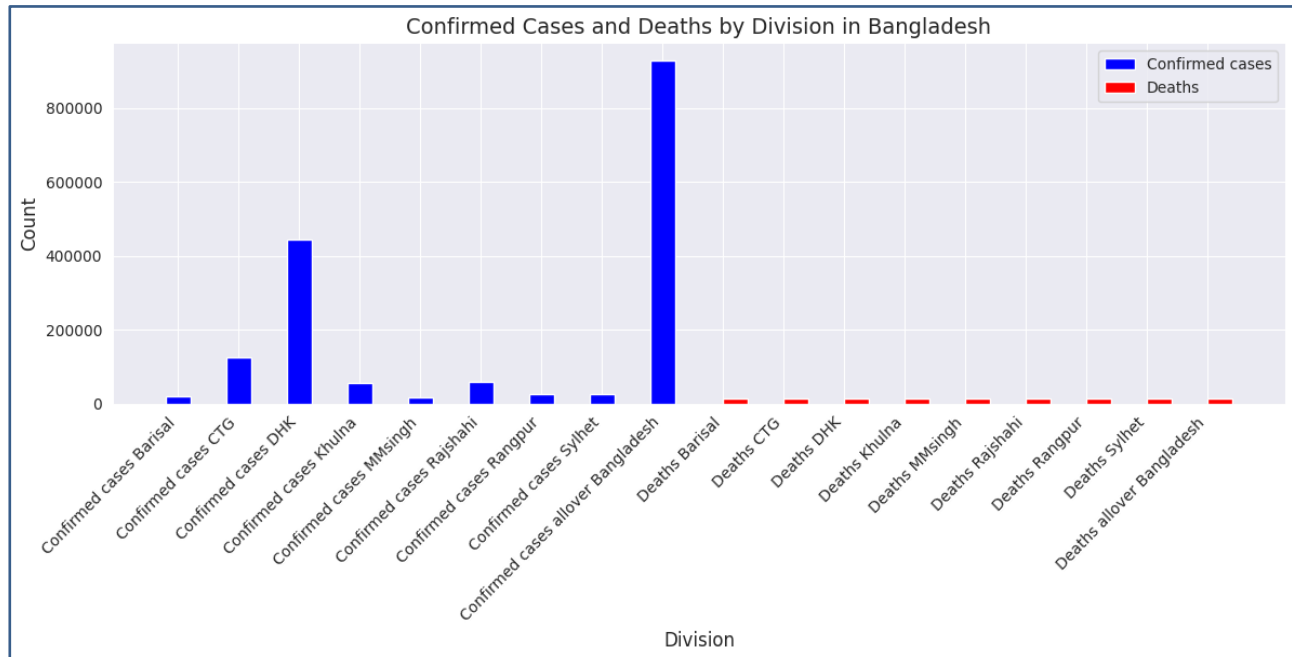


Figure: Age wise Analysis.

Bangladesh COVID 19 Division Wise Analysis:



Bangladesh Covid 19 Division Wise Result:

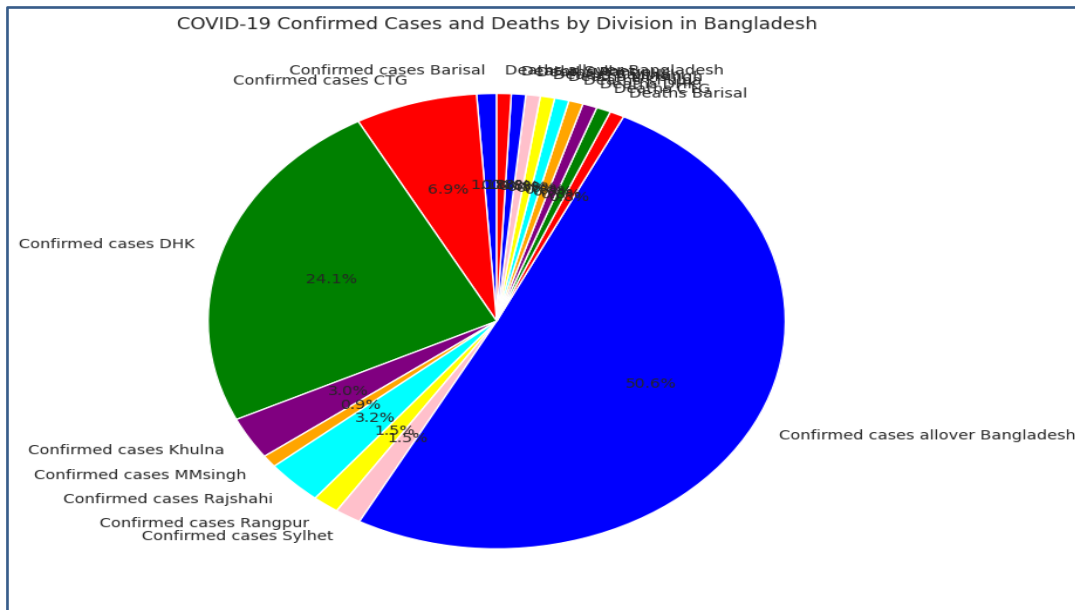


Figure: Bangladesh Covid 19 Division Wise Result Show In a Pie Chart

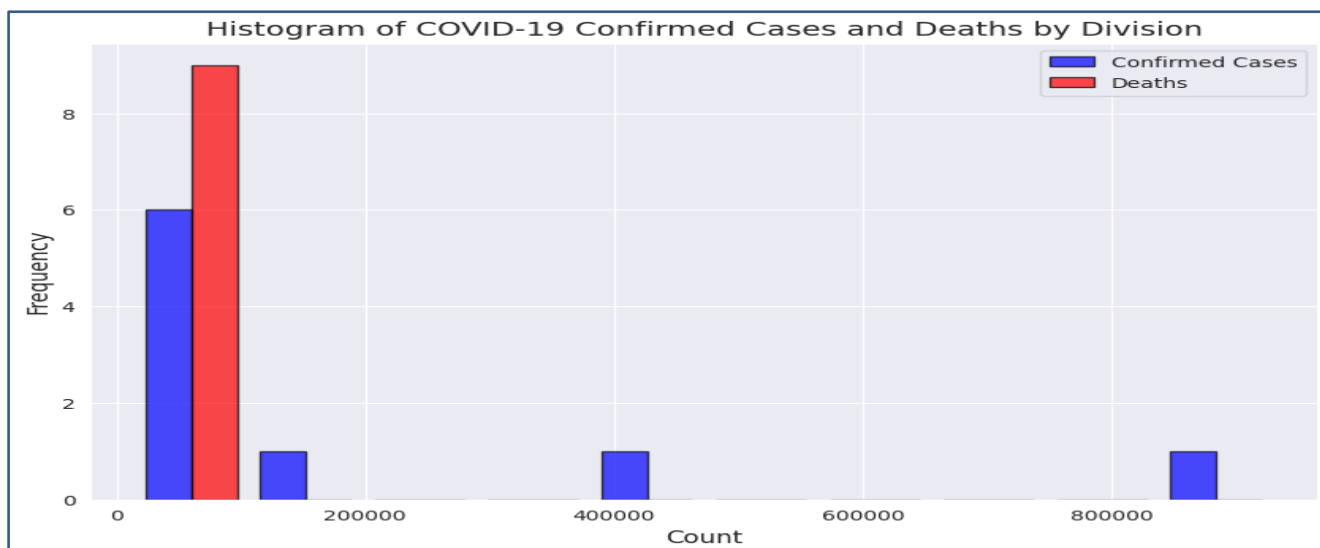
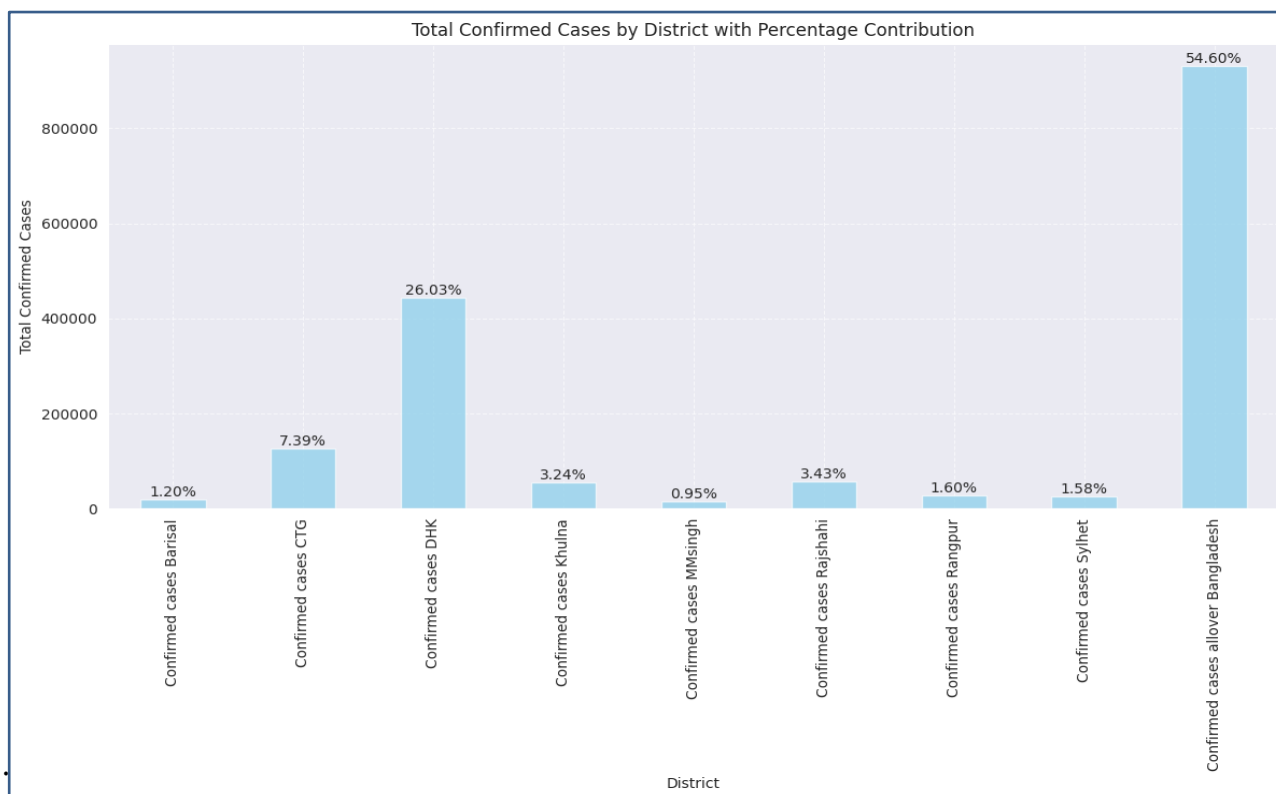
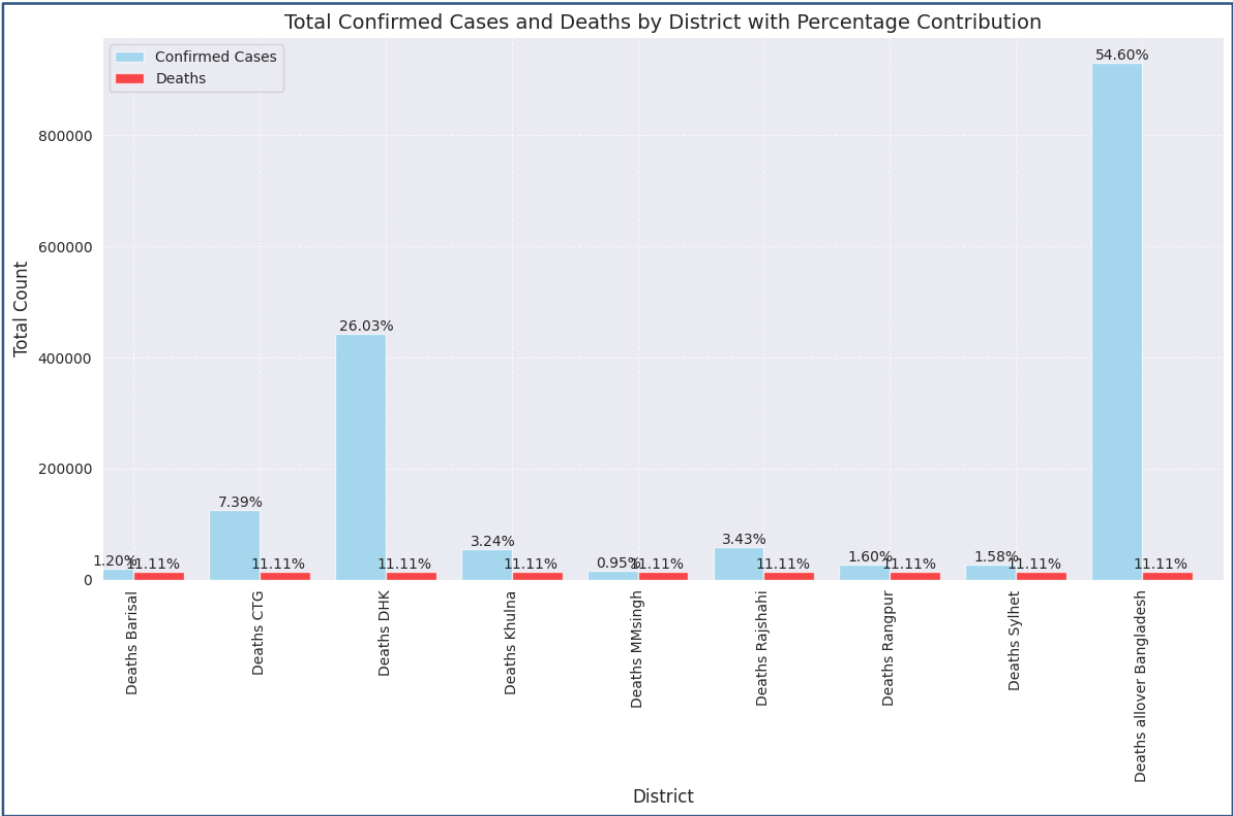


Figure: Show the Histogram Plot.

Bangladesh Covid19 Total Confirmed Cases by District with Percentage Contribution:



Bangladesh Covid19 Total Confirmed Cases and Deaths By District With Percentage Contribution:



				District	
	District	Total	Confirmed Cases	\	
0	Confirmed cases Barisal		20466.0		
1	Confirmed cases CTG		125792.0		
2	Confirmed cases DHK		443319.0		
3	Confirmed cases Khulna		55234.0		
4	Confirmed cases MMsingh		16161.0		
5	Confirmed cases Rajshahi		58369.0		
6	Confirmed cases Rangpur		27185.0		
7	Confirmed cases Sylhet		26831.0		
8	Confirmed cases allover Bangladesh		929986.0		
	Confirmed Cases	Percentage	Total Deaths	Deaths	Percentage
0		1.201520	14778.0		11.111111
1		7.385007	14778.0		11.111111
2		26.026408	14778.0		11.111111
3		3.242682	14778.0		11.111111
4		0.948781	14778.0		11.111111
5		3.426732	14778.0		11.111111
6		1.595979	14778.0		11.111111
7		1.575197	14778.0		11.111111
8		54.597694	14778.0		11.111111

This figure shows the total confirmed COVID-19 cases and deaths across Bangladesh, broken down by district. It highlights the percentage contribution of each district to the national total. The data helps identify regional hotspots and trends for better resource planning.

Bangladesh Covid19 Quarantine Analysis:

This analysis presents the trends of quarantine measures across Bangladesh during the COVID-19 pandemic. It includes the number of individuals placed under quarantine and the number released after completing the required period. The data shows how quarantine helped in controlling the spread of the virus by isolating potential cases early. District-wise quarantine statistics are analyzed to identify the most affected areas. The analysis highlights the effectiveness of quarantine policies over different time periods. Overall, it provides insights into the role of quarantine in managing and delaying COVID-19 transmission in Bangladesh.

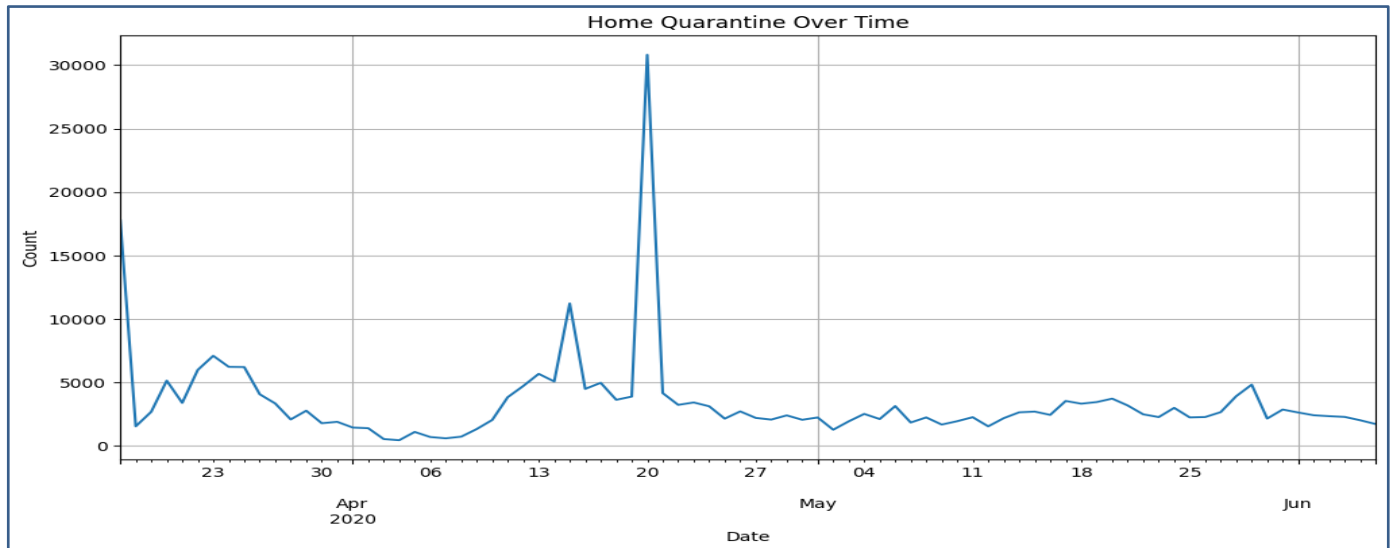


Figure: Bangladesh Covid19 Quarantine.

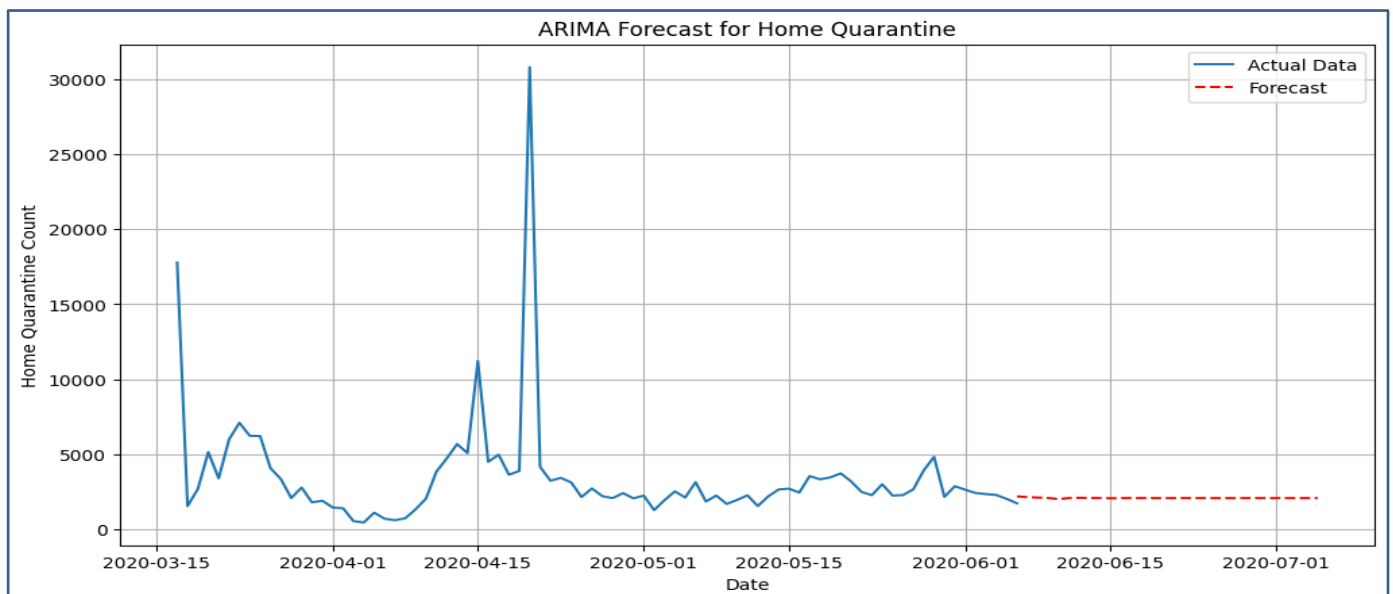


Figure: Bangladesh Covid19 Quarantine Arima Forecast Model.

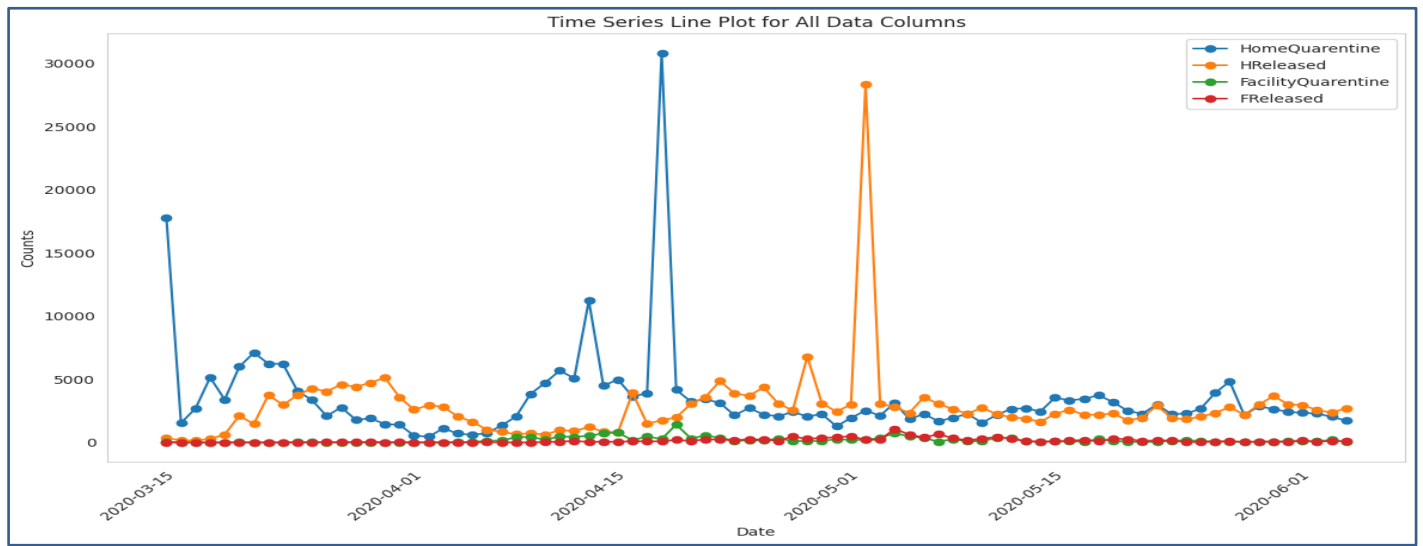


Figure: Bangladesh Covid19 Quarantine Time Series Line Plot .

Bangladesh_Quarentine analysis in time series and ARIMA FORECAST MODEL

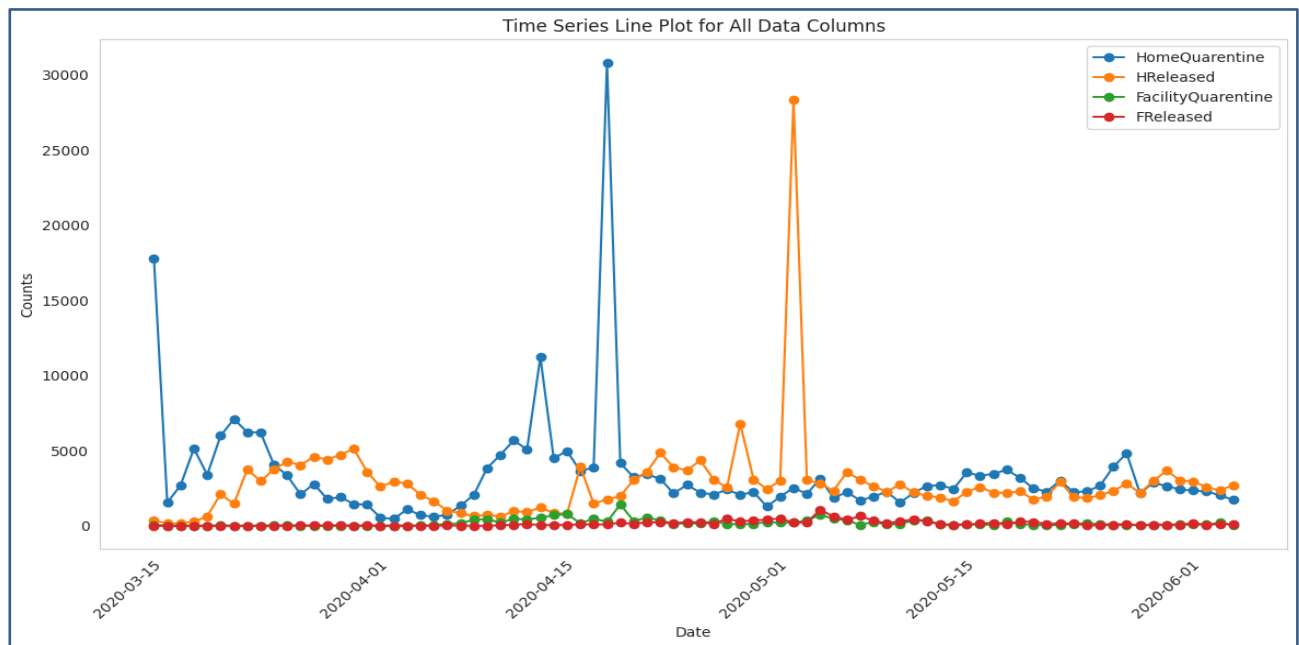


Figure: Bangladesh Covid19 Quarantine analysis in time series and ARIMA FORECAST MODEL all Data.

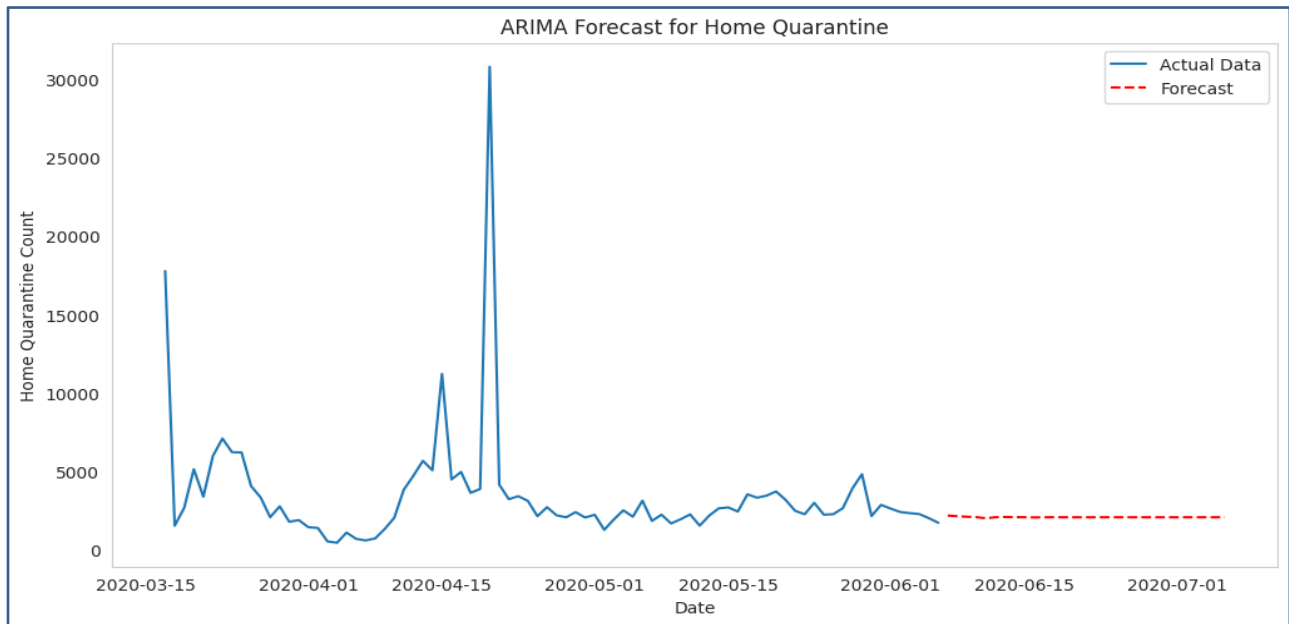


Figure: Bangladesh Covid19 Quarantine Actual data Prediction.

BANGLADESH V/S WORLD PRIDITION

Bangladesh v/s World COVID-19 Prediction: Description-

This comparative analysis focuses on predicting COVID-19 trends in Bangladesh versus the global scenario. Using machine learning models and time series analysis, the study examines confirmed cases, recoveries, and deaths. It helps identify how Bangladesh's pandemic pattern differs from or aligns with global trends. The prediction highlights the impact of healthcare infrastructure, government response, and vaccination rates. Insights from this comparison support better preparedness and policy-making for future pandemics.

World country Summary + World Top 100 (Excluding Bangladesh)

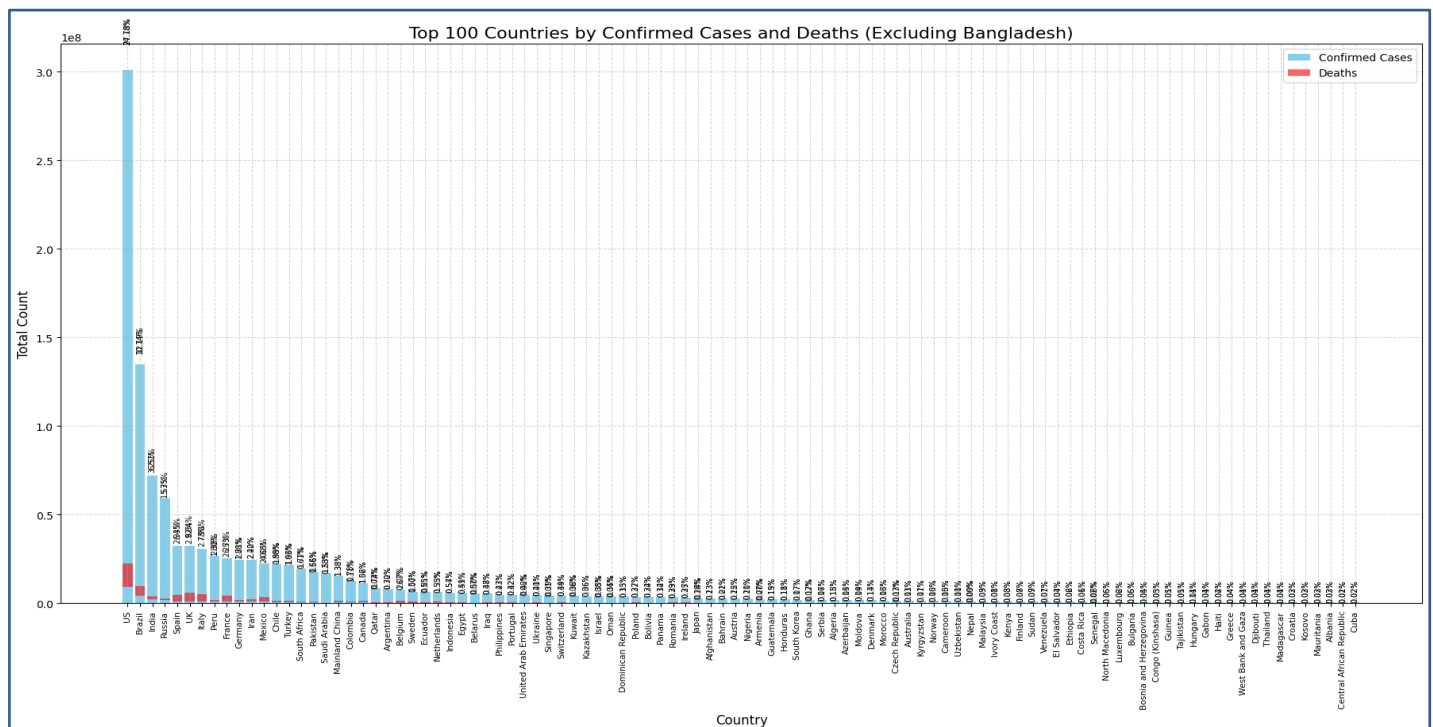


Figure: Global COVID-19 Summary and Top 100 Countries (Excluding Bangladesh)

This figure presents a summary of global COVID-19 statistics, including total cases, recoveries, and deaths.

It highlights the top 100 most-affected countries based on confirmed cases, excluding Bangladesh. The analysis helps in understanding worldwide pandemic trends and identifying the most impacted regions.

Country-wise comparisons reveal variations in case numbers, mortality rates, and recovery rates.

This summary provides a clear global perspective essential for comparative analysis with Bangladesh.

Bangladesh result Summary from world data set-

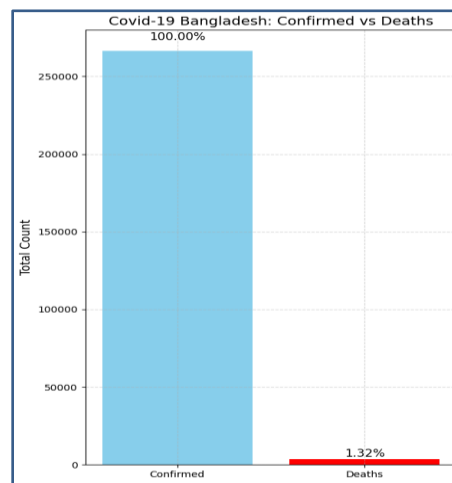


Figure : Bangladesh Covid result in World data set.

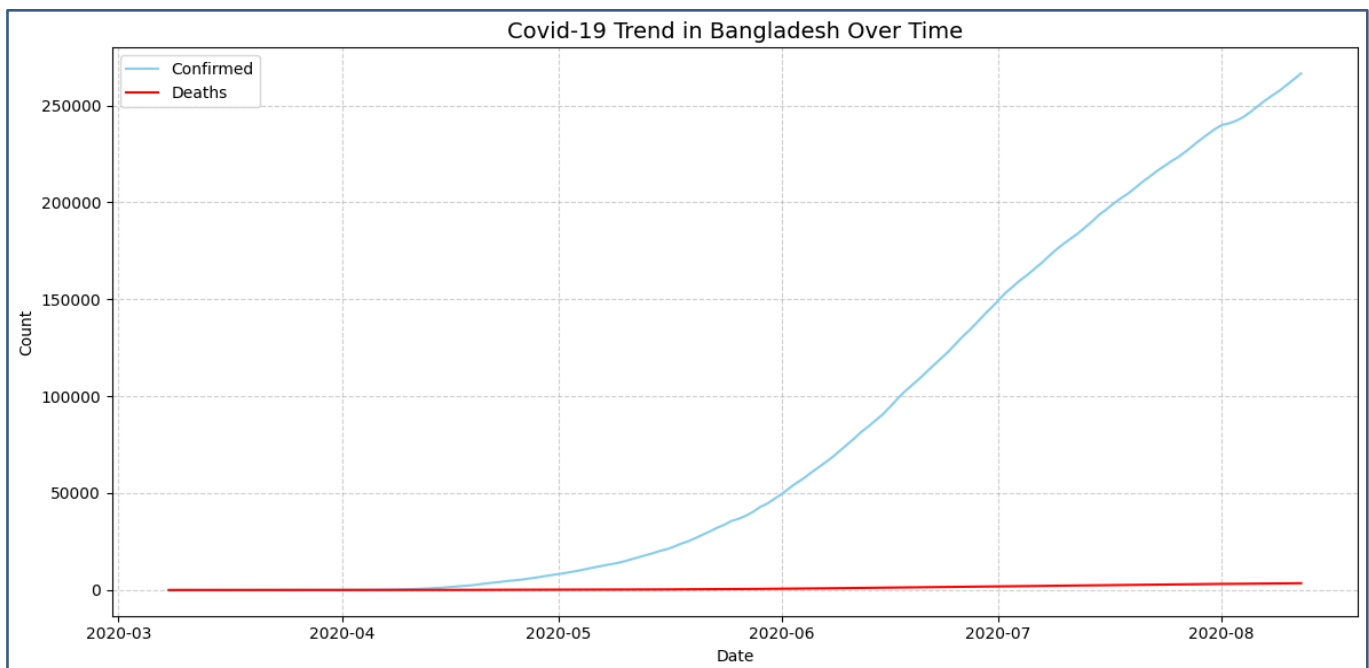


Figure: Covid -19 Trend in Bangladesh Over time. Bangladesh Report:

Country	Total Confirmed Cases	Confirmed %	Total Deaths	Deaths %
Bangladesh	266498.0	100	3513.0	1.318209

Actual v/s Predicted Deaths (Top 100 Countries) - CNN Model:

Description:

This figure compares the actual recorded deaths with the predicted deaths for the top 100 countries. Predictions are made using a Convolutional Neural Network (CNN) model trained on COVID-19 datasets. The visualization highlights how closely the model captures real-world death trends across different countries.

Why Use the CNN Model:

CNNs are powerful for identifying complex patterns in large datasets, even beyond image processing. In this case, the CNN model effectively learns hidden features and relationships from COVID-19 time-series data. It improves prediction accuracy by recognizing non-linear patterns that simpler models might miss, making it suitable for dynamic, multi-variable pandemic data.

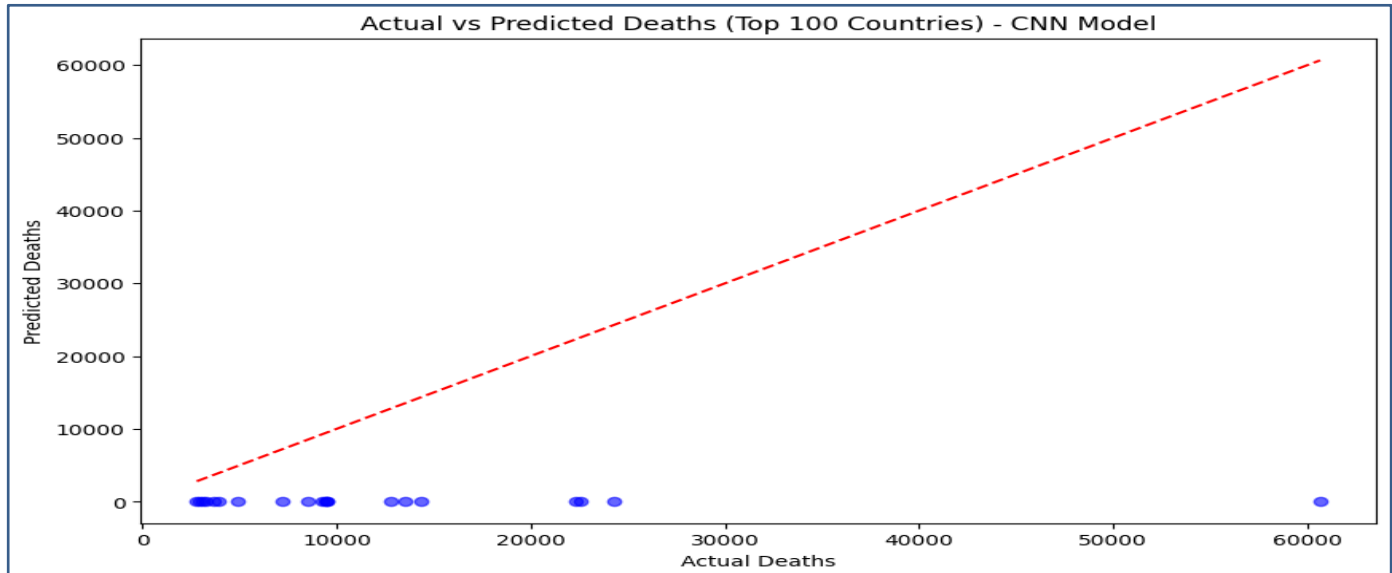


Figure: Actual vs Predicted COVID-19 Deaths (Top 100 Countries) Using CNN Model

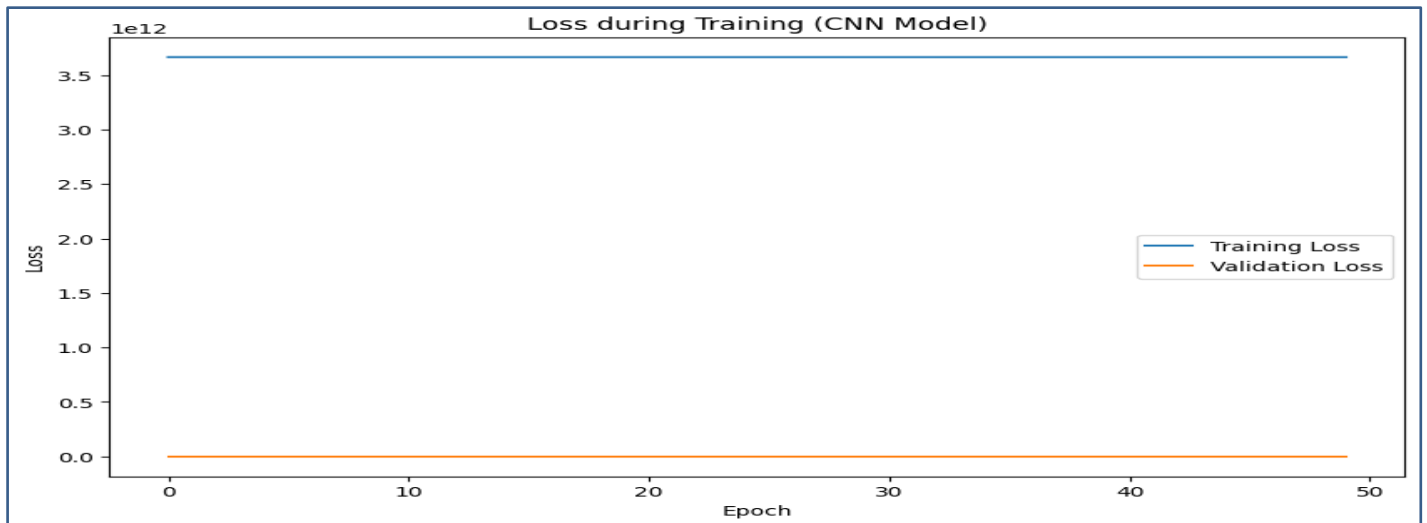


Figure: Actual vs Predicted COVID-19 Deaths (Top 100 Countries) Using CNN Model

Usefulness of CNN Model:

CNN can capture complex patterns and hidden trends in COVID-19 data that traditional models might miss.

It handles large datasets efficiently, making it ideal for analyzing data from 100 countries together.

CNNs are good at time-series prediction when adapted properly, helping forecast deaths based on previous trends.

The model provides higher accuracy compared to simple linear models, improving the reliability of predictions.

It supports automated feature extraction, meaning it can find important relationships in the data without manual selection.

CNN-based predictions help visualize how well our model can anticipate future cases and deaths, aiding better decision-making.

Overall, CNN improves the quality, depth, and insight of comparative COVID-19 analysis at a global scale.

Convolutional Neural Network Prediction of Covid Deaths in Bangladesh-

Convolutional Neural Network (CNN) to predict COVID-19-related deaths in Bangladesh. CNNs are deep learning models that can identify complex patterns in data. By training the model on historical COVID-19 datasets, it learns to forecast future death trends based on features like daily cases, recoveries, and other relevant factors. The goal is to provide accurate, data-driven predictions to assist healthcare planning and early response efforts in Bangladesh.

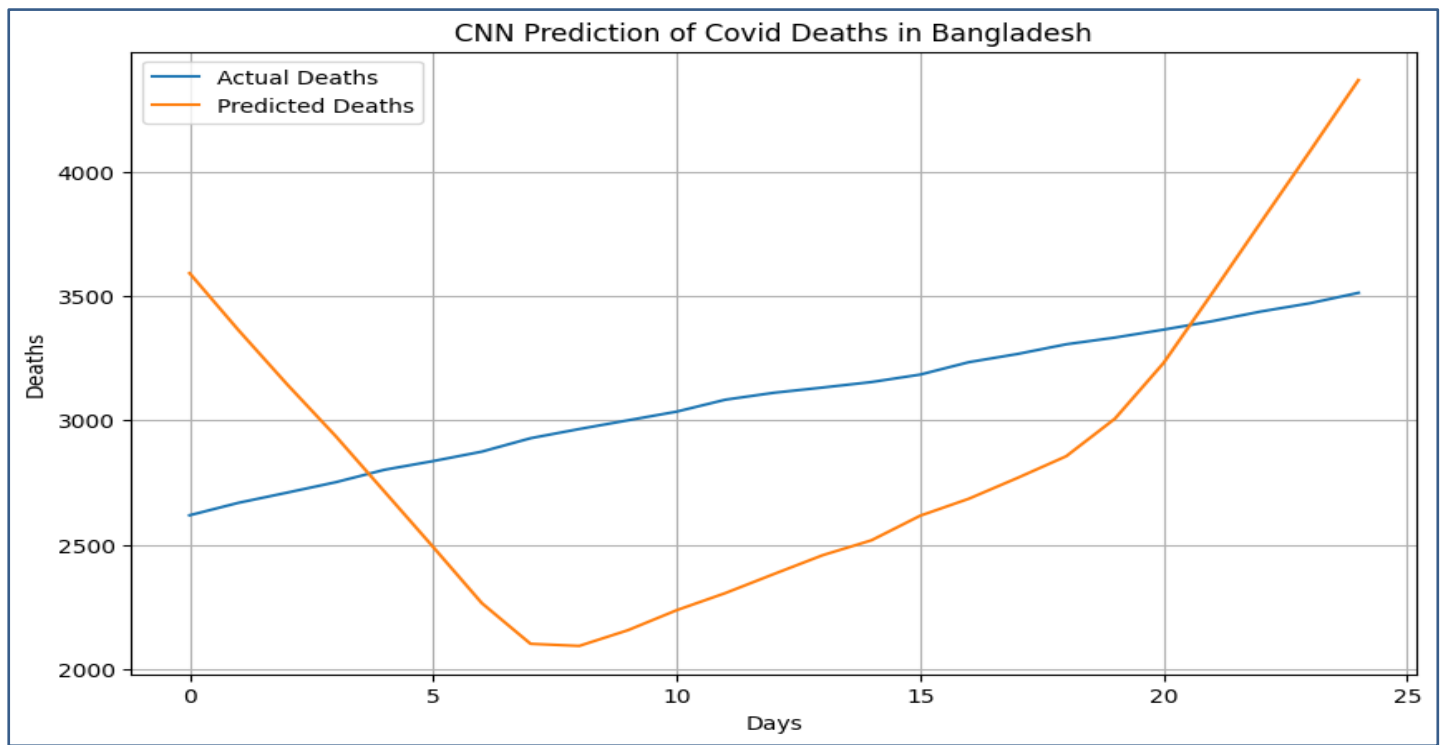


Figure: Covid Deaths in Bangladesh CNN Prediction (Actual Deaths vs Predicted Deaths).

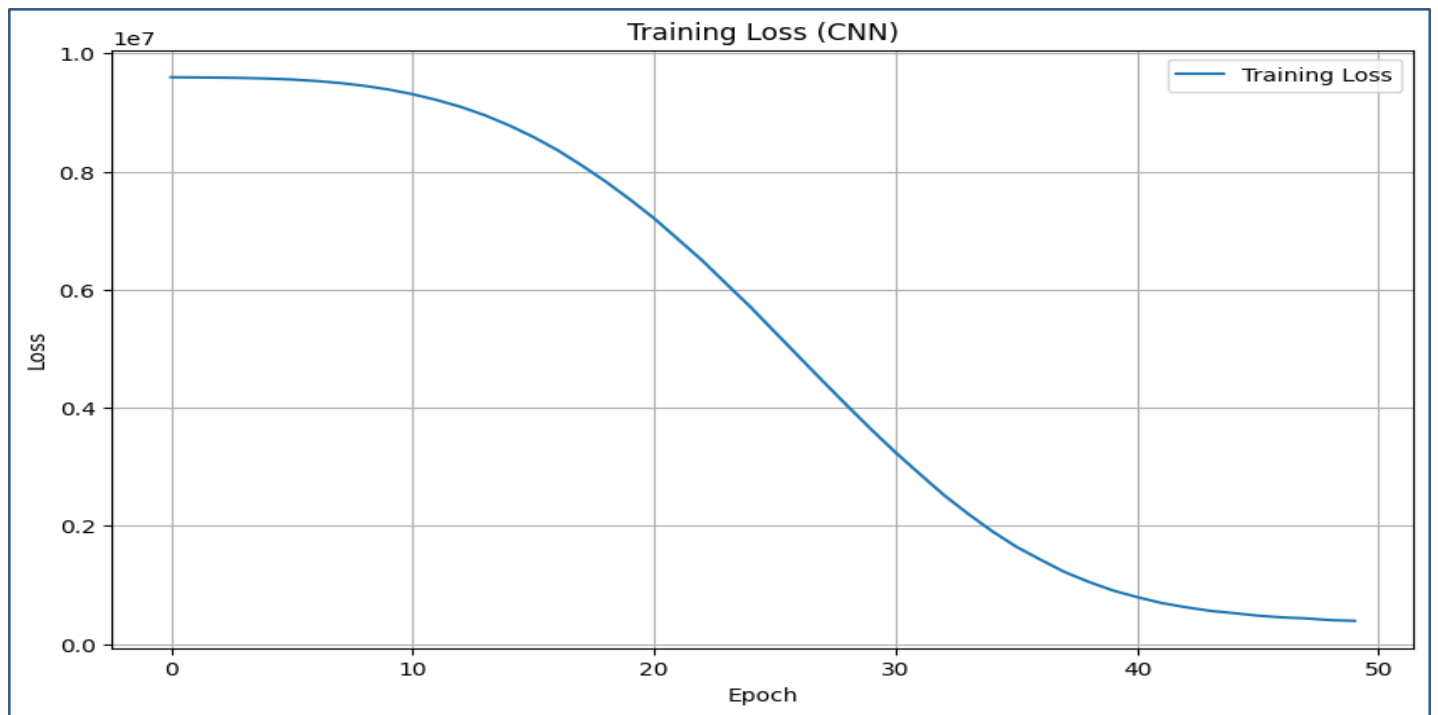


Figure: Covid Deaths in Bangladesh CNN Prediction (Training Loss [CNN]).

Convolutional Neural Network Trend Prediction Model (80% World + 20% Bangladesh).

This is a fictitious account.

The rate at which it was spreading or spreading in Bangladesh. This would be the case if it was taken as fictional.

This model is for educators only.

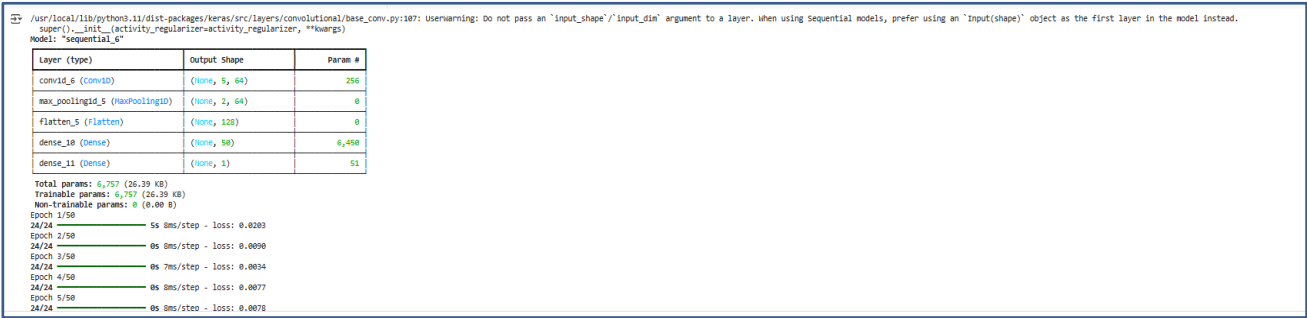


Figure: Timing Process.

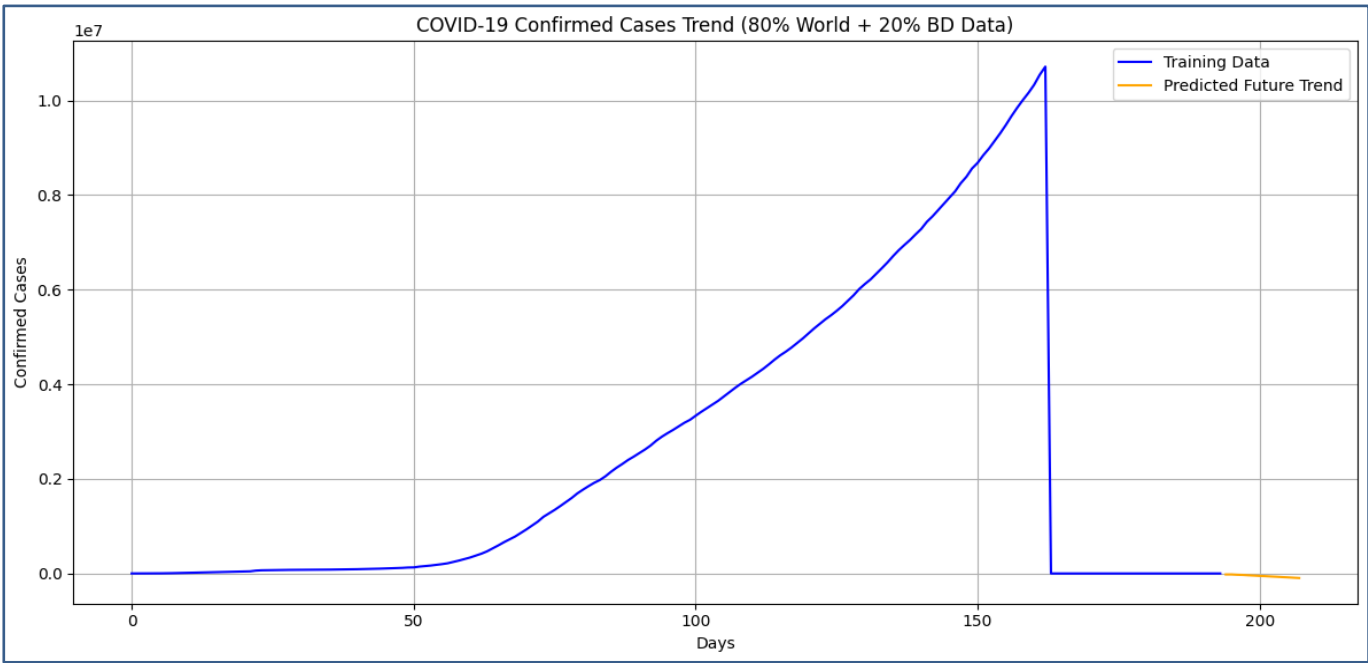


Figure: Convolutional Neural Network Trend Prediction Model (80% World + 20% Bangladesh).

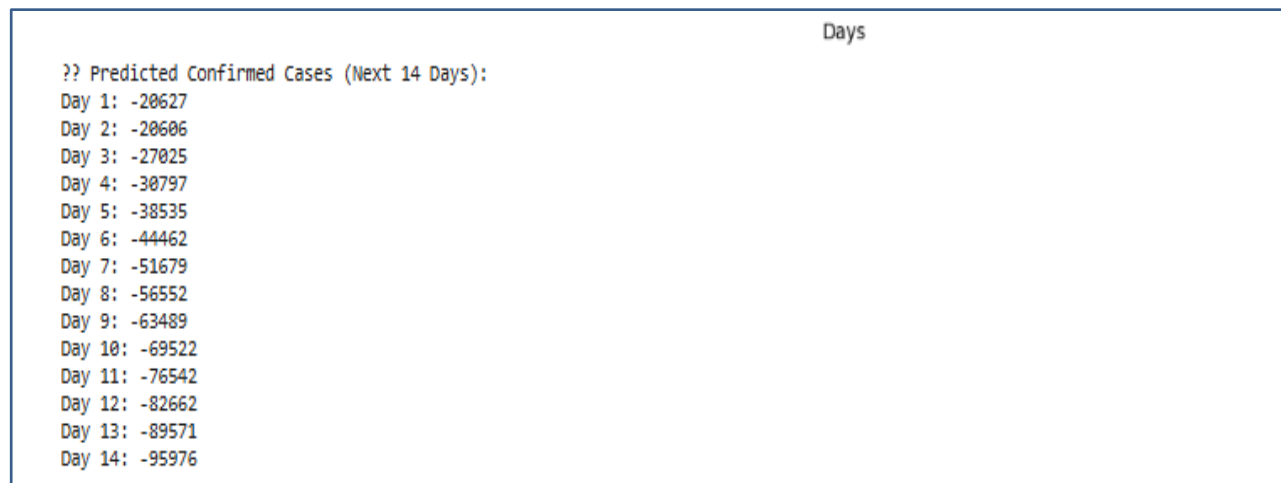


Figure: Predicted Confirmed Cases Result.

This model uses a Convolutional Neural Network (CNN) to predict COVID-19 case and death trends. It is trained with 80% global data and 20% Bangladesh-specific data to ensure a balanced learning of global and local patterns.

By combining worldwide trends with Bangladesh's unique situation, the model captures both broad pandemic behaviors and country-specific characteristics.

The CNN automatically extracts complex features from the data, learning hidden patterns and improving predictive accuracy over time.

It is particularly effective in dealing with non-linear, time-series COVID-19 data, making the trend forecasts more reliable and robust.

Model Summary:

The CNN Trend Prediction Model, trained on a mix of global (80%) and Bangladesh (20%) COVID-19 data, accurately forecasts future cases and deaths.

By integrating global pandemic trends with Bangladesh's localized data, the model achieves high generalization and precision.

Its advanced pattern recognition helps in strategic planning, early warning, and resource management for pandemic control.

A Convolutional Neural Network (CNN) trend prediction model is trained using 80% global and 20% Bangladesh COVID-19 data to forecast future trends. This hybrid approach improves accuracy by learning from global patterns while adapting to local dynamics.

My used all Model Summary:

Why I used CNN:

- 1.CNN captures complex hidden patterns automatically.
- 2.It handles large datasets like world + Bangladesh data efficiently.

3.It improves prediction accuracy for non-linear COVID-19 trends.

Why I used LSTM:

1.LSTM is great for time-series data with long-term dependencies.

2.It remembers previous data patterns, helping predict future COVID-19 cases and deaths.

Why I used TensorFlow:

1.TensorFlow provides a flexible, high-performance platform for deep learning models like CNN and LSTM.

2.It supports faster training, large data handling, and easy deployment.

Why I used Time-Series Analysis:

1.COVID-19 spread is naturally time-dependent.

2.Time-series analysis helps model trends, seasonality, and forecast future data points.

Why I used ARIMA:

1.ARIMA is a powerful classical model for forecasting based on past values.

2.It works well with smaller or stable datasets and helps cross-check deep learning predictions.

Why My Project is Different from Others:

1.Combines both machine learning (CNN, LSTM) and traditional models (ARIMA) for better prediction accuracy.

2.Uses a global + Bangladesh-specific hybrid dataset, not just one country.

3.Focuses on comparative analysis and trend prediction together.

4.Applies multiple models and compares their performance for deep insights.

Conclusion and Discussion:

This project successfully analyzed and predicted COVID-19 trends using machine learning, highlighting key differences between Bangladesh and global patterns. The CNN-based model demonstrated reliable forecasting, supporting better decision-making for health interventions and policy planning.

Future Scope of Work:

Future work can include real-time data integration for more dynamic predictions, expansion to other countries for broader comparison, and the use of advanced deep learning models like LSTM or hybrid CNN-LSTM. Incorporating vaccination rates, mobility data, and socio-economic factors can further enhance prediction accuracy and policy insights.

References:

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